Confined Nanoparticle Diffusion Imaging via iSCAT Microscopy

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Martin Jasinski

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Committee

Sabrina R. Leslie, Ph.D. — *Advisor*
Department of Physics and Astronomy,
University of British Columbia, Vancouver

Abstract

The understanding of characteristics of the thermal movement of particles within a confined space has been studied extensively in order to understand how biological processes, such as actin-network dynamics [1], or chemical processes, such as design of CO_2-capture devices [2]. The sizing of nanoparticles, which are particulates with diameters in the range of 20nm to 200nm, is a common measurement often intrinsically performed under confinement. However, despite the dynamics being theoretically established as a stochastic process, the role of electrostatic and hydrodynamic contributions remain an active field of interest.

By employing common techniques of nanoparticle tracking in tandem with a full theoretical model, the non-trivial effects of confinement on size-diffusivity were directly investigated and characterized. The work described in this thesis advances the method and tools of 3D nanoparticle analysis both theoretically and in application.

The first section of the thesis describes a theoretical model of Brownian motion based on the confined-Langevin model using a Debye-Hückel electrostatic potential. The model was implemented, based on literature [3], into a computer-based PYTHON simulation in order to parameterize and create probability distributions of predicted particle motion.

The second section describes a modified method of 3D tracking to obtain faster calculations of particle position from optical microscopy images. This was based on an algorithm described in Ref. [4]. The validation of the method was performed on analytical simulation data using a model in Ref. [5]. The model was evaluated in range of 1 nm of the ground truth.

The third section describes the acquisition of experimental data using a custom interferometric scattering microscope (iSCAT) with image pixel resolution 55 nm acquiring images between 400 and 1000 frames per second. The data consisted of images of scattering fringes from nanoparticle species and was analyzed using the algorithm developed in Chap. 7. The resultant information was accurate 3D particle tracks. The models developed in Chap. 2 were applied to the tracks to obtain infer particle size and charge as shown in Chap. 8.

Lay Summary

Physical objects are understood to be comprised of smaller subunits. A majority of physically important molecules, such as plastic subunits, medicines, and food particles, are known as nanoparticles, which have diameters ranging from 20 nm to 200 nm. To ensure nanoparticles that are used in products are correctly produced, it is important to be able to measure their functional properties, and their size.

This thesis develops a method to measure the 3D position of nanoparticles using the scattering of light and applies the method to learn about how nanoparticles interact in confined volumes.

This method has potential applications in nanoparticle production. The particle-surface interactions can be directly measured and quantified. Applications include lipid-nanoparticle based medicines, such as the COVID-19 mRNA-Vaccine [6], genetic targeted Onpattro vaccine [7], and most recently, novel cancer targeted mRNA-based immunotherapies vaccines. [8]

Preface

Before you lies the master's thesis *Confined Nanoparticle Diffusion Imaging via iSCAT Microscopy*, written to fulfill the graduation requirements of the Master of Applied Science in Engineering Physics at the University of British Columbia in Vancouver, British Columbia, Canada.

My interest in nanoparticle diffusion developed from prior nanoparticle research that I had worked on as an undergraduate thesis student at the Leslie lab. This study acted as a catalyst which inspired me to further investigate the physical laws and principles of confined single-particle diffusion.

At the time of the thesis start, the iSCAT microscope was recently assembled. The parts were ordered by Dr. Eric Boateng with help from the Grant lab in the department of chemistry at UBC. The system was assembled by Dr. Erik Olsén. The software, written in LabView, was programmed and set up by Dr. Erik Olsén.

The diagrams and plots in this thesis were illustrated by the author using INKSCAPE, PYTHON, and the MATPLOTLIB library [9–11].

In this thesis, the experimental plan and study outline was developed by the author. Simulations in Chap. 6, computational analysis method in Chap. 7, and experimental data in Chap. 8 are original work. Experimental data were acquired with help from Dr. Erik Olsén. Dr. Olsén aided in optical alignment, preliminary data analysis, and overall project guidance. The samples used in chapter 8 were selected by the author and formulation of the lipid-nanoparticles was carried out by Yao Zhang of the Cullis lab at UBC.

I would like to thank Dr. Erik Olsén for project guidance, mentorship, and support during my time as a master's student, Dr. Cynthia Shaheen for support and management support, and all of my colleagues at the Leslie Lab and the department of Physics. Lastly, I would like to thank Dr. Sabrina R. Leslie for the continued support towards academic and professional development. Without these individuals, this thesis would have not been possible.

GenAI* was used to aid in PYTHON plotting and style sheets for a portion of the figures as well as LaTeX formatting code.

*Claude Opus 4.8; Anthropic

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List of Symbols

Symbol	Description
B	Electrostatic Confinement Potential Amplitude
C_{onc}	Concentration
C_{str}	Contrast
C_{pw}	Electrostatic Amplitude Term
D	Diffusion coefficient
e_0	Elementary charge
f_{ocal}	focal length
h	Length
H_{PSF}	Point spread function
h_l	Optical transfer function
$\Im$	Imaginary part
j	Probability current
k	Wave vector
k_b	Boltzmann constant
m	Mass
M	Magnification
N	Number
n_m	Refractive index
r	Polar distance
$\Re$	Real part
R_l	Lens radius

Symbol	**Description**
Re	Reynolds number
t	Time
T	Temperature
V	Volume
$\vec{E}(\vec{x})$	Electric Field
$\vec{H}(\vec{x})$	Magnetic Field
$\vec{v}$	Velocity vector
W	Probability Transition Rate
W_t	Gaussian white-noise process
$\vec{x}$	Position vector
Z	Ion valence number
α	Polarizability
β	$1/k_bT$
γ	Velocity dependent friction
δ_{ij}	Kronecker delta
$\delta(\cdot)$	Dirac delta distribution
ε_0	Permittivity of free space
ε_r	Relative permittivity of the material
ζ_p	Particle zeta potential
η	Fluid viscosity
κ	Inverse Debye length
λ	Wavelength of light
λ_D	Debye-Length
μ	Mobility Coefficient
μ_0	Vacuum magnetic permittivity
$\mu_\parallel$	Parallel (axial) mobility coefficient
$\mu_\perp$	Perpendicular (transverse) mobility coefficient
ρ	Volumetric charge density

Symbol	**Description**
σ	Drift coefficient
σ_w	Surface charge density
τ	Camera exposure and lag-time
$\phi(\Delta)$	Probability of particle in space Δ
$\psi_g(z)$	Gouy Phase
ψ_w	Wall zeta potential
ω	Angular Frequency
Γ	Fluctuation–dissipation magnitude
Γ_i	Stern layer of species i
Ψ	Electrostatic potential
$\mathcal{O}(\cdot)$	Time complexity function

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The completion of this study would have not been possible without the help and resources from Dr. Sabrina R. Leslie, the supervisor of my thesis. I would like to thank Dr. Erik Olsén, a post-doc in the Leslie-Lab, who collaborated and provided guidance to me during this project. I would like to thank Yao Zhang for aiding in sample preparation with lipid-nanoparticles. A debt of gratitude is owed to my dear friends Aiyan Brown, Josh Collesso, and Daniel Currie for aiding in proof-reading and continued support through friendship. Lastly, I would like to thank my father for the unwavering support and interest.

Chapter 1

Introduction

Physical objects are fundamentally divided into approximately three length scales: the macroscopic, governed by classical dynamics, the microscopic, governed by quantum effects and statistical fluctuations, and the mesoscopic, governed by non-trivial dynamics often borrowing from classical, quantum, and statistical mechanics [12]. The classical comparison of length scales depicted by a comparison common objects found in nature is shown in Fig. 1.1. Mesoscopic objects are commonly referred to as nanoparticles where the largest dimension is at most $\sim$200 nm.

Macroscopic objects can be directly measured with a ruler or a photograph. However, as the size drops below the wavelength of visible light, the object's diameter approaches that of the light itself, becoming unresolvable due to a phenomena known as Abbe diffraction [13]. Atomic methods such as x-ray diffraction, neutron scattering, or super-resolution fluorescent microscopy are able to obtain size and structural information with the requirement of an unnatural preparation or artificial perturbation to the sample. Visible light has the unique property of scattering off of matter without altering it's structure, allowing for observation of mesoscopic objects in their natural, unperturbed state.

Due to the diffraction limit, the smallest optically-resolvable size of a mesoscopic object is half of the wavelength of light [14, 15]. Recovering the dimensions of anything smaller, known as sub-diffraction, requires indirect measurements.

The Abbe diffraction limit plays a significant role in observing biological systems with sizes that are sub-diffraction. The sizes of biological matter such as proteins, enzymes, and organelles are not only a structural property but they are also directly relate to functional properties [16]. Considering the lipid nanoparticle, LNP: a drug-delivery pharmaceutical nanoparticle designed to encapsulate a payload such as doxorubicin or mRNA within a lipid-shell to be safely delivered to the liver within the body [6]. Lipid nanoparticles

containing mRNA strands, or mRNA-LNPs, have a therapeutic efficacy dependent on how efficiently the mRNA is released into the cellular environment. The release of mRNA into the cytoplasm depends on a size-selective biological process known as endosomal escape, where therapeutic cargo or pathogens escape from vesicles known as endosomes in order to reach the cellular cytosol and, in the case of mRNA, express into proteins [17–21]. The size-selectivity in this process is an active field of research, however, the size-dependence of the mRNA-LNP is experimentally observable [22, 23].

Since the potency of the mRNA-LNP varies with a parameter related to the size of the particle, observing it in the biological environment gives incredible insight into how it is chemically and structurally regulated [24]. Thus, the precise measurement of particle size is essential to reliable pharmacological production, quality control, and rational design.

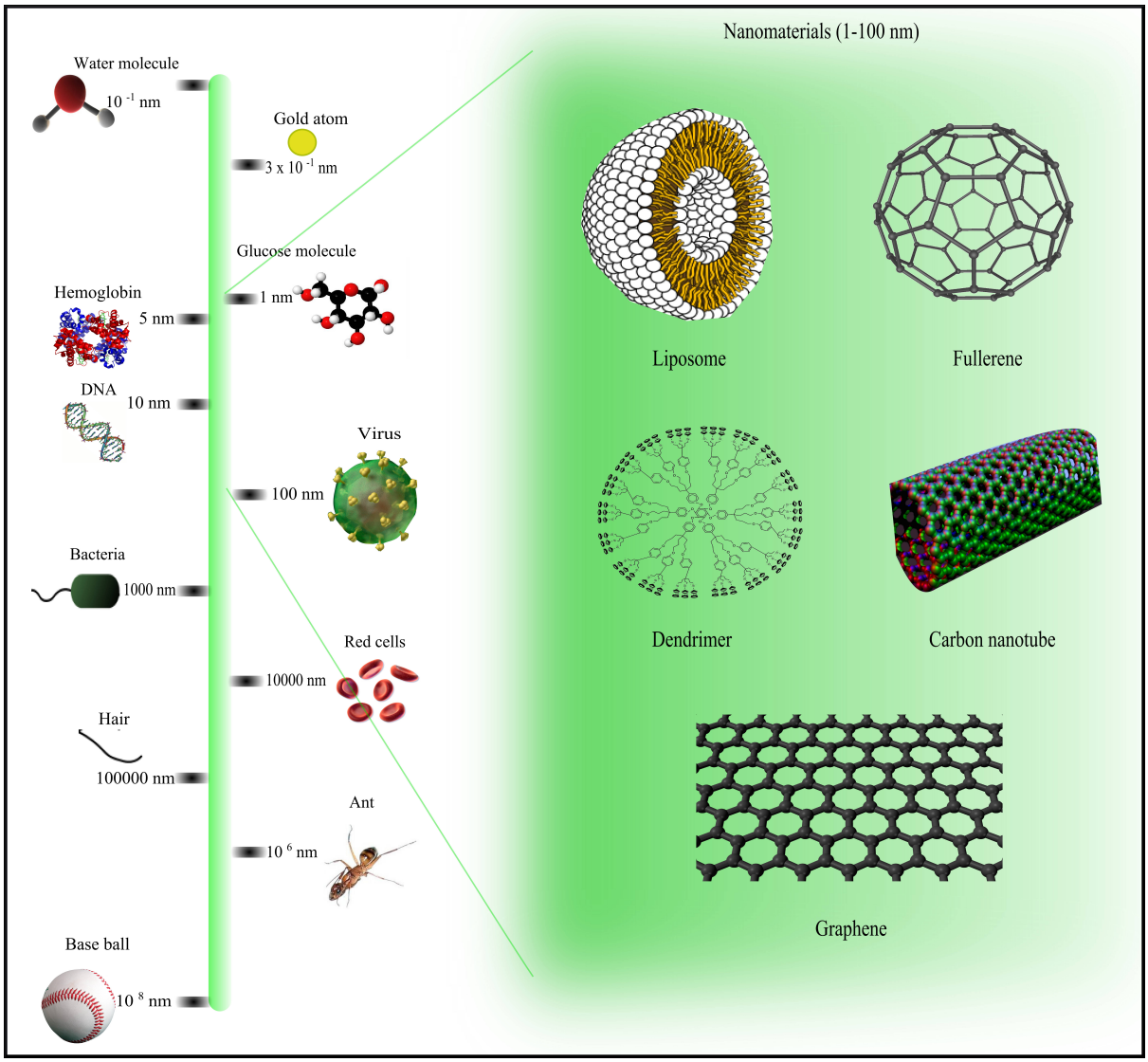

Figure 1.1: Size comparison of various particles. Nanoparticles are defined as particles where the largest dimension is at most ∼200 nm. Figures taken from "Comparison of nanomaterials sizes" by Sureshbup, available under CC BY-SA 3.0 at URL

One universal property of sub-micron particles is that they undergo an apparent size-related stochastic motion when suspended in media [25–27]. At the nano-scale, within most biological systems, thermal energy, defined by the product of the Boltzmann constant and temperature $\sim k_b T$, is the dominant force that acts on the particle. The many external forces, such as gravity and inertia, that typically govern macroscopic dynamics, are negligible in this length-scale. In liquid, which one can reasonably consider as a continuous media, the viscous drag dominates, constituting a random motion within particle motion [*]. The emergence of this thermal motion is typically studied under highly-predictable fluid conditions where mixing is time-reversible. This occurs when the ratio of the inertial forces to viscous forces in a particular media, known as the Reynold's number, Re, is much lower than one, $\mathrm{Re} \ll 1$ [28].

The relation between the particle's size and its motion was first explained in Einstein's seminal paper on Brownian motion [26]. Einstein showed that the diffusion coefficient, a tensor D that describes how fast a particle will randomly move away from its starting point, and the diameter of a spherical particle, can be extracted from the statistical information of a diffusing particle trajectory [27]. In order for the size estimate obtained from a particle trajectory to be statistically relevant, the number of data points, N, in the positional trajectory must be sufficient. Since the statistical uncertainty of the diffusion coefficient scales as $\sim\sqrt{N}$, a higher number of particle locations yields a more precise size inference [29].

Long particle trajectories are typically obtained by imaging the particle position over a period of time using optical microscopy. One common method of optical microscopy is to use fluorescence microscopy. Fluorescence microscopy utilizes fluorophore dyes, often covalently bonded to lipid membranes or proteins[†], to produce optical signal under laser excitation. Although fluorescence signal is high, meaning that the fluorescent molecule is highly distinct from other parts of the image, the fluorophores have finite lifetimes during which they oxidize and stop producing light in a process known as photobleaching [31]. Alternatively to fluorescence microscopy, scattering microscopy is a technique that utilizes the scattering of light off of particle boundaries to obtain signal. This technique requires computational processing to realize images. This process can be imaged indefinitely without loss in signal as particles are not damaged by the light alone.

In order to obtain the requisite large number of particle positions while maintaining independent particle specificity, particles are often spatially confined within the field of view of an optical imaging system for an extended period of time [16]. In the case of a wide-field[‡] microscopy, consisting of an objective collecting light from a single plane

[*] This is also a function of temporal resolution [27].
[†] Sometimes the protein itself, such as in GFP [30].
[‡] As opposed to a confocal microscope.

of focus known as the focal plane. Without axial confinement, particles rapidly diffuse in the axial direction in and out of the focal plane. In recent scientific publications, convex-lens induced confinement, a method of artificially confining nanoparticles into micro-geometries of volume $\sim 1.5\,\mu m^3$ using a convex shaped lens, was used in order to image nanoparticles for temporally extended measurements via fluorescence microscopy [32–34]. In complement, nanoparticle tracking analysis, NTA, measurements follow the same principles using an imaging technique known as side-scattering microscopy. Fluid flow is applied to nanoparticles suspended in a fluid and confined between two plates with a gap on the order of 300 µm. The particles are exposed to an incident laser from a perpendicular angle to the flow. The scattered light from the particles are collected and saved as digital images. Using a time-series of these images, a size reconstruction can be then performed on each track of nanoparticles. In NTA, however, particles are still able to diffuse in and out of the focal volume [35].

Although experimentally feasible, confinement in small spaces influences the motion of particles near surfaces. This property intrinsically modifies the size-diffusivity relation. This effect is non-trivial, introducing a position dependent viscous drag that has been shown to significantly alter the measured diffusivity [27, 36].

To account for near-wall effects, the particle must be tracked in three-dimensions with respect to the confinement barriers. Two-dimensional x, y tracking, which is commonly performed with optical microscopy alone, is theoretically insufficient to recover the true size and to find spatially varying diffusion coefficient [37]. Since the particle's drag friction is dependent on its axial position with respect to nearby surfaces, a true size-estimate requires the axial-position dependent drag friction accounted for in each measured position; only possible in a three-dimensional reconstruction of the trajectory. Without tracking the axial dimensions, the spatially dependent correction is uncertain, which causes size estimates to collect statistical bias.

In hopes of more accurately measuring physical properties of nanoparticles, such as diffusivity and size, this thesis presents an advancement of the iSCAT technique allowing for rapid and accurate three-dimensional tracking reconstruction. From the three-dimensional trajectory, the diffusivity, size, and environment conditions can be easily determined. Additionally, the confined axial size-measurement technique is shown to be a media insensitive approach; complementary to trajectory-based methods.

The thesis consists of 8 chapters. Chapter 2 provides a description of the literature of confined Brownian motion over the last century. Chapter 3 describes the theoretical framework of interferometric scattering microscopy, iSCAT, and physical limits that apply in optical theory. Chapter 4 discusses the application of Brownian motion and iSCAT in a biological context, with details regarding lipid-nanoparticles and why they need to be

studied on the single-molecule level. Chapter 5 consists of exploration of confined diffusion using Langevin simulations. A simulated system was developed and the size-diffusivity relation in an electrostatically screened system was investigated. Chapter 6 describes the 3D iSCAT tracking technique developed during this work. Chapter 7 describes the method applied to experimental data and utilizes various nanoparticle species, highlighting their varied diffusions. Chapter 8 concludes the work and provides a future outlook on the topic and steps which need to be advanced.

Chapter 2

Brownian Motion

The physical phenomenon of irregular movements of nanoparticles in liquid is known as Brownian motion. The observation of this motion originated around the year 1827 when Robert Brown observed a stochastic motion of clay particles originating from plant pollen under his microscope [25]. This stochastic motion, coined as Brownian Motion, was discovered to be inversely related to the size of the particle by Albert Einstein in 1926 [26].

This chapter aims to establish the theoretical background underlying the relationships between the diffusivity, particle size, and spatial coordinate in order to be applied to experimental data analysis in further chapters. Stochastic processes, are generally not fully describable within a single displacement. However, as one observes the evolution of the process for a longer period of time, the randomness tends to converge such that it conforms.

Einstein's diffusion equation argument and Langevin's stochastic equation of motion are discussed and shown to have implication* through the Fokker-Planck formalism. In the case of diffusion under confinement, the hydrodynamic wall effects modify the mobility. In addition, electrostatic interactions, described by the Debye-Hückel potential, govern the particle-wall dynamics in the presence of an ionic solution. Finally, equilibrium distributions are discussed with regards to spatial probability distributions arising in confined diffusion processes.

*Meaning that the Langevin formulation converges to the Einsteinian formulation.

2.1 Einstein's Brownian Motion

The physical phenomenon of diffusion can be rigorously described by the emergent behavior that occurs in particles of microscopic length when undergoing random collisions with members of the surrounding environment when the particles are large relative to their surroundings. In the microscopic limit, where the radius of an approximately spherical particle is sufficiently small causing the gravitational and surface drag contributions to be negligible,the thermal fluctuations of nearby fluid collection dominate the particle dynamics. This erratic motion is referred to as Brownian motion [25].

The classical treatment of Brownian motion describes how the molecular-kinetic theory of heat relates with microscopic particle suspensions in liquid or gas [26, 38]. Einstein's argument is one of macroscopic motion. Using the original notation of Ref. [26], suppose that $\phi(\Delta)$ denotes the probability to find a particle a volumetric space Δ subject to the normalization

$$\int \phi(\Delta) d\Delta = 1. \tag{2.1}$$

Assuming that the medium is isotropic, such that $\phi(\Delta) = \phi(-\Delta)$ is even, let $f(x,t)$ denote the number of particles at the point x at time t. Einstein then takes a finite slice of this volume and equates it to a probability space:

$$f(x, t+\delta t) = \int f(x+\Delta, t)\phi(\Delta) d\Delta \,, \tag{2.2}$$

where δt is an infinitesimal increment in time. This integral states that the density of particles at some infinitesimal time $t+\delta t$ can be measured by adding the contributions of the density of particles nearby the point $x+\Delta$. Thus, the evolution of particle number in time can be calculated by adding all possible spatial displacements Δ.

By utilizing a Taylor-expansion of the left-hand side of this equation, dominant terms of the dynamics can be approximated:

$$f(x, t+\delta t) \approx f(x,t) + \delta t \frac{\partial f(x,t)}{\partial t} \,. \tag{2.3}$$

Similarly, the change in number of particles can be Taylor-expanded as an approximation of the spatial dimension:

$$f(x+\Delta, t) \approx f(x,t) + \Delta \frac{\partial f(x,t)}{\partial x} + \frac{\Delta^2}{2} \frac{\partial^2 f(x,t)}{\partial x^2} \,. \tag{2.4}$$

Adding together these terms and cancelling out the opposites, the resultant equation is

known as the classical diffusion equation, and is shown to nearest order;

$$\frac{\partial f(x,t)}{\partial t} = D\frac{\partial^2 f(x,t)}{\partial x^2}, \tag{2.5}$$

where the diffusion coefficient is associated with the second moment of the probability distribution associated with the particle's density:

$$D = \frac{1}{\delta t}\int \frac{\Delta^2}{2}\phi(\Delta)d\Delta\,. \tag{2.6}$$

Additionally, Einstein had shown, by describing the osmotic pressure ascribed to suspended particles in fluids, that the diffusion coefficient was shown to be inversely proportional to the friction coefficient, γ [26];

$$D = \frac{k_b T}{\gamma}\,. \tag{2.7}$$

The friction coefficient is physically related to the solution of the Navier-Stokes equations at a low Reynolds number† $\mathrm{Re} \ll 1$. Specifically, when the inertial forces are negligible and $\mathrm{Re} \ll 1$, then the drag force experienced by a particle is linearly related to the velocity, as stated by Stoke's equation:

$$F_d = 6\pi\eta a v = \gamma_0 v, \tag{2.8}$$

where $\gamma_0 = 6\pi\eta a$ is the drag coefficient, η is the fluid viscosity, and a is the spherical particle's radius. This relation substituted can be directly applied, by letting $\gamma = \gamma_0$, to the diffusion coefficient relation to recover Einstein's famous formula:

$$D = \frac{k_b T}{6\pi\eta a}\,. \tag{2.9}$$

This equation states that the diffusion, or rate of spread, of a particle in space is directly proportional to is size. This fact allows one to postulate that the measurement of sub-diffraction particles can be done by reconstructing their trajectory and utilizing their motion to calculate the expected size.

The solutions to Eqn. 2.5 describe the time evolution of the probability density of the Brownian particles. The solutions can be generated by using the Green's functions for the d-dimensional Laplacian with Neumann boundary conditions. In the example of an infinitely far boundary, known as the free-space boundary condition in d dimensions, the probability density of displacements of a particle centered around $\vec{x} = 0$ is described by a

†Which describes the relative inertial forces to the viscous forces.

Gaussian distribution with zero mean and finite variance,

$$f(\vec{x},t) = \frac{1}{\sqrt{4\pi dD\Delta t}} \exp\left(-\frac{|\vec{x}|^2}{4dD\Delta t}\right), \tag{2.10}$$

where d is the number of dimensions, and Δt is the difference from a known initial condition. The variance of a diffusion process distribution is referred to as the mean-squared displacement and is described as the variance of the displacement probability distribution,

$$\langle x^2 \rangle_c = 2dD\Delta t\,. \tag{2.11}$$

The mean-squared displacement is an intuitive picture that relates the statistics of a particle trace and it's diffusion coefficient. As the particle jumps around, the amount of variability in the length of these jumps will eventually be directly observable with enough measurements. This variability is then directly related to the physical coefficient of diffusion, which describes the spread in the length of the displacements. Experimentally, one can use the definition of sample variance to measure $\langle x^2 \rangle_c$:

$$\langle x^2(t) \rangle_c = \frac{1}{N-1} \sum_{n=1}^{N} |x(t) - x(0)|^2\,, \tag{2.12}$$

where $x(t)$ is the full time-series of positions, $x(0)$ is chosen as the starting reference position, and N is the total number of displacements [‡].

Einstein's theory of diffusion instantiates a partial differential equation that describes how the size of a particle relates to a thermodynamic motion as described in Eqn. 2.5. The Stokes-Einstein relation in Eqn. 2.9 which describes the relation between diffusion coefficient and particle size, the experimental form of the mean-squared displacement to measure the experimental variance in Eqn. 2.12, and the forms of the mean-squared displacement for various boundary conditions as found in Eqn 2.11 and further in Ref. [39], the size of particles can be directly measured from time-series position information.

Einstein's original paper directly motivated the sizing of particles using statistical estimates of their random motion, specifically with the existence of the mean-squared displacement relation. This relation allows one to utilize a particle trajectory, or single-particle motion as an indirect measurement of size; one of the cornerstone concepts of single-molecule optical measurement.

[‡]For a more in-depth discussion around calculating this quantity, see Ref. [27]

2.2 Langevin Dynamics

In single-particle movement-based experiments, such as observations of Robert Brown's particulates or tracking of diffusing lipid-nanoparticles, the observed Brownian particle moves within a fluid[§] volume. If one assumes that this particle exists in the thermodynamic equilibrium, it should obey the equipartion theorem [38, 40]. However, the observed motion appears to have more energy than predicted by the equipartion at certain times, indicating that an additional contribution to the dynamics must be considered [41, 42]. This missing component of motion can be described by the bombardment of the surrounding particles in the liquid. In particular, the system experiences a random force on it at every time step,

$$m\ddot{\vec{x}} = -\frac{\dot{\vec{x}}}{\mu} + \vec{F}(t)\,, \tag{2.13}$$

where μ is the fluid mobility represented as a viscous friction term, $-\frac{\dot{\vec{x}}}{\mu}$. Lastly, $\vec{F}(t)$ is the stochastic force term, which strictly speaking is a random function, taking a random value from a distribution, p_F for each time step, t. This model, although easily written, is quite profound and its analytical treatment is non-trivial. Here, $x(t)$ is a stochastic process, where each realization of $F(t)$ produces a unique trajectory $x(t)$.

The statistical content of $\vec{F}(t)$ is determined by the moments and auto-correlation. Notably, this relation is physically described by the fluctuation-dissipation theory, which relates the drag force to the random force [38]. In Brownian motion, this takes the form

$$\langle\vec{F}(t)\rangle = 0, \quad \langle\vec{F}_i(t)\vec{F}_j(t')\rangle = \Gamma\delta_{ij}\delta(t-t') \tag{2.14}$$

which is known as delta-correlated noise; in particular, that of a Wiener process. Notably, Γ is constrained by the equipartion theorem, as can be shown by calculating the velocity autocorrelation function and choosing the large t limit to agree with the classical thermodynamics [38]:

$$\Gamma = \frac{2k_bT}{\mu}\,. \tag{2.15}$$

One can take the ensemble average of this equation and, notably, extract correlation information. Following the derivation of Ref/ [38], it is common to rewrite Eqn. 2.13 in terms of velocity, and assuming zero drift velocity:

$$m\dot{\vec{v}} = -\frac{\vec{v}}{\mu} + \vec{A}(t), \quad \int_{-\infty}^{\infty} A(t)dt = 0\,. \tag{2.16}$$

This represents a linear differential equation with a solution that defines the velocity

[§]This thesis assumes Newtonian fluids as they are experimentally relevant.

function. It is interesting to see how the velocity of the particle changes as it evolves in time. To obtain this information, Eqn. 2.16 can be averaged by a reference velocity function, obtaining the velocity auto-correlation function:

$$\langle \vec{v}(0) \cdot \vec{v}(t) \rangle = \frac{1}{m\mu} e^{-t/(m\mu)} \,. \tag{2.17}$$

The velocity auto-correlation function and the MSD are closely related. A single spatial displacement can be calculated from the velocity function of the particle,

$$x(t_2) - x(t_1) = \int_{t_1}^{t_2} v(t')dt' \,. \tag{2.18}$$

This equation can be squared and averaged to obtain the mean-squared displacement on the left-hand side,

$$\langle |x(t_2) - x(t_1)|^2 \rangle = \left\langle \left(\int_{t_1}^{t_2} v(t')dt' \right) \left(\int_{t_1}^{t_2} v(t'')dt'' \right) \right\rangle \,. \tag{2.19}$$

The right hand side can then be evaluated and written in terms of the velocity auto-correlation, defined as the average over a product of two velocity functions evaluated at different times.

$$\langle |x(t_2) - x(t_1)|^2 \rangle = \int_{t_1}^{t_2} \int_{t_1}^{t_2} \langle v(t')v(t'') \rangle dt'dt'' \,. \tag{2.20}$$

Let $t_1 = 0$, and evaluate the right hand side integral substituting Eqn. 2.17. Utilizing the substitution of $\langle v^2 \rangle = \frac{3k_bT}{m}$, then the solutions of the position mean-squared displacement are evaluated

$$\langle x^2 \rangle = 6k_bTm\mu^2 \left[\frac{t}{m\mu} - (1 - e^{-t/(mB)}) \right] \,. \tag{2.21}$$

The limits of this equation can be evaluated independently. The short-time limit, known as the ballistic limit, is a statement that describes the motion of a particle before the drag starts to slow it down,

$$\xrightarrow{t \ll m\mu} \frac{3k_bTt^2}{m} = \langle v^2 \rangle t^2 \,. \tag{2.22}$$

The long-time limit is a statement of mean-square displacement. As the particle diffuses for a long time, we expect the MSD to converge to that of the ensemble limit or Eqn. 2.11.

$$\xrightarrow{t \gg m\mu} \frac{6k_bTm\mu}{m} t = (6\mu k_bT)t \,. \tag{2.23}$$

Noting Eqn. 2.9, Einstein's relation is recovered where the mobility, being the inverse of

the friction, can be substituted by the Stokes friction,

$$\mu = \frac{1}{6\pi\eta a}\,. \tag{2.24}$$

The unique advantage of the Langevin formulation is that it can be readily simulated. [3, 43]. It can be shown that using an Euler-Maruyama scheme[¶], a mathematical technique used to iterate stochastic differential equations in time-steps, the stochastic differential equation can be integrated over realizations of the stochastic force $F(t)$ using a random number generator, recovering the dynamics of an ideal Brownian system [3].

[¶]The EM scheme is chosen due to relevance in Ref. [3].

2.3 From Master Equation to Fokker-Planck

To show that the Langevin formulation implies the Einstein's formulation, the master equation can be shown that the Langevin equation has the same equilibrium probability distribution as the Einsteinian formulation. Suppose that $f(x,t)dx$ represents the probability of an arbitrary particle in the ensemble making a displacement between x and $x+dx$. Let W represent the traditional probability transition rate between positions, then the master equation is written as

$$\frac{\partial f(x,t)}{\partial t} = \int_{-\infty}^{\infty} \left[f(x',t)W(x'|x) - f(x,t)W(x|x') \right] dx'. \tag{2.25}$$

This equation is a statement about the probability of positional transitions, which is fundamental to the Langevin equation[‖] [44]. Since the Langevin equation is memory-less and Gaussian, the right hand side can be expanded using the Kramer-Moyal expansion and truncated at second order, yielding the Fokker-Planck Equation [44]

$$\frac{\partial f(x,t)}{\partial t} = -\frac{\partial}{\partial x}\left[\sigma(x) f(x,t)\right] + \frac{1}{2}\frac{\partial^2}{\partial x^2}\left[D(x) f(x,t)\right] \tag{2.26}$$

where the diffusive and drift terms are parameterized over the transition probabilities

$$\begin{aligned} \sigma(x) &= \int_{-\infty}^{\infty} \xi W(x|\xi) d\xi \\ D(x) &= \int_{-\infty}^{\infty} \xi^2 W(x|\xi) d\xi \,. \end{aligned} \tag{2.27}$$

In the case of zero drift and homogenous diffusivity, the form of Eqn 2.26 is equal to the diffusion equation, Eqn 2.5, of the Einsteinian formulation. Thus, the Langevin formulation implies the Einsteinian formulation in the long-time limit [45]. This important property allows us to investigate solutions and dynamics of the two systems interchangeably. Additionally, the results from either formulation are intrinsically the same, which implies that the probabilistic distributions of the Einsteinian formulation can be recovered from running Langevin simulations over a long time-course for an equilibrium Markovian chain. Thus, non-analytical systems, such as non-homogenous cases of the diffusion equation, can be investigated in this context using simulations without a loss of applicability.

[‖] Given a Markovian system.

2.4 Confined Diffusion Processes

The effect of diffusion near an interface with a nearby particle is non-negligible [27]. The effect of confinement can be first understood through the lens of the effects due to from friction from fluid flow. Notably, the particle being immersed in a fluid invites a Stokes friction which affects the perpendicular and parallel motion. Swedish physicist Hilding Faxén derived the mobility relations with a no-slip boundary conditions near a Brownian particle [27, 46]:

$$\begin{aligned} \mu_{\parallel}\left(\frac{a}{h}\right) &\approx \mu_{\infty}\left[1-\frac{9}{16}\left(\frac{a}{h}\right)+\frac{1}{8}\left(\frac{a}{h}\right)^{3}-\frac{45}{256}\left(\frac{a}{h}\right)^{4}-\frac{1}{16}\left(\frac{a}{h}\right)^{5}\right]; \\ \mu_{\perp}\left(\frac{a}{h}\right) &\approx \mu_{\infty}\left[1-\frac{9}{8}\left(\frac{a}{h}\right)+\frac{1}{2}\left(\frac{a}{h}\right)^{3}\right]. \end{aligned} \tag{2.28}$$

The above equations describe the change in mobility from a single planar wall as a Brownian particle approaches it. However, in the case of non-uniform surfaces, or multiple surfaces, the single-wall approximation cannot account for the non-trivial fluid dynamics that originate from the Navier-Stokes equations. Thus, for more elaborate geometries, further work must be done.

In the case of a double-wall confinement, there is much theoretical discussion on the confinement coefficient in hydrodynamic boundary conditions [47–49]. In particular, the two-wall confinement is a symmetric potential where the Navier-Stokes equations can still be evaluated. In application, it has been shown that the Onseen drag shows experimental agreement [50] and can be expressed in a linear-superposition of linear mobility coefficients

$$\gamma_2(z) \approx \gamma_0 + [\gamma_1(z) - \gamma_0] + [\gamma_1(H-z) - \gamma_0], \tag{2.29}$$

where γ_2 is the overall drag, which is inverse to the mobility coefficient, μ. Additionally, $\gamma_1(z)$ is the inverse one of the Faxén mobility coefficients $\mu_{\perp}$ or $\mu_{\parallel}$, and γ_0 is the Stokes drag coefficient found in Eqn. 2.9. This linear-superposition of terms allows for an experimentally motivated drag that derives from a linear-superposition of in-plane or out-of-plane mobility coefficients.

In context of single-particle sizing measurements, the distance between the particle and nearby surfaces is often small [33, 34, 51]. Thus, in applications of confinement-based particle sizing, the confined particle friction term must be applied in order to obtain a non-biased estimate. It has been shown in literature this form of a system can be simulated using a Langevin equation [3].

2.5 Self-Consistent Confinement Coefficient

Particle sizing measurements in confinement will be biased due to the non-homogenous nature of their diffusion coefficient in a confined environment. To correct for this and obtain an approximation of the bulk diffusion coefficient of such measurements, the mobility relations presented in Eqn. 2.28, which relates the bulk diffusivity to the confined axial or transverse diffusion coefficients, must be used. By substituting $\gamma_0 = \mu_\infty^{-1}$ and substituting it into Eqn 2.9, the bulk diffusion coefficient can be related to the confined diffusion coefficient by a scalar value,

$$\frac{D_\infty}{\gamma_0} = \mu_\parallel^{-1} D_\parallel = \mu_\perp^{-1} D_\perp \,, \tag{2.30}$$

where D_∞ is the bulk diffusivity, γ_0 is the Stokes drag, $D_\parallel$ is the confined diffusivity in the transverse x, y plane, $mu_\parallel$ is the transverse confined mobility coefficient along the wall. $D_\perp$ is the confined diffusivity in the axial x, y plane into the wall, and $mu_\perp$ is the transverse confined mobility coefficient. The values of the corrections are, however, functions of the particle radius a. For the Stokes-Einstein relation and the Faxén mobility corrections to be consistent, the following relation must hold.

$$\frac{k_b T}{6\pi\eta a} = \mu_\parallel^{-1}(a,h)\,\hat{D}_\parallel = \mu_\perp^{-1}(a,h)\,\hat{D}_\perp, \tag{2.31}$$

where the $\hat{\cdot}$ indicates measured values, a is the particle radius, and h is the width of the two-wall confinement. This condition is self-consistent with a general mobility that is dependent on the particle radius if the particle height is known**. One useful property of linear-superposition approximations in Eqn. 2.29, opposed to the Faxén relations, is that it is monotonically decreasing, illustrated in Fig. 2.1.

In order to ensure that the drag term $\mu^{-1}(a,h)$ is consistent with the particle size estimate that is obtained from evaluating Eqn. 2.9, the self-consistency metric can be constructed as shown in App. B. This relation has been previously used in analytical single-molecule measurements for size measurements in a confined environment [33]. The self-consistency scales the estimated particle size $\hat{a}$ by a scalar that minimizes the difference between the size recovered from the Stokes-Einstein relation and that of final scaled product:

$$a^* = \arg\min_{a^*} \left| \frac{\hat{a}}{\mu_{\parallel,\perp}} - a^* \right| , \tag{2.32}$$

where a^* is the self-consistent particle size, $\hat{a}$ is the estimated size from the Stoke-Einstein, $\mu_{\parallel,\perp}$ is the linear-superposition inverse drag using one of the Faxén mobility relations. To demonstrate the self-consistency of a single particle evaluation, a grid search was

**As was applied in Ref. [33]

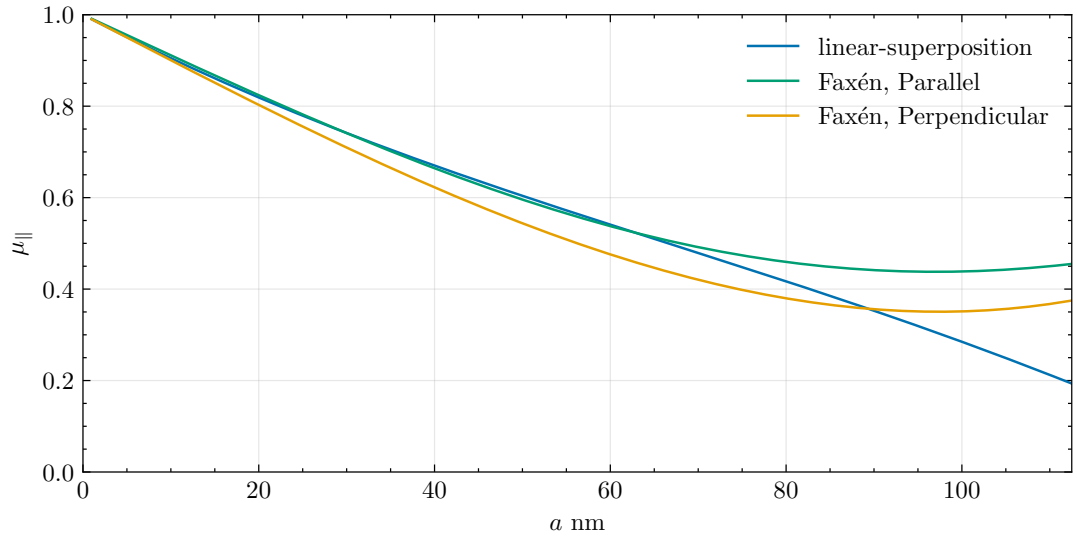

Figure 2.1: Linear-superposition mobility curve for linear-superposition and Faxén relations with $h = 225\,\text{nm}$.

performed on an example of as illustrated in Fig. 2.2.

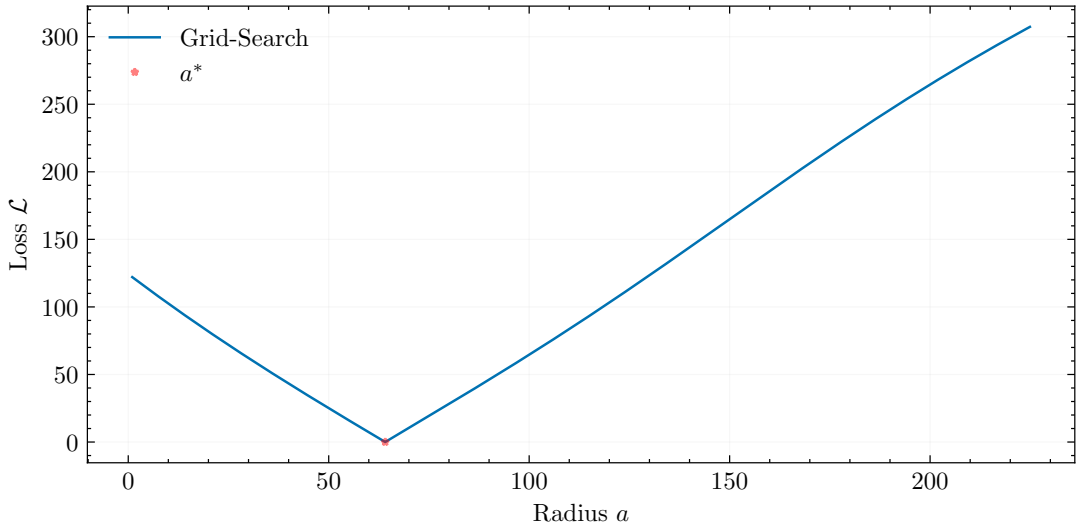

Figure 2.2: The y-axis plots the right-hand side of Eqn. 2.32 for the self-consistent diffusion estimation in confinement. An $a =\sim 50\,\text{nm}$ polystyrene bead was imaged and tracked in the transverse plane as provided by the manufacturer. The initial size estimate before confinement analysis was $\hat{a} = 124.30\,\text{nm}$. Post self-consistency, applying a linear superposition theory estimate of the confinement effects, $a^* = 64.1\,\text{nm}$.

As shown in Fig. 2.2, the sizing estimate from the self-consistency relation is significant. The manufacturer provided estimate of $a = 50\text{nm}$ is far away from the initial estimate of the size from the Stokes-Einstein relation. However, after the confinement correction is evaluated and the mobility relation is balanced with the measurement of the diffusion coefficient, the estimation agrees much better with the initial manufacturer claims and with the ensemble of particle sizes within the experiment.

2.6 Electrostatic Potentials

In experiments, such as a charged plastic bead diffusing between two charged glass plates, it takes little but a class in high-school physics to wonder if the charges on the bead and glass will cause each other to repel. In the case of a highly sensitive size measurement, the same question still lingers.

A particle diffusing near a charged wall, specifically a surface with a charge density, in an ionic solution, a fluid that contains Coulomb-like[††] charged molecules referred to as ions, will feel a screened electrostatic interaction from the walls due to the arrangement of ions. To appropriately describe the electrostatic effects of these non-trivial interactions, the Debye-Hückel theory can be used. The Debye-Hückel theory describes the electromagnetic interactions between ions within a solution [52, 53]. Additionally, the electrostatic potential due to these ions can be solved in terms in the presence of other charges, described by Poisson's equation [54]:

$$\nabla^2\Psi(\vec{x}) = -\frac{\rho(\vec{x})}{\varepsilon_0\varepsilon_r},. \tag{2.33}$$

where $\Psi(\vec{x})$ is the potential for all space, $\rho(\vec{x})$ denotes the volumetric charge density of the system, and ε are the standard permittivity of space notation.

In a general solution of ions, there will be i species of ions. The spatial concentration distribution of the ions in a single-ion solution can be thought of as the solution's ionic charge distribution. Instead of counting all ionic positions, the Nernst relation can be drawn and write the ion concentration in a Boltzmann form,

$$\rho(\vec{x}) = \sum_i C_i e_0 Z_i \exp\left[-\beta Z_i e_0 \Psi(\vec{x})\right], \tag{2.34}$$

where C_i is the concentration of ions, Z_i is the ionic charge number of ions, β is the inverse of k_bT, e_0 is the elementary charge, and $\Psi(\vec{x})$ is the potential.

The charge density can be substituted into the Poisson equation and is referred to as Poisson-Boltzmann equation [55, 56],

$$\nabla^2\Psi(\vec{x}) = -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} C_i e_0 Z_i \exp\left[-\beta Z_i e_0 \Psi(\vec{x})\right]. \tag{2.35}$$

There is a first-order linearization approximation to this equation where the potential is small,

[††]Infinitesimally small, charged molecules.

$$\begin{aligned}\nabla^2\Psi(\vec{x}) &\approx -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} C_i e_0 Z_i \left[1-\beta Z_i e_0 \Psi(\vec{x})\right] \\ &= -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty}\left[C_i e_0 Z_i - \beta C_i Z_i^2 e_0^2 \Psi(\vec{x})\right] \\ &= -\sum_{i=1}^{\infty}\left[\frac{C_i e_0 Z_i}{\varepsilon_0\varepsilon_r} - \kappa_i^2\Psi(\vec{x})\right],\end{aligned}$$

where κ_i^2 is inversely related to the Debye-length λ_D of the ith species interaction, a unit of the electrostatic screening caused by the ionic interactions.

When the bulk solution is electro-neutral, $\sum_i C_i Z_i = 0$, then the constant terms vanish, leaving only the approximation;

$$\nabla^2\Psi(\vec{x}) = \kappa^2\Psi(\vec{x}), \quad \kappa^2 = \frac{\beta e_0^2}{\varepsilon_0\varepsilon_r}\sum_i C_i Z_i^2 \,. \tag{2.36}$$

This equation has an obvious solution which are the real exponential functions;

$$\Psi(\vec{x}) = Ae^{\kappa\vec{x}} + Be^{-\kappa\vec{x}} \,. \tag{2.37}$$

The coefficients A and B are specified by the strength of the electric potential. In the case of ionic potentials, the strength of the coefficient is a measure of the number of ions and their charge density near the surface. To proceed generally, it is necessary to define boundary conditions and geometries to fully obtain the coefficients, however, one can make approximations to save time.

In this thesis, a double-electric potential for a sphere-plate system was assumed in accordance with Ref. [57]. The solution of this potential is a superposition of the decaying portion of Eqn. 2.37 such that the potential is infinite at the wall and decays to zero at infinity. Additionally, a superposition of two walls can be constructed such that there are two infinitely high barriers and a decaying potential between them as demonstrated in [3]. The form of the amplitude of this potential, shown in Ref. [57],

$$B = C_{pw}\Gamma_p\Gamma_w \,, \tag{2.38}$$

where C is a fixed amplitude due to charge density, and Γ_w, Γ_p are terms associated with the Stern layer, the layer of ions closest to the surface of the geometry, of the wall and the particle [53, 57]. The values of the amplitudes are chosen based on literature. Firstly, the amplitudes Γ are shown as the plate-plate interactions between two plates at a fixed

zeta-potential, which is the potential at the Stern layer. This is written for the ionic strength of the sphere and wall

$$\begin{aligned} \Gamma_w &= \tanh\left(\frac{e\psi_w}{4k_bT}\right) \\ \Gamma_p &= \tanh\left(\frac{e\zeta_p}{4k_bT}\right) \\ \psi_w &= \frac{2k_bT}{e}\text{asinh}\frac{e\lambda_D\sigma_w}{2\varepsilon_0\varepsilon_r k_bT}\,. \end{aligned} \tag{2.39}$$

where Γ_w is related to walls zeta-potential ψ_w. The zeta-potential is taken from Ref. [53] and is chosen to be related to the surface charge density of the wall for a more intuitive measurable of a wall-like structure. Here Γ_p is the amplitude from the particle, where ζ_p is the measured zeta-potential from an experimental device[‡‡]. The charge density amplitude, written as C, is approximated by the Grahame equation written on a sphere which describes the potential of an isolated charged surfaced in the presence of ions [53];

$$C_{\text{pw}} = \frac{64\pi\varepsilon_0\varepsilon_r a k_bT}{e_0^2} \tag{2.40}$$

These electrostatic potentials are important for estimation and simulation of confined Langevin dynamics in environments with highly charged particles and boundaries, however, should be used with caution as they are chosen to be within orders of magnitude rather than analytically exact.

Additionally, the hyperbolic form of the drift of Eqn. 2.26 allows one to rewrite the equation into a gaussian form and solve for an estimation of the drift coefficients. In the presence of a surface-charge dependent drift, depending on the the surface charge and particle charge, these quantities can be inverted using the descriptions written above. These equations are necessary for extraction of information from three-dimensional particle tracking information.

[‡‡]See Chap. 8 for device discussion.

Chapter 3

Sizing of Biological Particles

As described in Chap. 2, the diffusivity of a particle, intuitively as the random motion it undergoes when in a thermal bath, is mathematically described by the diffusion coefficient [26, 38]. The property of diffusion is largely universal in the length-scale that it occupies itself in; particles with a non-negligible diffusion coefficient will randomly move around their surroundings.

It happens that in many biological systems this diffusive motion is a driver for many reactions such as enzyme-protein coupling, protein-protein coupling, and vesicle binding [58, 59]. It is even the case that many viruses and bacteria use their thermal diffusion as a mechanism of mass-transfer, meaning moving from one area to another [60]. However, as one looks closer inside of a cell, it is apparently that a cell is not free space. The cellular cytosol is filled with proteins, organelles, and vesicles which are constantly rubbing into each other [58]. To fully understand the cellular dynamics within such a crowded environment, it is utmost importance to adequately understand the confinement effects on diffusion of biological particles.

In addition to the nature of cellular biology, the confinement of nanoparticles is directly used in synthesis and developmental research. Nanoparticle synthesis requires quality control, ensuring that the nanoparticles generated, whether they be for food science, pharmacology, or scientific research, is that they are up to standard. As introduced in Chap. 1, confinement is a natural part of these methods as it allows for extended measurements of the same particle. However, in the case of quality control, it is paramount that the effects of confinement do not hinder the final measurement of size or diffusion.

3.1 Size Measurement Techniques for Biological Particles

As previously shown in Chap. 2, the Stokes-Einstein diffusion relation is used to directly infer the size of a biological diffusing particle. In particular, this relation can be explicitly formalized for the hydrodynamic radius of a spherical nanoparticle. Using Eqn. 2.9 and inverting the relation to calculate the particle radius;

$$a = \frac{k_b T}{6\pi\eta D}\,, \tag{3.1}$$

where a is the particle's hydrodynamic radius, η is the fluid viscosity, D is the particle's potentially spatially dependent diffusion coefficient, and k_bT is the Boltzmann factor multiplied by the temperature. Methods such as nanoparticle-tracking analysis, NTA, or single-particle tracking, SPT, utilize this equation to extrapolate the size of particles after measuring the diffusion coefficient from the particle trajectories [35, 61].

The SPT and NTA method offer a fast and reliable measure of the ensemble size estimate from sampling many particles for a short amount of time. These techniques not only use low amounts of sample volume, but also often utilize light scattering microscopy to track particles, free of labels, meaning there is no chemical fluorescent tag attached. The common application of SPT requires an objective and a single coverslip, on which a droplet of sample is placed. The droplet is observed and particles are tracked over time. This technique, although sampling many particles at a time, cannot guarantee the observation of the same particle for a extended periods due to their diffusion in and out of the objective's focal plane. Similarly in NTA, a wide fluid channel is used in tandem with a microscope objective to capture particle diffusion in the presence of a controlled fluid drift, particle diffusion is measured and inverted to particle radius. As the particles drift along the chamber, they are free to diffuse in and out of the objective's focal plane, ultimately obtaining ensemble information.

To accurately measure single-particle dynamics, other methods can be used to measure extended dynamics of biological particles. Fluorescence-based measurements, such as fluorescence correlation spectroscopy, FCS, or fluorescence single-particle microscopy allow for the detection of biological reactions in real-time. These techniques utilize a phenomena known as Förster resonance energy transfer*, FRET, where fluorophore A is energetically coupled with fluorophore B such that the excitation of A will excite B if they are in close proximity [62]. However, fluorescence-based measurements suffer from the inevitable loss of efficiency known as photo-bleaching. This chemical decay process occurs as the fluorophores enter energetic states known as triplet states, and are oxidized

*For an in-depth discussion, see Ref. [62].

into non-fluorescent molecules [62]. Additionally, the addition of a fluorophore onto a biological molecule, such as a protein or LNP, causes significant conformational changes, which are often not in the label-free case [63].

In compliment to single-particle techniques, bulk methods, such as dynamic light scattering, DLS, allow the measurement of particle size based on their scattering properties [64]. In particular, the light scattering intensity recovered from measuring the light can be weighted to the inverse of the particle size to the sixth power; measuring the size of the sample. However, these methods are designed for a homogenous sample. Samples, such as LNPs, which are highly non-homogenous, have significant structural limitations in which these methods can be used.

Lastly, to address label-free, extended temporal based size measurements, one can turn their attention to interferometric light scattering microscopy, iSCAT. The iSCAT technique, discussed extensively in Chap. 4, allows for extended observation of a general class of nanoparticles without the use of labels. This is done by scattering light onto a sample and mathematically calculating the three-dimensional position of the particle using the scattering pattern [31, 65]. In addition, iSCAT can be paired with an artificial confinement system, such as convex-lens induced confinement, which uses two coverslips to create a chamber of nano-volume, to trap particles and observe them for long periods of time [66]. However, since iSCAT uses scattering as it's main signal, two particles of the same shape and mass are very difficult to distinguish. For example, two proteins with different amino acid sequences would look identical on the camera of an iSCAT system. Thus, this limitation poses experimental design challenges regarding applicability.

3.2 Lipid Nanoparticles

One of the seminal discoveries of nanoparticles in biology were the liposome in 1958 [67]. The liposome is a spherical particle composed of lipid subunits, each carrying a charge. Due to the hydrophobic nature of the subunits, the liposome self-assembles itself into spherical shapes when in the presence of water. The size of a single sub-unit is around 19 Å [68, 69]. Liposomes, however, are found to be between 20 nm − 3000 nm [70]. Naturally, seeing as liposomes and cellular vesicles, which are used in the cell to carry material, are very similar chemical molecules, posed the question of if one can engineer these particle to be able to carry drugs into a living system [71]. The liposome was a semi-successful particle used to carry drugs into the body [72], however, ultimately were succeeded due to storage stability problems caused by lipid oxidation and hydrolysis, drug leakage from the vesicle, rapid clearance during blood circulation, and the absence of cancer cell-specific targeting [73].

The lipid nanoparticle, LNP, is the current gold-standard for drug delivery in high-risk drug targets such as cancer and mRNA based drug. This class of particles consist of a set of phospholipids, ionizable lipids, and cholesterols which come together in a stable spherical sphere [6]. It has been shown that this molecule can be produced to hold a various drug cargo, such as mRNA [74, 75]. The LNP has already had application in drug delivery; for example, in the SARS-Covid vaccine in 2020 as a drug deliverer of the mRNA encoded for the spike protein [6, 76–78]. A diagram of a standard LNP, as well as CRYO-EM photographs are shown in Fig. 3.1 and Fig. 3.3.

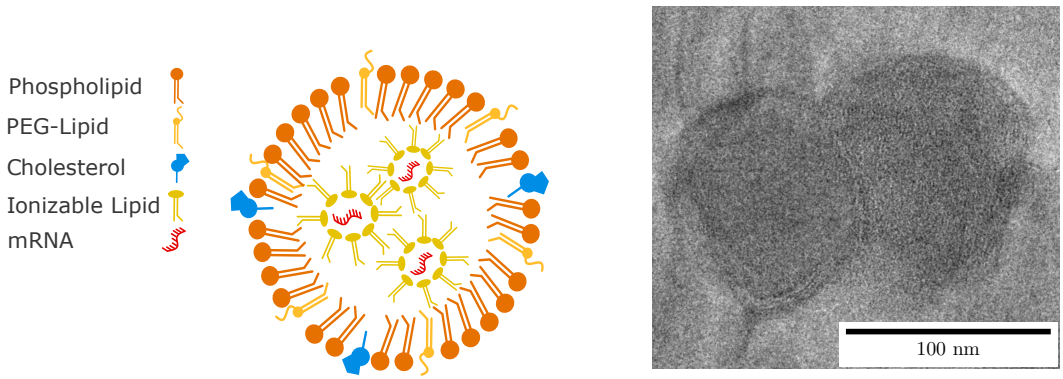

Figure 3.1: Lipid nanoparticle diagram and CRYO-EM image. CRYO-EM image obtained from the NIST Segmentation of Lipid Nanoparticles dataset [79]. Shown are KC2 LNPs at pH 7, with RNA taken at 0.223 nm per pixel.

Recent research in LNP function is focused on the LNP topologies [80]. The topology of an LNP is described as the structural makeup that the LNP uptakes during the initial synthesis, which is a stochastic process of mixing an organic solution containing lipids with an aqueous buffer. These topologies have various types, however, the topologies are

grouped into three main types: liposomal, spherical, and bleb-like, visualized in Fig. 3.2 and imaged by Ref. [81] in Fig. 3.3.

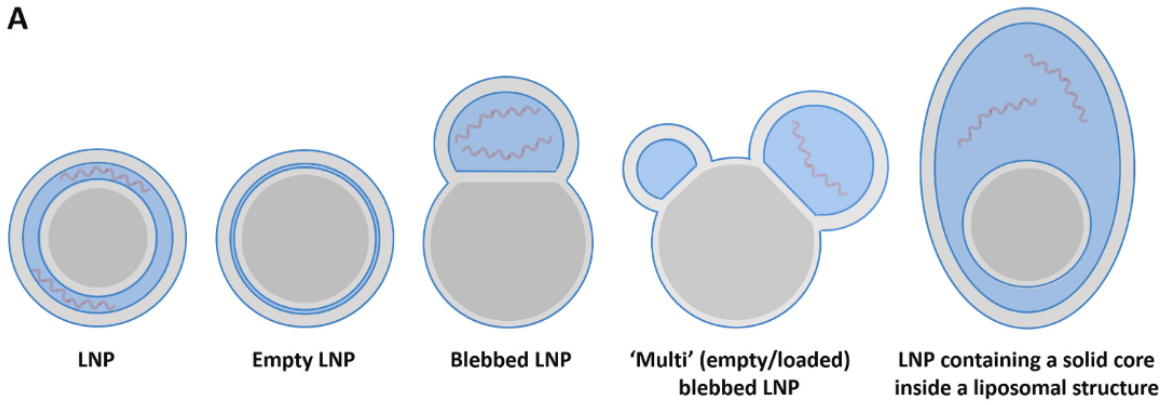

Figure 3.2: mRNA-LNP types diagram. Black is lipids, grey is hydrophobic region, red is mRNA, orange is oil-core. Taken from Ref. [80] under a creative-commons license 4.0

Due to the different shapes of these LNPs, the measurement of LNPs must account for the population poly-dispersity. For instance, bulk methods such as DLS cannot reasonably account for this without filtering for various shapes, which is costly in both time and monetary value. For size-based measurements, it is logical to think that single-particle based methods, such as iSCAT or fluorescence microscopy, are best suited as each particle can be individually measured one at a time, alleviating concerns of mixing up two particles. Recent publications in LNP measurement discuss CLiC as a high-throughput, accurate, and reliable method to characterize the size and payload copy number of LNPs containing a fluorescently labelled payload [32–34, 51]. CLiC utilizes the confinement of particles to observe their untethered, random motion [66].

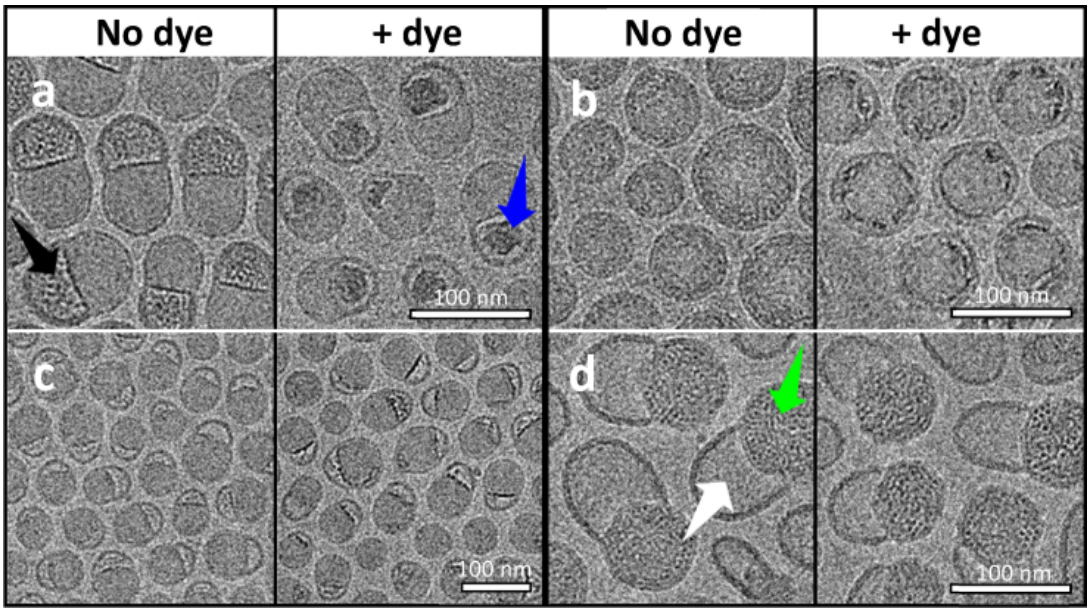

Figure 3.3: Locating encapsulated mRNA in LNPs by CRYO-EM. The effect of thionine on mRNA-LNPs of different morphologies reveals that lipid-dissociated mRNA may reside in bleb compartments. (a) or may be more lipid-associated in spherical (b) or less prominently blebbed particles (c). (d) Charge-driven migration of mRNA from the bleb into the body of the LNP. Modified from Ref. [81] under a creative-commons license 4.0.

Chapter 4

Interferometric Scattering Microscopy Principles

To determine the physical properties of nanoparticles suspended in a fluid media, tracking their movement and reconstructing their spatial trajectories is typically performed. To achieve this, a particle localization method that maintains a consistent signal over extended measurement periods is required. One method that enables such localization of particle traces is interferometric light scattering microscopy, iSCAT [31]. iSCAT is an optical imaging technique that utilizes light scattering contrast to localize particle signal [31]. The particle's transverse position can be directly rendered by utilizing the scattering pattern's angular symmetry, which assumes that the particle's refractive index differs that that of water, has a non-zero polarizability, and is a structurally spherical mass such that the rotation of the particle with respect to the incoming light is negligible [4, 31, 65, 82]. Similarly, utilizing the phase difference in the scattering pattern between series of images, the relative axial distance can be measured [83]. In short, the 3D positional information is encoded in the scattering pattern in iSCAT.

4.1 Light as a Casual Field

While electromagnetic fields are fundamentally composed of photons [84], the wave-particle duality permits the use of the theory of electromagnetic waves. Light, however, can be represented by the Maxwell's equations all the same. Let $E_{\text{elec}}(\vec{x}, t)$ represent the electric field. The optical field is then defined as the classical field by a traveling photon with a wavelength in the visible light spectrum;

$$\vec{E}_{\text{optic}}(\vec{x}, t) = \sqrt{\varepsilon_0 n_m c/2}\vec{E}_{\text{elec}}(\vec{x}, t)\,, \tag{4.1}$$

where where the normalization constant, $\sqrt{\varepsilon_0 n_m c/2}$, affects how the electric field is related to the measured light intensity [85]. The intensity of light is defined by the modulus of the optical field,

$$I = |\vec{E}_{\text{optic}}(\vec{x}, t)|^2\,. \tag{4.2}$$

This formulation of light as a classical field allows to discuss the dynamics of light fields in time. In the context of light scattering microscopy, the evolution of the classical light field after hitting a dielectric body will determine the image that the camera detector will eventually measure. Thus, it is paramount to understand how this signal forms for particles that are smaller than the wavelength of light, around $\lambda = 405\,\text{nm}$, such as nanoparticles, $a = 20\text{-}100\,\text{nm}$.

4.2 Electromagnetic Scattering of Spheres

When a homogenous particle is subjected to light produced by a monochromatic laser, a diffractive pattern is produced upon scattering [14, 54, 86]. These patterns can be captured CMOS detectors to extract positional information from this pattern. However, to correctly relate the detected signal to a quantifiable, three-dimensional position, it is necessary to first mathematically describe it.

To solve for the expected field that is produced when light is scattered off of a particle, such as a sphere, the Maxwell equations must be solved. The electromagnetic wave solutions must obey the wave equation,

$$\nabla^2\vec{E} + k^2\vec{E} = 0\,, \quad \vec{H} = \frac{1}{i\omega\mu_0}\nabla\times\vec{E}\,. \tag{4.3}$$

where $k = 2\pi/\lambda$ is defined as the wave-vector, λ being the wavelength of the wave. Here, ω is defined as the angular frequency of the wave and μ_0 is the magnetic permeability of free space. The electric field is defined as $\vec{E}$ and the magnetic field is defined as $\vec{H}$.

The eigenfunctions of the wave equation are the harmonics of the Laplacian operator, ∇^2, in an arbitrary coordinate basis [54, 87, 88]. For isotropic, radial scattering, the spherical basis has been used in the literature [86]. The solutions of the vector-wave equation are known as the vector spherical harmonics which are derived by first solving the scalar wave equation and taking vectorial derivatives to form an orthogonal basis. Let ψ be a solution to the scalar wave equation,

$$\nabla^2\psi + k^2\psi = 0\,, \tag{4.4}$$

where ∇^2 is the three-dimensional Laplacian and k is the wave-vector. Using the method of separation, ψ can be rewritten in spherical coordinates,

$$\psi(r,\theta,\phi) = \Phi(\phi)\Theta(\theta)R(r)\,, \tag{4.5}$$

where Φ represents the azimuthal component, Θ the angular, and R the radial component. The solutions of each of these functions can be found in Ref. [54],

$$\begin{aligned}
&\Phi(\phi) \in \{\cos(m\phi), \sin(m\phi)\}\,,\\
&\Theta(\theta) = P_n^m(\cos\theta),\\
&R(r) \in \{j_n, y_n, j_n + iy_n, j_n - iy_n\}\,.
\end{aligned} \tag{4.6}$$

where the indices m, n represent indices for appropriate sums. The symbols j_n, y_n represent the spherical Bessel functions. Their complex sums, $j_n \pm iy_n$, are the Hankel functions, h_n which are not written. Lastly, P_n^m represents the associated Legendre functions of the

first kind.

These separable solutions are then used to form generating functions that satisfy the scalar wave equation in the spherical basis. These are forms such that the subscript o defines an odd function, and e defines an even function;

$$\begin{aligned} \psi_{emn} &= \cos(m\phi)P_n^m(\cos(\theta))z_n(kr)\,, \\ \psi_{omn} &= \sin(m\phi)P_n^m(\cos(\theta))z_n(kr)\,, \end{aligned} \tag{4.7}$$

where $z_n(kr)$ is any one of the four solutions of the radial solution $R(r)$, $j_n, y_n, j_n + iy_n$, or $j_n - iy_n$.

The vector spherical harmonics can then be generated from ψ_{emn} and ψ_{omn} as written in Eqn. 4.17-4.20 of Ref. [86],

$$\begin{aligned} \vec{M}_{emn} &= \vec{\nabla} \times (\vec{r}\psi_{emn}), \quad \vec{M}_{omn} = \vec{\nabla} \times (\vec{r}\psi_{omn}), \\ \vec{N}_{emn} &= \frac{1}{k}\vec{\nabla} \times \vec{M}_{emn}, \quad \vec{N}_{omn} = \frac{1}{k}\vec{\nabla} \times \vec{M}_{omn.}\,. \end{aligned} \tag{4.8}$$

These equations are a complete basis for solutions of scattered waves off of spheres. In physical terms, all combinations of possible scattered waves will consist of a sum of these equations. This is an incredible achievement as scattered optical fields can be decomposed into a set of vector spherical harmonics, fully describing the position and shape of the particle that it had scattered off of in the process.

However, to calculate the scattered pattern that the camera would measure, it is wise to first calculate what such a scattering field will be in principle. Although the solution will consist of a sum of Eqns. 4.8, it is still unknown which ones, and how much. To solve this, the scattering problem can be stated and solved.

A scattering event will consist of incident light hitting a spherical particle and scattering off in many directions. The incident light, $\vec{E}_i$, is assumed to be a polarized plane wave along the x-axis without loss of generality. The scattered light, $\vec{E}_s$, exists as a solution of the spherical vectorial wave equation as a sum of Eqns. 4.8. As the scattered light hits the surface of the sphere, there is a portion which can transmit through it, forming an electric field inside of the sphere, $\vec{E}_1$. One can assume that each of the fields, being the scattered light when subjected to the energy conservation boundary condition written as

$$(\vec{E}_i + \vec{E}_s - \vec{E}_1) \times \hat{r} = (\vec{H}_i + \vec{H}_s - \vec{H}_1) \times \hat{r} = 0\,, \tag{4.9}$$

where $\hat{r}$ denotes the radial unit vector. The exact solution of these conservation equations can be found in Chapter 4 of Ref. [86]. The resulting solutions, known as the Mie theory

of scattering, is written as

$$\begin{aligned}\vec{E}_i &= E_0 e^{ikr\cos\theta}\hat{x}\,,\\ \vec{E}_s &= \sum_{n=1}^{\infty} E_n(ia_n\vec{N^{(3)}}_{e1n} - b_n\vec{M^{(3)}}_{o1n})\,,\\ \vec{E}_1 &= \sum_{n=1}^{\infty} E_n(ic_n\vec{M^{(1)}}_{o1n} - d_n\vec{N^{(1)}}_{e1n})\,,\end{aligned} \tag{4.10}$$

where the constants a_n, b_n, c_n and d_n are amplitude factors and the superscript (1) labels the use of the radial function j_n, and the superscript (3) labels the use of the Hankel function $j_n + iy_n$, meaning that $z_n(kr) = j_n(kr)$ and $j_n + iy_n$ respectfully. Intuitively, the scattering field consists of an outward radial scattering with a phase that modulates. This is represented in the difference between the even solution, $\vec{N}_{e1n}$ and the odd solution $\vec{M}_{o1n}$.

The amplitude factors a_n and b_n represent the scattering amplitudes that must be solved for a set of boundary conditions. For scattering in free-space, a full derivation of these is offered in Ref. [86]. The amplitude factors of interest are

$$\begin{aligned}a_n &= \frac{m\Psi(mx)\Psi'(x) - \Psi(x)\Psi'(mx)}{m\Psi(mx)\xi'(x) - \xi(x)\Psi'(mx)}\,,\\ b_n &= \frac{\Psi(mx)\Psi'(x) - m\Psi(x)\Psi'(mx)}{\Psi(mx)\xi'(x) - m\xi(x)\Psi'(mx)}\,,\end{aligned} \tag{4.11}$$

where Ψ and ξ are the Riccatti-Bessel functions [86]. These are largely unintuitive solutions and written for the sake of clarity.

The formulation of the scattering of light off of a sphere allows for projected solutions in any dimension one chooses. Theoretically, Mie theory can solve for the analytical scattering of large spheres, where $kr > 1$. For instance, a 1 µm polystyrene bead can be fully tracked in three dimensions which high accuracy using Mie theory [5, 65, 89]. However, as written in Eqn. 4.10, these solutions require the computation of an infinite sum, which is very computationally expensive. Thus, there is a need to simplify the mathematics by applying approximations.

4.3 Rayleigh Scattering

As particle size decreases and is observed from far away, approximations can be made to simplify the form of the scattering equations. As commonly done in theoretical physics, approximations can be taken without degrading the quality of approximation in the length-scale of interest [54, 88]. Notably, as the product, $kr \ll 1$, the Hankel function, previously denoted $h_n = j_n + iy_n$, approach the form of a spherical wave function as shown in Eqn. 4.42 and 4.43 in Ref. [86],

$$h_n(kr) \xrightarrow{kr \ll 1} (-i)^{n+1} \frac{e^{ikr}}{kr} \,. \tag{4.12}$$

This approximation allows one to re-write the scattered field in terms of angle-dependent functions π_n and τ_n. A full derivation is offered in Ref. [86] as well as the supplementary information of Ref. [4]. Letting $\pi_n = P_n^1/\sin(\theta)$ and $\tau_n = dP_n^1/d\theta$, the scattering field is

$$\vec{E}_s = \frac{iE_0 e^{ikr}}{kr} \left[\hat{\theta} \cos\phi \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} (a_n \tau_n + b_n \pi_n) + \hat{\phi} \sin\phi \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} (a_n \pi_n + b_n \tau_n) \right] , \tag{4.13}$$

where a_n and b_n are the scattering amplitudes and is k is the wave vector of the scattered light. After substituting the dominant terms, represented as a far-field approximation, the sum reduces significantly to the equations for transverse and axial scattering fields,

$$\begin{aligned} \vec{E}_{s,\perp} &= \vec{E}_{i,\perp} \frac{k^2 \alpha e^{ikr}}{r}, \\ \vec{E}_{s,\parallel} &= \vec{E}_{i,\parallel} \frac{k^2 \alpha e^{ikr}}{r} \cos\theta \,. \end{aligned} \tag{4.14}$$

Using that the wave vector k of a wave of wavelength λ passing through a medium of refractive index n can be expressed as $k = \frac{2\pi}{\lambda/n}$, the scattering field of spheres on the order of the length of the wavelength of the incident light is written in the commonly referenced form,

$$\vec{E}_s = \vec{E}_0 e^{ikr} \frac{\sqrt{8}\alpha\pi^2 n_m^2}{\lambda^2 r} \sqrt{1 + \cos\theta} \,. \tag{4.15}$$

This equation is the scattering field of an ideal Rayleigh particle and has been used in the literature for axial estimation of iSCAT particles [4]. In addition, it provides a easily computable form of the scattering pattern for a general Rayleigh scatterer, plotted for a various set of conditions as shown in Fig. 4.1.

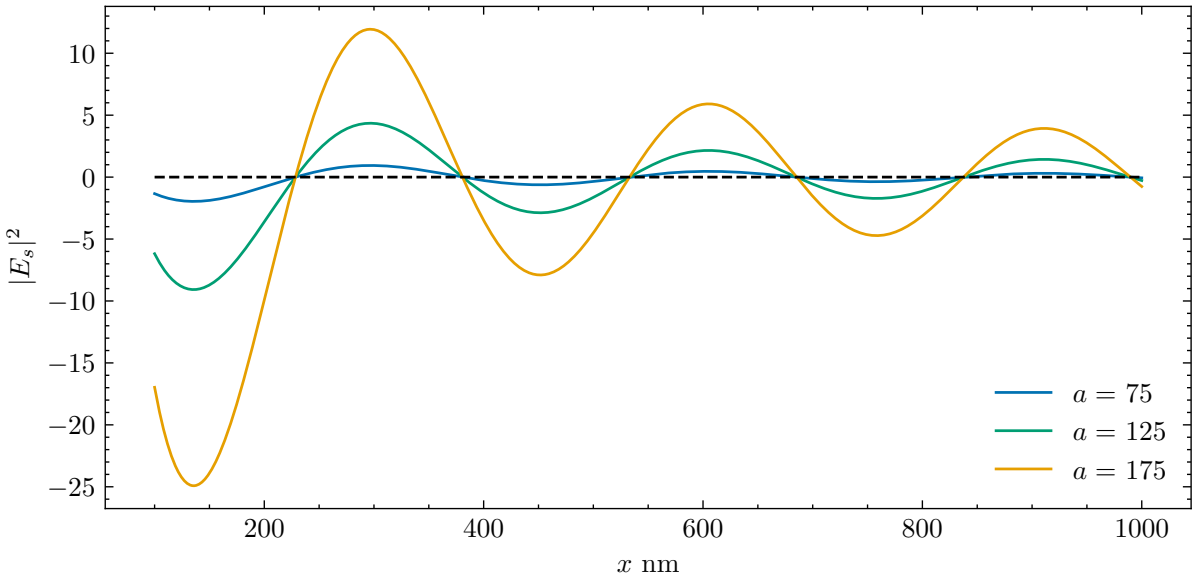

Figure 4.1: Rayleigh scattering pattern, Eqn. 4.15, plotted for various sized particles. The y-axis, plotted in units of optical intensity, denotes the amplitude of the modulus of the scattered field. The x-axis plots the distance from the center of the scatterer in nm.Parameters fixed such that $\lambda = 405\,\text{nm}, n_m = 1.33, E_0 = 1, \theta = 0$. Particle parameters set as $n_p = 1.59, \alpha \propto V$. As the particle volume increases, the scattering pattern increases in amplitude.

4.4 Interferometric Light Scattering

In application, the scattered light can be collected from a variety of angles. In literature, these are referred to as scattering schemes and include the forward scattering, side-scattering, and back scattering schemes [90]. The forward scattering scheme works best for large particles, in the range of $a = 250\text{-}500\,\text{nm}$, as it collects a weak amount of scattered light, which is dominated by the reference beam. The side-scattering scheme has been used in NTA and can resolve position, however, it has not been favored as the the main method in three dimensional tracking due to experimental challenges with small volumes. Lastly, back-scattering allows for the viewing of small particles with a high SNR due to only reflected laser light being collected.

In the back-scattering scheme, the light collected is the sum of the incident light and the scattered light,

$$I(x, y) = |\vec{E}_r + \vec{E}_s|^2 \,. \tag{4.16}$$

Given that the incident wave is known as a polarized plane wave, this equation can be rewritten for the scattered light;

$$I = E_r E_r^* + E_s E_s^* + 2\Re\left[E_r E_s\right] \,. \tag{4.17}$$

The incident field will be modulated by a scattering term, which is the Rayleigh scattering amplitude, $|\vec{E}_s| = S$, as well as the reference term, $|\vec{E}_r| = R$. Thus, the total intensity will be the sum of the scattering amplitude, reference amplitude, and the phase interference between the two fields, $\Delta\varphi = \varphi_s\varphi_r$;

$$I = |E_i|^2 \left[|R|^2 + |S|^2 + 2|R||S|\cos\Delta\varphi\right] \,. \tag{4.18}$$

To accurately detect the particles, implying that the scattering term is neglected, the contrast metric can be used. This metric relates the total field to that of a reference field;

$$\begin{aligned} C &= \frac{I - |R|^2}{|R|^2} \\ &= \frac{|S|^2 + 2|R||S|\cos\Delta\varphi}{|R|^2} \,, \end{aligned} \tag{4.19}$$

where I is the total intensity. In the limit where the the scattered field is much smaller

than the reference field, the asymptotic

$$C \xrightarrow{|S|\ll|R|} \frac{2|S|}{|R|}\cos\Delta\varphi\,, \tag{4.20}$$

can be taken, which relates the contrast metric to a phase difference between the two signals. The phases in the interference field consists of a sum of optical fields. These fields acquire phase differences as they traverse through space. Aberration effects from optical transformations are also acquired as spatial and phase terms. The total phase is dominated by the Gouy phase, which is the sum of accumulated phase as the light passes through a converging or diverging lens [31]. The de-focusing phase is a relative phase shift due to misalignment in lens position which is described by the imaging condition equation [14]. Lastly, the phase difference from the inference term, which is compared to a Michelson-interferometer type system, is a dominant signal present directly in the iSCAT equation [31].

The final image consists of the collected phase modulated optical fields and the aberrations from optical misalignment. These parameters must be considered when performing analysis on iSCAT images.

4.5 3D Position Information from Optical Signal

To obtain information from the iSCAT technique, the holographic images obtained from the contrast metric C must be related to the theoretical scattering pattern derived in Eqn. 4.15. To do this, the raw CMOS image is first converted into the contrast metric. The interferometric point spread function is defined as the contrast metric of the scattering off of a sub-diffraction sized particle,

$$\text{iPSF} \equiv \frac{2RS\cos\phi_d}{R^2}\,. \tag{4.21}$$

The naming convention is a bit tricky, however, the iPSF is essentially what a particle is viewed as when a raw image of it is converted by the contrast metric.

The Rayleigh scattering pattern has several parameters that have a degenerate effect on the iPSF, including the phase, axial position, light wavelength, and polarizability which cause the pattern to become more or less dense. To avoid problems in model fitting, the parameterization developed in Ref. [4] is used. This parameterization redefines the axial height and phase,

$$z_p' = z_p - z_f\,, \tag{4.22}$$

$$\phi_0 = -\phi_{\text{ref}} + 2n_m k z_f + \arg(\alpha)\,, \tag{4.23}$$

$$E_0 = \frac{E_i 4\sqrt{2}\pi^3|\alpha|}{E_r(\lambda/n_m)^3}\,, \tag{4.24}$$

where z_p is the particle position, z_f is the focal position, ϕ_{ref} is the laser reference angle, n_m is the refractive index, k is the wave vector of the incident light, λ is the wavelength of the incident light, and α is the polarizability of the sphere. This re-parameterization has been shown to have good agreement to complex models as demonstrated in Ref. [4]. The final model is presented as the following equation,

$$\text{iPSF} = 2E_0\frac{1}{n_m k r_p}\sqrt{1+\left(z_p'/r_p\right)^2}\cos\left[-\left(\phi_0 + n_m k z_p' + n_m k r_p\right)\right]\,. \tag{4.25}$$

This function allows one to obtain the z position of a particle with respect to the objective from image data.

4.6 Resolution Limits of Optical Microscopy

There is a limit of resolution that can be achieved through optical methods [13]. The limit is described by the Abbe diffraction limit which states that the minimum distance between two resolvable particle is fixed by the light wavelength and objective's numerical aperture [13, 91].

Optical fields consists of a sum of plane waves that construct and deconstruct. The electric field can be decomposed into a Fourier series of exponential harmonics. [14]

To understand how the optical field modulates as it travels, the propagation equation can be written in a plane-wave basis [92],

$$\vec{E}(x,y,z) = \iint_{\Omega} \vec{E}_0(k_x,k_y) e^{-i\left(xk_x+yk_y+z\sqrt{k^2-k_x^2-k_y^2}\right)} dk_x dk_y\,, \quad \Omega = \{x \in \mathbb{R} : x^2 \leq k^2\}\,. \tag{4.26}$$

The energy of the wave is fixed by the trigonometric relation of the transverse wave vectors, fixed by the relation $|k| = \frac{2\pi}{\lambda}$.

The propagated wave passes through an objective which can be modeled by a lens. Using the standard ABCD matrix formulation, the imaging condition must be satisfied,

$$\frac{1}{z_1} + \frac{1}{z_2} = \frac{1}{f_{\text{lens}}}\,, \tag{4.27}$$

where z_1 and z_2 are the positions of the observer and the object, and f_{lens} is the focal distance of the lens [14]. The magnification of the lens is defined as $M = z_2/z_1$.

To define a field formulation for this system, the propagation equation can be used. The incoming light that passes through the finite area of a lens is defined a pupil function $P(x,y)$ which is unitary if $r \leq R$, where R is the radius of lens. The point spread function of the object is then defined as a delta-function through a two lens system. The measured optical field is given by the Fourier transform of the product of the pupil function and the delta-function [92],

$$H(x,y) = \iint_{\Omega} P(x',y') e^{\frac{ik}{2}\left(\frac{1}{z_1}-\frac{1}{f_1}-\frac{1}{f_2}+\frac{1}{z_2}\right)\sqrt{x'^2+y'^2}} e^{-\frac{ik}{z_2}(xx'+yy')} dx'dy'\,. \tag{4.28}$$

Expanding this field, one can use a zero-order Hankel transform to integrate over the radius and find that the point-spread function, H, is defined in terms of the numerical aperture and the magnification;

$$H(r,z) = \int_0^1 e^{-\frac{ik}{2}\frac{\text{NA}^2}{n}\rho^2 z} J_0\left(\frac{k\text{NA}}{M}\rho r\right)\rho d\rho\,, \tag{4.29}$$

where ρ is the rescaled radial vector $\rho = r'/R$, $\mathrm{NA} = nR/f_1$, and $M = nf_2/f_1$ [92]. Here, z is the axial position which defines the position of the image with respect to the plane of focus defined by the lens.

In perfect focus, $z = 0$, and the image is the point spread function. For $z \neq 0$ there is a defocusing aberration which is defined as the secondary exponential. For a point source, which is diffraction limited, the following is the limit of the optical microscope,

$$H(r, 0) = \left| \frac{J_1(k\mathrm{NA}r)}{k\mathrm{NA}r} \right|^2 . \tag{4.30}$$

Solving for the minimum distance between the peak of the PSF and the minimum, the relation $k\mathrm{NA}r = x_0$ defines the minimum transverse imaging condition, where x_0 is the first root of $J_1(x)$. This is known as the transverse Abbe diffraction limit,

$$r_{\min} = 0.61 \frac{\lambda}{\mathrm{NA}} . \tag{4.31}$$

Similar results are found for the minimum axial imaging criterion by setting $r = 0$ and integrating.

$$z_{\min} = \frac{2\lambda n}{\mathrm{NA}^2} . \tag{4.32}$$

These equations impose fundamental resolution limitations to general optical microscopes and must be taken into account when interpreting microscopy data.

4.7 Sources of Noise in Optical Imaging

In imaging, there is a number of sources of aberrations that one must account for. The sources of noise depend on the optical system used. For a general imaging system, one only requires a single lens for imaging. This simple setup is shown in Fig 4.2.

For such a system, the sources of noise are the point spread function noise, Gouy phase aberrations, background noise, electronic noise, and spherical aberrations [93, 94].

The optical point spread noise originates from the transformations of light passing through optical components. Each lens used in an optical path will collect light from the object plane, U_o to a transformed plane, U_i. The propagation of that wave will collect aberration from the lens point spread function, as well as phase accumulation. The traversal over the point spread function is given by the equation,

$$U_i(x_i, y_i) = \iint_{\mathbb{R}^2} h(x_i, y_i, x_o, y_o) U_o(x_o, y_o) dx_o dy_o, \tag{4.33}$$

where $h(x_i, y_i, x_o, y_o)$ is known as the optical transfer function, or the response of a lens

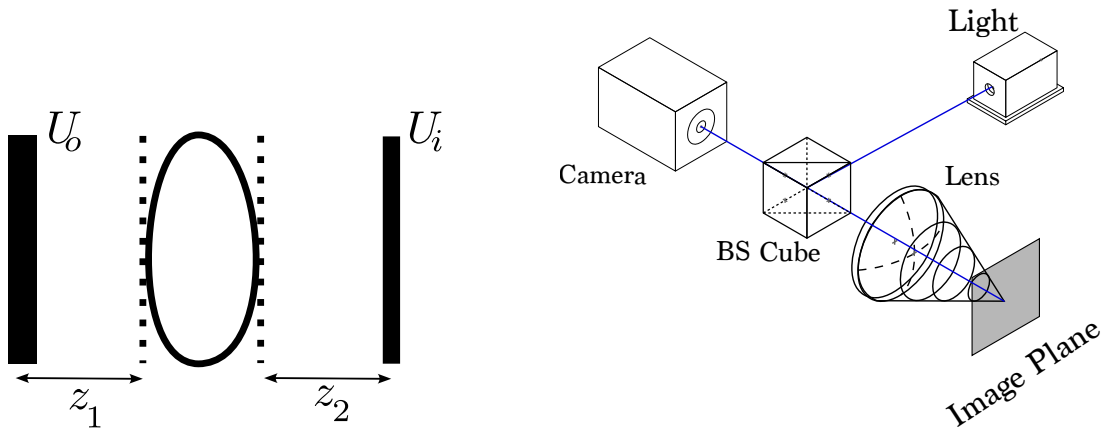

Figure 4.2: Simple imaging setup. Left, a lens is placed before an object. The light focuses on an imaging plane. The schematic consists of a feasible setup. Right, an image consists of a plane wave illumination, a lens, and a camera.

to a point-source. The lens magnification is incorporated into this using the following approximation;

$$h(x_i, y_i, x_o, y_o) \approx K\delta(x_i \pm Mx_o, y_i \pm My_o) \tag{4.34}$$

where K is a complex constant and M is a real-valued magnification. Each lens causes a small amount of aberration and misalignment which spatially limits the final optical field.

The phase accumulation, known as the Gouy phase, originates from plane waves converging to a gaussian beams when passed through a lens [14]. The plane wave is tightened and forms a beam-waist around the lens focal point. As this focusing occurs, there is a phase that is accumulated and equal to

$$\psi_g(z) = \arctan\left(\frac{z}{z_R}\right), \quad z_R = \frac{\pi w_0^2 n_m}{\lambda}, \tag{4.35}$$

where w_0 is the waist of the final beam, n_m is the refractive index, and λ is the wavelength of light. In imaging, as long as the refractive index is constant, the phase is shift is equal to π.

In optical microscopy where a reference or excitation field is used, a background beam or illumination is present. The background noise profile will take the shape of the illumination profile alongside the intensity of the power density [4, 31].

The camera noise consists of electronic noise associated with the model and electronic design. For general CMOS cameras, the noise present will consist of photon noise from the detectors and readout noise from the data-transfer. These noises can be modeled as Poisson noise from the photon detections and a Gaussian for the readout noise [93].

In the case of using an imperfect lens, or a set of lenses as one has with an objective as one does in high magnification microscopy, there is a risk of spherical aberration. Spherical

aberrations are caused by imperfections and misalignments in the lens itself. Additionally, this can be caused by having a beam waist too large for the lens, where the approximations that lead to a Gaussian beam break down. This cases the focal point of the lens to be slightly different depending on which area the light is focused by lens [95]. For imaging purposes, spherical aberrations cause various parts of the image to be out of focus, while others are in focus, creating non-uniform images.

Chapter 5

Interferometric Scattering Microscope Optical System

To observe single nanoparticle diffusion, an iSCAT microscope system was developed*. A scattering microscope is able to resolve nanoparticle three-dimensional position without a loss of signal over time, as presented in Chap. 4. The table in Tab. 5.1 shows the iSCAT microscope component list.

Component	**Product ID**	**Manufacturer**
Laser head	LDM-405-500-C	Lasertack GMBH
Mirror	BB1-E02P	Thorlabs
SF Lens 1 ($f = 40\,\text{mm}$)	AC254-040-A-ML	Thorlabs
SF Pinhole ($20\,\mu\text{m}$)	P20K	Thorlabs
SF Lens 2 ($f = 250\,\text{mm}$)	AC508-250-A-ML	Thorlabs
Objective focusing lens	BB1-E02P	Thorlabs
Polarizing beam splitter	05BC15PH.13	Newport
Beam splitter mount	9411	New Focus
Quarter-wave plate	WPQ10E-405	Thorlabs
60×/1.42 Oil-immersion objective	MRD71670	Nikon
Tube lens ($f = 40\,\text{mm}$)	AC254-040-A-ML	Thorlabs
sCMOS camera	PCO.EDGE 4.2	Excelitas

Table 5.1: Optical components of the iSCAT microscope.

*The iSCAT microscope was built as a collaborative effort as part of two CFI grants before the start of the author's thesis period. The first CFI grant was written in 2020 and awarded in 2021. The second was written in 2022 and awarded in 2023. The microscope parts were ordered by Dr. Eric Boateng as based on Ref. [96–98]. The assembly and software was completed by Dr. Erik Olsén with assistance from the Grant lab at UBC. Schematics and figures in this thesis were developed by the author.

5.1 Optical System Diagram

The iSCAT microscope system was designed based on prior work by the Kukura lab [97, 98]. The iSCAT microscope is a wide-field optical system that illuminates the whole field of view with a coherent laser light. The iSCAT diagram was shown in Fig. 5.1.

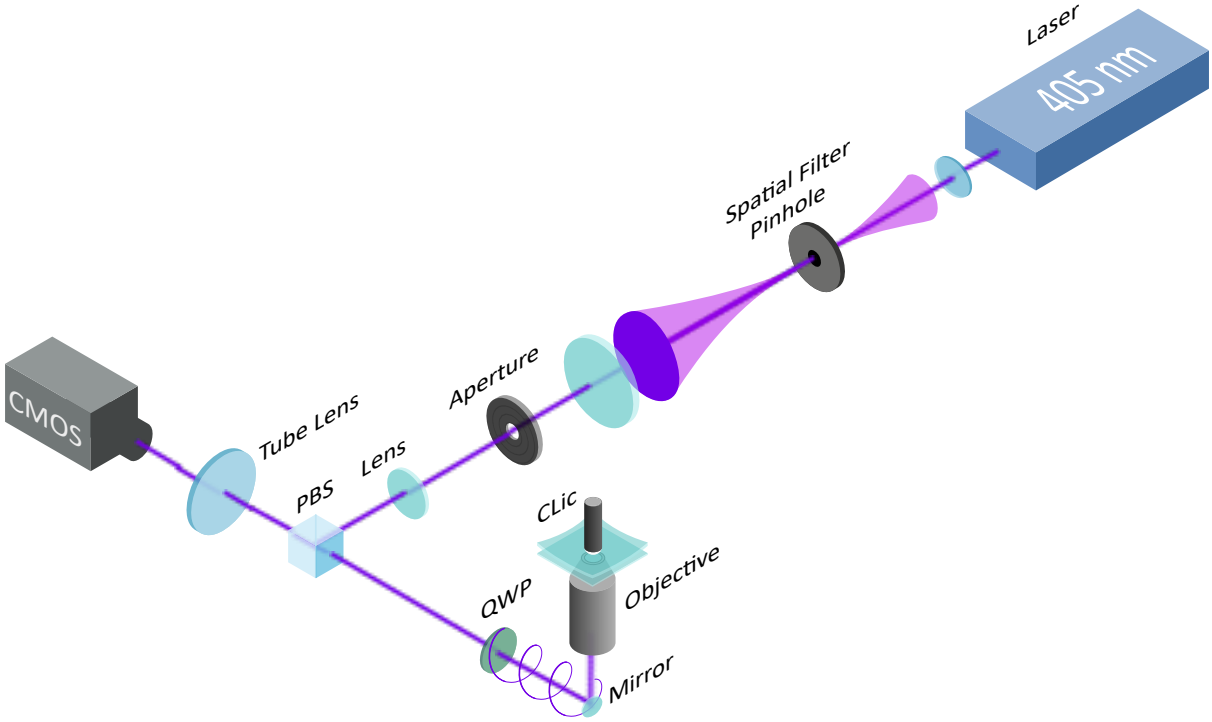

Figure 5.1: Isometric iSCAT Schematic. A laser is first shaped using a spatial filter for a gaussian beam profile. The gaussian beam is sized by an aperture and focused into the back-plane of an objective. During the focusing, the light is circularly polarized. At the sample plane, two coverslips are placed for sample containment. To control the gap between the two coverslips, the convex-lens induced confinement (CLiC) system is used. The scattered and reflected light is collected through a polarizing beam-splitter and the final image is focused on a CMOS camera array.

The laser illumination is provided by a 405 nm laser with a beam-width of 1 mm. The laser beam travels through a $4f$-lens system, consisting of a spatial filter, to optimize beam shape, and a beam expander, to enlarge the beam diameter. The shaped beam is then passed through an aperture, which is used to adjust for various camera array sizes. The beam then is passed into a focusing lens, which directs it towards the objective. The beam first passes through a polarizing beam splitter, which partially reflects the linearly polarized light. The reflected light passes through a quarter-wave plate, which circularly polarizes the light. The circularly polarized light is then focused on the back plane of the objective. The objective focuses the light onto the sample plane, and the sample scatters the light back into the objective. The scattered light is then passed back through the quarter-wave plate, linearly polarizing it. The light is then transmitted through the beam splitter and focused onto the CMOS camera.

The sample is contained between two glass cover-slips separated by a plastic spacer. To control the gap between the two coverslips, the convex-lens induced confinement (CLiC) system is used. The CLiC device is manually controlled with a control-box, which is not shown on the diagram. The control-box connects to a piezo controlled actuator and regulates the pushing force with fine precision.

5.2 Experimental Optical Table Overview

The laboratory implementation shown in Fig. 5.1, was assembled on an optical table. Fig. 5.2 and 5.3 show the lab setup.

The 405 nm laser, fabricated by Coherent, was controlled by a Thorlabs laser driver, which varies the output power. The laser driver was set to 850 mW which outputted a laser light intensity of ~ 220 mW.The Thorlabs power meter was able to measure the output light intensity with high accuracy. The laser driver is shown in Fig. 5.4.

Imaging was performed with an oil-immersion objective with magnification factor $60\times$ and numerical aperture NA = 1.4. The immersion oil was used required to match the refractive-index of the coverglass to the objective. The tube lens, located in front of the CMOS camera, was selected with a magnification factor of $2\times$. The total optical magnification factor was $120\times$. The final image was digitally zoomed such that each pixel represents 55×55 nm. Since the camera pixel array is 6.5 µm in size, it produces a total magnification of $118\times$.

The camera used was the PCO.EDGE 4.2. This camera has a pixel array of 2048×2048 with pixel size 6.5×6.5 µm. The camera can produce images of the full field of view at a maximum rate of 40 fps. When a smaller field of view is used, the camera can reach up to 1000 fps. The camera saves the images with the timestamps to provide precise frame-rates.

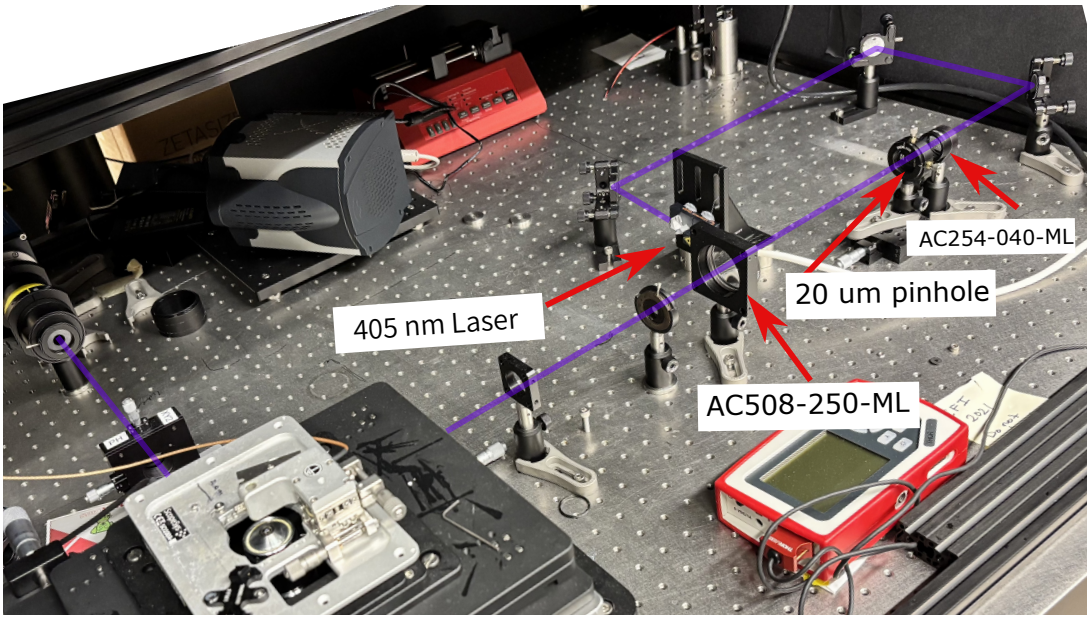

Figure 5.2: Optical table for the iSCAT microscope. Photographed at the Leslie lab, UBC.

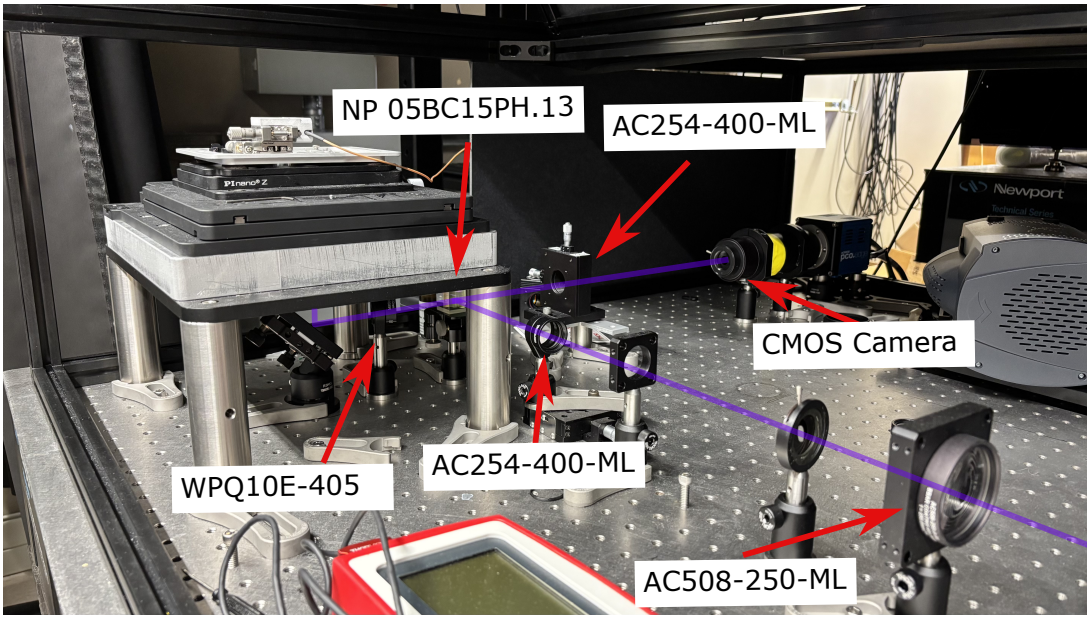

Figure 5.3: Optical table for the iSCAT microscope. Photographed at the Leslie lab, UBC.

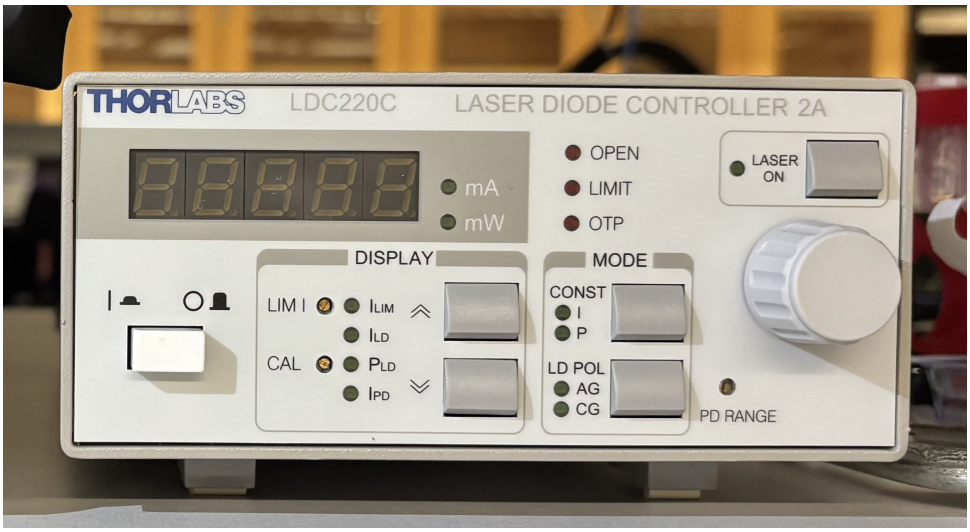

Figure 5.4: Laser Driver for the 405 nm laser diode.

5.3 System Parameters

The optical system parameters consist of technical specifications that are used to compare and contrast optical systems. These parameters are listed in Tab. 5.2.

Quantity	**Symbol**	**Value**
Wavelength	λ	405 nm
Power at Source	I_{Input}	240 mW
Power into QWP	I_{Out}	120 mW
Illumination Angle	θ_{ill}	45°
Objective magnification	M_{Obj}	60 ×
Objective numerical aperture	NA	1.4
Tube lens focal length	f_{TL}	40 mm
CMOS camera array length	l_{px}	6.5 µm
Digital pixel length	l_{dg}	55 nm
Camera frame-rate	fps	$40 - 1000$ fps
CMOS array type	—	sCMOS
Array ROI	ROI	2048×2048
Exposure time	τ	0.0001 sec
Temporal resolution	fps · ROI	$(401 \cdot 250) \cdot 1000$

Table 5.2: System Parameters for iSCAT Illumination, objective, and camera settings.

The laser power measured at the objective is sufficient. The contrast between the background and the particle signal is a function of the exposure time and laser power. The current laser power allows for clear contrast between the particles and background at a low exposure time, allowing for a high frame rate.

The speed of the camera is sufficient. For a large ROI, 401×450, the frame-rate is 400 fps. For smaller ROI, 201×250, the frame-rate is 1000 fps. The mean step length of a 200 nm particle in water at these conditions is ~ 100 nm. For axial explorations, this frame-rate is sufficient. For transient measurements, the viscosity can be added to reduce this step length by a factor of $10 - 100\times$. This frame-rate allows for resolutions of ~ 1 nm.

The pixel size of this setup is sufficient for scattering data. Since the scattering pattern is much larger than the particle, clear iPSF images can be recovered.

5.4 Alignment Procedure

The system's alignment is divided into three portions, the laser input, objective illumination, and the camera detection.

The alignment of the illumination does not need to be adjusted once aligned. The system is aligned to a spatial filter, which consists of a 20 µm pinhole. This pinhole constrains

the position and angle of the laser beam. The alignment is thus easy to diagnose as a misaligned laser beam at this stage would significantly drop the laser power into the objective.

The alignment of the objective is quite stable, however, requires maintenance when significantly altering the objective positions. The mirror directing the laser beam into the objective is fixed with screws and is placed 1 cm from the optical table, minimizing thermal expansion.

The alignment of the quarter wave-plate is stable and can be regularly altered to adjust for on-the-fly measurements. Specifically, to adjust the reflectance into the camera, the QWP is angularly adjusted and spatially rotated during measurements. Once in place, the QWP does not need to be adjusted. Thus the QWP is often adjusted during the start of measurement to optimize the SNR.

5.5 Future Improvements

Future additions and improvements can be made to optimize the usability and frame-rate. The camera frame-rate, although sufficient for work done in this thesis, can be improved upon to obtain transient information in water. To do this, the cameras can be replaced to faster frame-rate models. In other methods, more cameras could be added and synced to have a frame-rate offset that allows for sequential image capture. This additional camera acquisition would effectively double the frame-rate, which would significantly improve the effective temporal resolution of the microscope system.

Chapter 6

Simulations of Diffusion of Nanoparticles in Double-Layer Walls

Langevin dynamics are a framework for computational simulations that model the model of particles by accounting for both random thermal fluctuations arising from surrounding environment and deterministic forces such as external fields or interactions with other particles.

This chapter describes a simulation model of nanoparticle brownian motion within a confined space subject to electrostatic field generated by two parallel charged surfaces.

Python software was written based on prior Langevin python implementation published on Github [3]. Langevin algorithms were implemented and parameterized to account for physical electrostatic forces and charge quantities.

Using the software, statistical averages of the particle motion was calculating for a range of electrostatic screenings length presented in the sections below.

6.1 Simulation Equations

The general Langevin equation, discussed in Chap. 2, consists of a classical equation of motion with a stochastic form term. In a general system, the equation is written as

$$m\ddot{\vec{x}}(t) = -\gamma\dot{\vec{x}}(t) + \vec{F}(t) + \Gamma\vec{W}(t)\,, \tag{6.1}$$

where γ is the drag coefficient, Γ is the magnitude of fluctuation, $\vec{F}(t)$ is a deterministic force, and $\vec{W}(t)$ is a vector random variable.

The Langevin equation programmed into the software is parameterized in terms of the position dependent friction coefficient, $\gamma(z)$; specifically an axially dependent hydrodynamic

friction depending on how close the particle is to a surface. Additionally the stochastic magnitude, Γ, is substituted with a relation associated by the fluctuation-dissipation theorem. The equation is written as

$$m\ddot{\vec{x}}(t) = -\gamma(z)\dot{\vec{x}}(t) + \vec{F}(\vec{x}(t)) + \sqrt{2k_bT\gamma(z)}\vec{W}(t)\,. \tag{6.2}$$

In this form, the diffusion coefficient becomes axially dependent due to the axially dependent friction. The Stokes-Einstein relation then reads as,

$$D(z) = \frac{k_bT}{\gamma(z)}\,. \tag{6.3}$$

In over-damped case, where $m\ddot{\vec{x}} \ll 1$, the the equations of motion can split into the transverse and axial motion as they are independent, coupled solely by the diffusion. Here, we assume that the inertia term is negligible, meaning the velocity does not affect dynamics. The transverse motion will have no external forces, only being modulated by the diffusion coefficient.

$$\frac{dx}{dt} = \sqrt{2D_x(z)}W_x(t)\,, \tag{6.4}$$

where the diffusion coefficient, from Eqn. 6.3, is substituted inside of the magnitude of stochastic force. The axial equation of motion is modulated by the external force associated with the electrostatics and hydrodynamics of approaching any given surfaces. Without defining boundary conditions, this is written as

$$\frac{dz}{dt} = \sqrt{2D_z(z)}W_z(t) + \frac{F_z(z)}{\gamma(z)}\,, \tag{6.5}$$

as described in Eqn. 7 of Ref. [3]. The validity of this model

Since $\vec{W}(t)$ is a random, non-analytical function, classical ODE methods of time evolution do not apply. To simulate the stochastic portion, the Euler-Maruyama method was used.*

The Langevin form of an SDE can be expressed in terms of differential increments known as the Itô form. This convention is preferred due to the Itô form being more computationally reasonable as it allows for calculation based on the current time-step. The Langevin approach, assuming a midpoint approximation of time-derivatives, requires back calculation, which is not computationally preferred as stability issues can arise.

The equations in Eqn. 6.4 can be written into the Itô form by converting the derivatives

*The Euler-Maruyama method was chosen due to use in prior publications such as Ref. [3] and a lack of time during laboratory work. Further considerations should be made towards higher order techniques such as the Milstein or Itô-Taylor approach, however, these only dictate convergence rates, which was not of concern in this work.

into differentials,

$$\begin{aligned} dX_t &= \sqrt{2D_x(z)}dw_x\,, \\ dZ_t &= \sqrt{2D_z(z)}dw_z + \frac{F_z(z)}{\gamma(z)}dt\,, \end{aligned} \tag{6.6}$$

where dw denotes the Wiener increment, dX_t denotes the transverse positional increment, dZ_t the axial positional increment.

In the Itô form, the Euler-Maruyama equations are written as a pair of discrete time-step evolutions as the differentials can be expressed by a finite-difference scheme. This scheme then leads to the Euler-Maruyama approximation. The equations, written in terms of their finite-differences, are expressed as

$$\begin{aligned} x_{n+1} &= x_n + \sqrt{2D_x(z_n)\Delta t}w_{x,n}\,, \\ z_{n+1} &= z_n + \left[D_z'(z_n) - \beta D_z(z_n)V'(z_n) + \sqrt{2D_z(z_n)}w_{z,n}\right]\Delta t\,, \end{aligned} \tag{6.7}$$

where $V(z)$ is the electrostatic potential, $\beta = 1/k_bT$, and Δt is the discretized time. For clarity, Δt is an incremental time step, $w_{x,n}$ and $w_{z,n}$ are independent Gaussian noises with zero mean and $1/\Delta t$ variance. It should be noted directly that there exists a convention-dependent spurious drift term, $D'(z)$ originates from the conversion from the Langevin approach to the Itô forward time convention. This correction is directly applied such that the simulated z_t positions satisfy the Gibbs-Boltzmann distribution in the equilibrium limit [3]. The equations are programmed into PYTHON software, and are referenced to as MJ_LANGEVINSIMULATIONS.

6.2 Electrostatic Input Parameters

The simulation electrostatic conditions can be chosen to mirror those found in the laboratory. The laboratory conditions are defined by the salt concentration, which is directly related to the ionic concentration. This motivates the formulation of electrostatic contributions in simulation to accurately mirror that which will be observed in experiment.

The deterministic portion of the Langevin equation in the simulations are determined by the electrostatic and gravitational forces acting on the particle. In the model of nanoparticles, the gravitational forces are negligible, leaving the electrostatic forces as the driving non-spurious drift. A diagram of the simulation geometry is shown in Fig. 6.1.

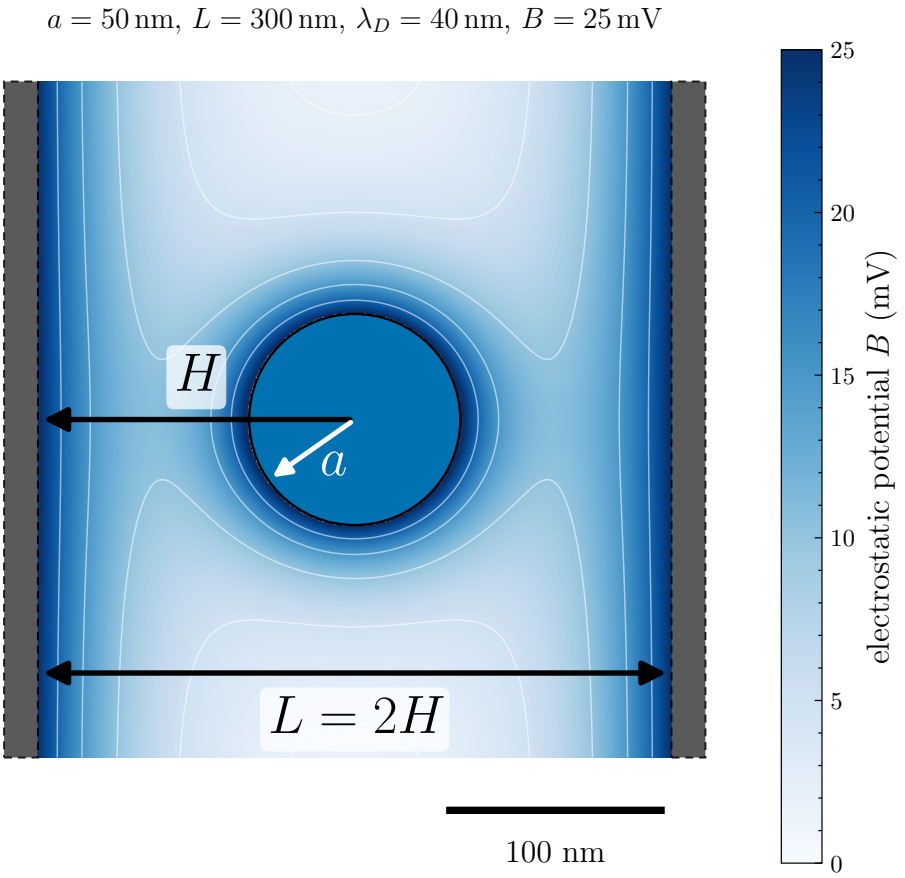

Figure 6.1: Model diagram of simulation geometry. A spherical particle of ζ-potential 25 mV4 and radius $a = 50\,\text{nm}$ is modeled between two charged parallel plates spaced by $L = 300\,\text{nm}$. The electrostatic potential, modeled by the Debye-Hückel potential, are drawn in a blue gradient and fixed at a Debye-length of 40 nm. The values are chosen to model a semi-realistic system, such as a plastic bead in a salt solution while still providing an artistic rendering for intuition.

The electrostatic forces that drive the dynamic motion are described by the Debye-Hückel potential, as described by Eqn. 2.37:

$$U(z) = B\left[e^{-(H-z)/\lambda_D} - e^{-(H+z)/\lambda_D}\right], \tag{6.8}$$

where the amplitude term, B, is described by the electrostatic contributions due to the charges present in the solution and on the surfaces of the geometries, recalled as the Stern layers. The electrostatics of this system are described in depth in Chap. 2. To ensure that the equations represents the sample of interest, the electrostatic amplitude is calculated from laboratory measurable quantities. The function is described the by the Gouy-Chapman equations in a linear-superposition approximation of a double-layer potential described by Eqn. 2.39 and Eqn. 2.40 [53, 57]. These equations are physically constrain the electrostatic field amplitude to ensure physical motion during the simulation process.

6.3 Python Simulation Implementation

The Langevin equations are implemented using PYTHON. The PYTHON version used for simulations was 3.13.5. The libraries NUMPY version 2.4.6, SCIPY version 1.18.0, and NUMBA version 0.99.0 were used as the main drivers for the simulations [10, 99, 100].

Since NUMBA uses a pre-compiled set of functions, it was more CPU efficient to write a parameter class and a simulator class separately.

The parameters class, PARAM holds the main parameters of the simulation. These parameters are shown in Tab. 6.2

Quantity	**Symbol**	**Default Value**
Particle Radius	a	50×10^{-9} m
Confinement Length	L	225×10^{-9} m
Debye Length	λ_D	10 nm
Electrostatic Amplitude	B	30
Temperature	T	300 K
Viscosity	η	1.0×10^{-3} Pa·s
Particle Permittivity	ε_r	80
Particle Density Difference	$\Delta\rho$	50 kg/m^3
Boltzmann Length	ℓ_B	None
Time Step	Δt	None

Table 6.1: Variables stored in the PARAM class.

This PARAM class has functions that calculates the current viscosity and forces based on some input z. These functions are described by the confined mobility relations and Debye-Hückel electrostatic forces in Chap 2.

Quantity	**Symbol**	**Input parameters**
Transverse Viscosity	γ_x	a, L, η, z
Axial Viscosity	γ_z	a, L, η, z
Transverse Diffusion coefficient	D_x	T, γ_x
Axial Diffusion coefficient	D_z	T, γ_z
Derivative of Axial Diffusion coefficient	D_z'	T, z, η, a
Gravitational Force	F_g	$z, a, l_B, \Delta\rho$
Electrostatic Force	F_{elec}	B, z, λ_D, T

Table 6.2: Methods of the PARAM class.

To iterate the simulation in time, a STEPTRAJ function is written that operates on elements of the PARAMS class. This is done to utilize NUMBA's pre-compiled library of functions. This function, which is written in a NUMBA JIT decorator, takes in elements from PARAMS class, the current axial position z, the number of sub-steps and a NUMPY randomizer seed for reproducibility. The function loops over the number of sub-steps.

First, the viscosities, spurious forces, and external force magnitudes are calculated in both the transverse and axial directions. The Euler-Maruyama equation is then calculated and the position of the particle is updated in both axises. The returned parameters are the time array and the axial and transverse positions as indexed by the time array.

The time complexity of this function is $\mathcal{O}(n)$, where n is the number of time steps. The number of simultaneous trajectories, M, does not affect time-complexity in applications used in this thesis. The time complexity of the model for one single trajectory of length τ is measured in Fig. 6.2. To illustrate this further, the case of two parallel trajectories are computed in Fig. 6.3.

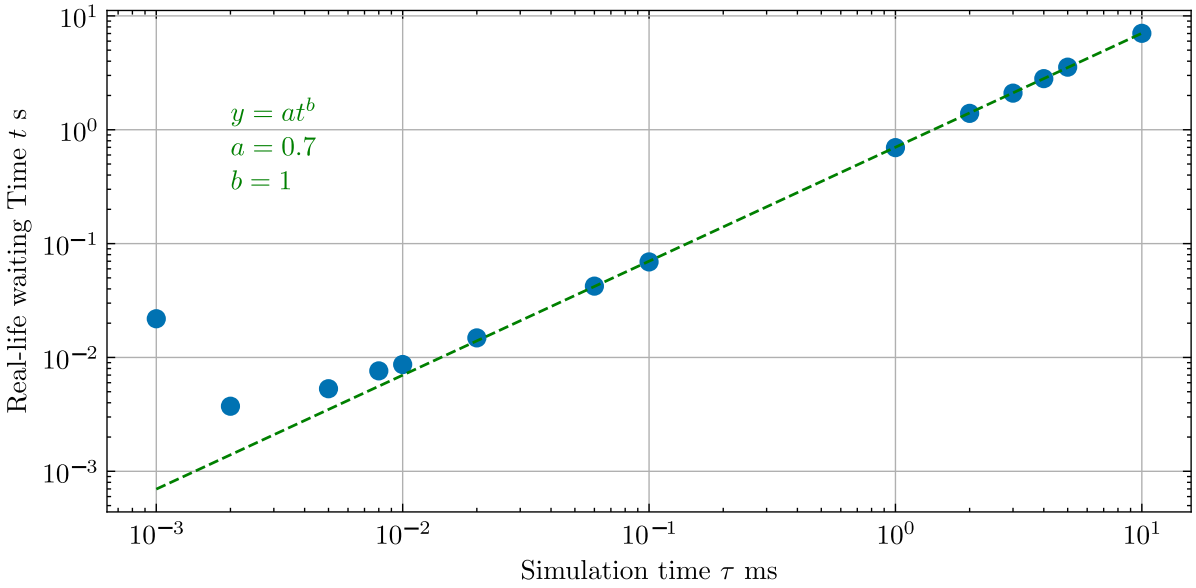

Figure 6.2: Langevin simulation function StepTraj time complexity. Simulation parameters are set as default with $dt = 2 \cdot 10^{-7}$. The number of simultaneous particles is 1, and the number of recorded positions consists of all of the points in the trajectory. In big-O notation, the time complexity is $\mathcal{O}(n^1)$. Simulations performed on a 2020 MacBook Pro with ARM M1 processor.

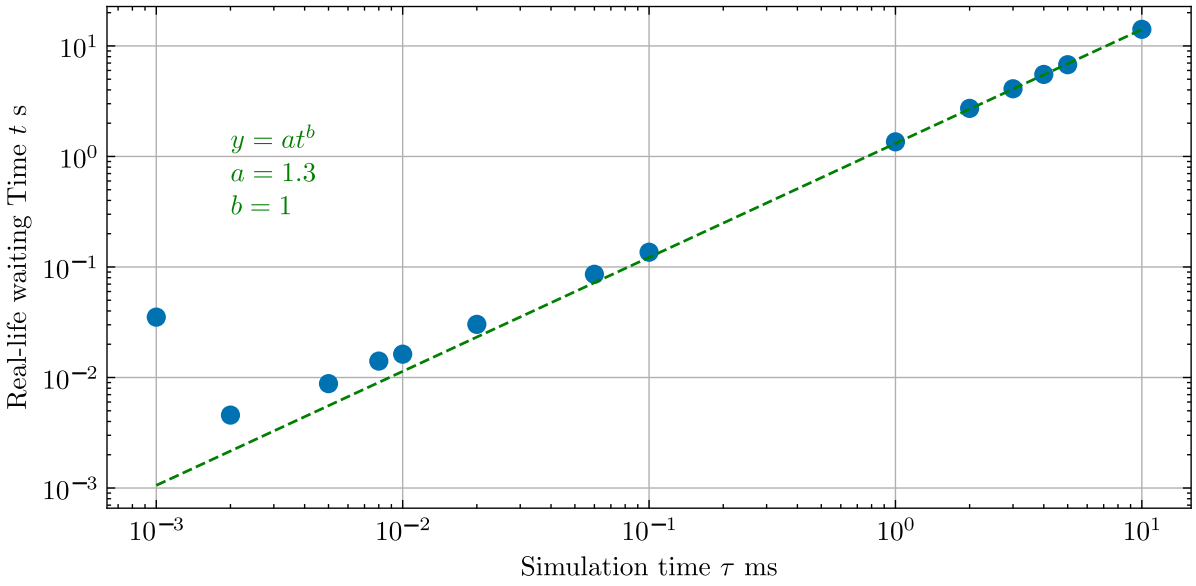

Figure 6.3: Langevin simulation function StepTraj time complexity. Simulation parameters are set as Fig. 6.2 with two parallel simulations.

6.4 Mean-Squared Displacement in Confinement

From the trajectories, the mean-squared displacement, MSD, can be calculated. The simulation results provide three-dimensional trajectories of particles. An example of this simulation can be found in Fig. 6.4 and Fig. 6.5. Mathematically, the MSD is written in terms of the sample variance of the displacements of the transverse and axial displacements for a given time lag j:

$$\mathrm{MSD}[j] = \frac{1}{N-1}\sum_{n=1}^{N} |x_{i+j} - x_i|^2 \,, \tag{6.9}$$

where $1 \leq j < N$ denotes. Here, MSD$[j]$, also labelled as $\langle x \rangle_c$ and $\langle z \rangle_c$ for consistency of notation for the transverse and axial MSD, denotes the MSD with a time lag for j steps. Intuitively, this is the variance of displacement after waiting $j\Delta t$ time after starting a clock and observing the particle diffusion.

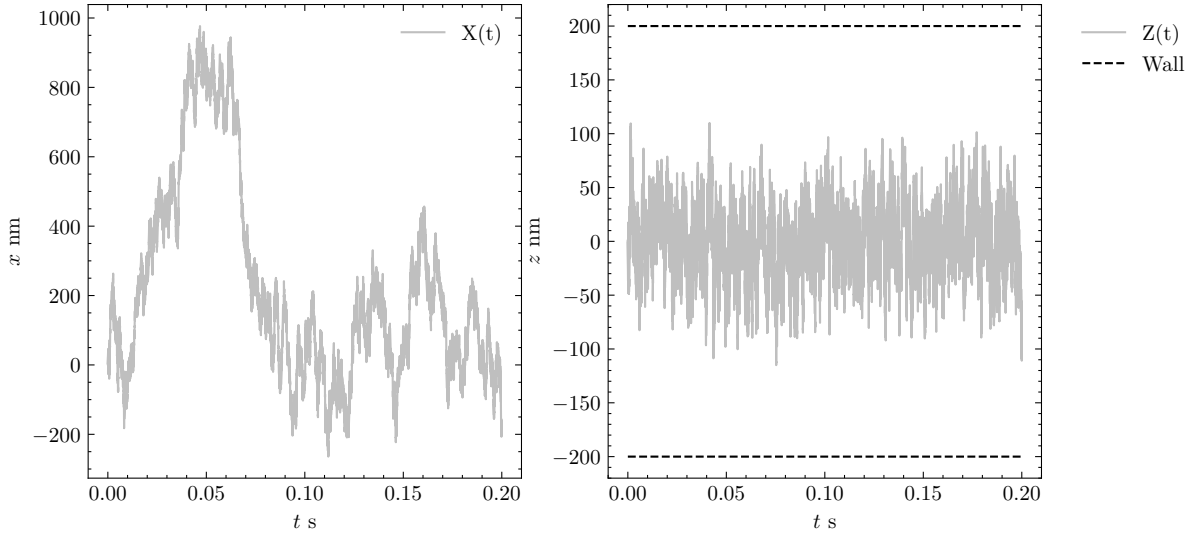

Figure 6.4: Confined Brownian Particle realization of z diffusion in electrostatic potential. The step-histogram is plotted in green. Simulation input parameters are $a = 50\,\mathrm{nm}, \zeta_p = -3.5\,\mathrm{mV}, \sigma_w = -2\,\mathrm{mC/m^2}, L = 225\,\mathrm{nm}, \lambda_D = 100\,\mathrm{nm}, T = 300\,\mathrm{K}, \eta = 1\,\mathrm{mPa\,s}$.

For each simulated particle trajectory, the MSD can be found. The results are statistically inferred parameters that can be compared to the baseline inputs. The size of a particle can be measured from the mean-squared displacement. In confinement, an addition correction must be applied due to the axial friction. In the case an electrostatic field, the electrostatics will add an additional confinement if the walls repel the particle further. Analytically, this is not solvable as the equilibrium solution is non-analytical. Specifically this involves solving the one dimensional diffusion equation with a hyperbolic sin external force, which originates from the derivative of the Debye-Hückel potential;

$$\frac{d^2p(x)}{dx^2} = -\sinh(x)p(x) \,, \tag{6.10}$$

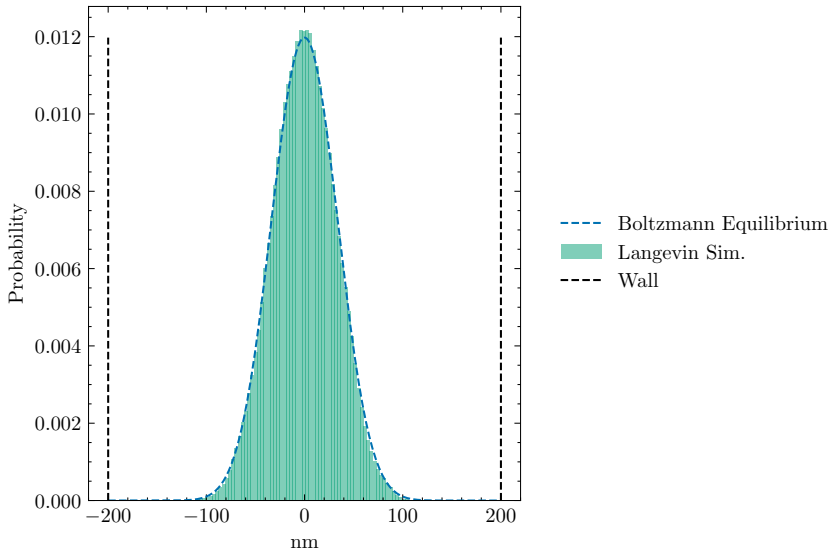

Figure 6.5: Gibbs-Boltzmann equilibrium of a confined Brownian particle. The green histogram is the experimentally measured distribution while the blue line is the theoretical distribution. Simulation input parameters are $a = 50\,\text{nm}, \zeta_p = -3.5\,\text{mV}, \sigma_w = -2\,\text{mC/m}^2, L = 225\,\text{nm}, \lambda_D = 100\,\text{nm}, T = 300\,\text{K}, \eta = 1\,\text{mPa s}$.

which has no analytical solutions [101].

To quantify if effective confinement length L, which is the asymptotic plateau of the MSD in a confined environment, changes as a function of the electrostatic field, the MSD of the axial and transverse trajectories are compared in two cases. In the electrostatic ally neutral case, this asymptotic is equal to $L^2/6$ in a one dimensional, linear confinement [39]. The simulation of a highly negatively charged particle and the second for a neutrally charged particle was performed, shown in Fig. 6.6.

The simulations demonstrated that as the particle charge becomes greater in magnitude, there is a large apparent change in the MSD plateau. This phenomena is expected, as the electrostatic field focuses the particle into the center of the two plates. This relationship is a function of the zeta-potential of the particle and can be explained using Coulomb's law as explained in Chap. 2.

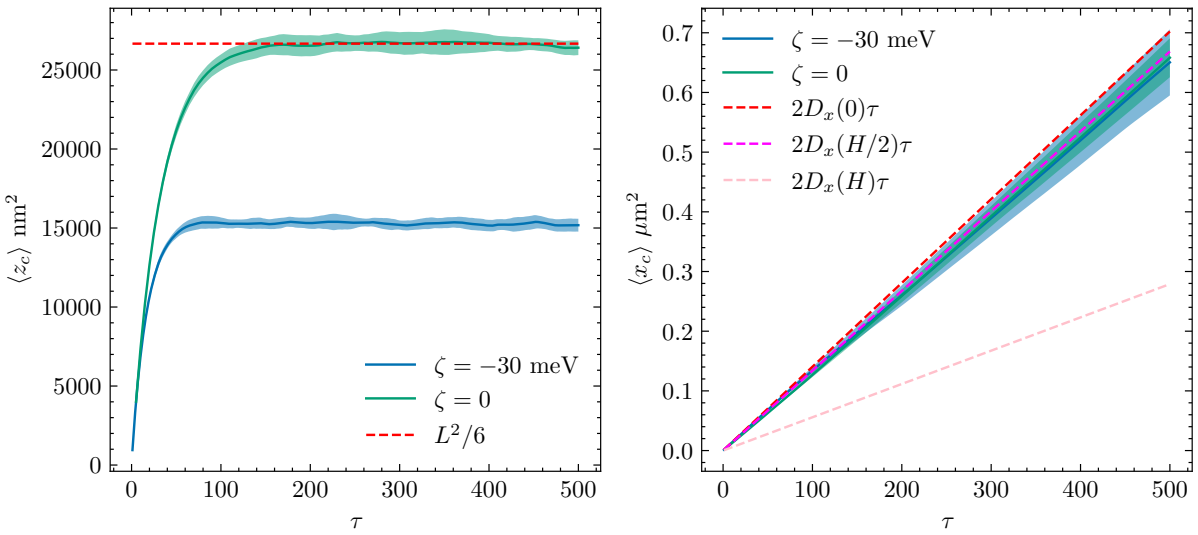

Figure 6.6: Comparison between MSD curves of two different charged particles. The green trace represents a neutrally charged particle with a maximal MSD of $\sim 26\,000\,\mathrm{nm}^2$. The blue trace represents a negatively charged particle with a maximal MSD of $\sim 15\,000\,\mathrm{nm}^2$. The chart on the right shows the transverse MSD of the two particles. Each unit of τ consists of on exposure time of 200 µs. The Debye-length was fixed to $\lambda_D = 10\,\mathrm{nm}$.

6.5 Simulation Results

Using the developed software, the relationship between the effective confinement length L, which as mentioned is the horizontal asymptote of the mean-squared displacement, MSD, can be numerically determined as the software numerically approximates the system. To mirror laboratory experiment, the electrostatic field must be varied as the zeta-potential of the particle is constant. In the simulation, the electrostatic field can be easily manipulated and, as previously discussed, the laboratory experiment can achieve similar conditions by varying the ionic concentration of the solution. Changing the electrostatic amplitude in laboratory experiments is possible as changing the Debye-length is inversely proportional to the changing, ionic concentration [53];

$$\lambda_D \propto \frac{1}{\sqrt{C}}, \tag{6.11}$$

where C is the ionic concentration of a dilute electrolyte compound.

The simulation for the geometry described above was conducted for various Debye-lengths. The conditions of the simulation are such that the zeta potential of the particle and glass are chosen to resemble glass and a polystyrene bead. The simulation results are shown in Fig. 6.7

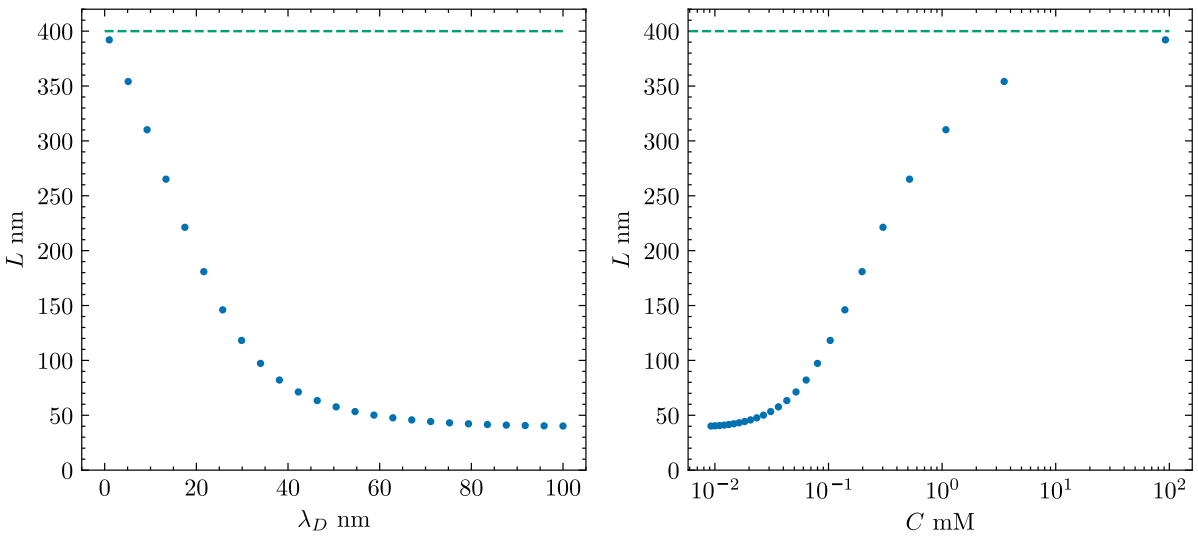

Figure 6.7: Effective confinement length L, or the minimum and maximum trajectory axial position, plotted (left) against the solution's Debye-length λ_D and (right) expressed in terms of ion concentration C. The blue dotted line represents the confinement lengths inferred from the average plateau values of 1000 simulated confined diffusions. The green dashed line represents the theoretical $L - 2a$ confinement length, which results from the total length subtracted by the diameter of the particle. The particle parameters were chosen to be $a = 50\,\text{nm}, \zeta = -30\,\text{mV}$. The geometric conditions are: $\sigma_w = -2\,\text{mC/m}^2, L = 500\,\text{nm}, T = 300\,\text{K}, \Delta t_i = 0.2\,\mu\text{s}, \tau = 200\,\mu\text{s}, \eta = 1\,\text{mPa s}$.

The figure obtained from the simulation shows that as the Debye-length increases, the confinement length decreases. When the ion concentration increases, the particle is less screened which allows it to move closer to the walls. With increased concentration of the ions, the particles can more easily explore the volume. Once the concentration reaches a sufficient saturation, the particle move from wall to wall. As the ionic concentration approaches 0, the confinement length approaches a minimum of $\sim 50\,\text{nm}$. Classically, without fluctuations, this should reach 0 nm. Due to the stochastic force and simulation integration artifacts, the minimum confinement appears to plateau at around $1a$ of diffusion volume. This extra motion is attributed to having a finite time-step during the evolution of the Langevin equation, adding extra force.

As a result from the changing effective confinement length, the diffusion coefficient changes as a function of ionic concentration. Since diffusion is dependent on the axial position of the particle with respect to the wall, there is a relative diffusion decrease with addition of ions. The exact relationship can be investigated using the developed simulations.Using the simulated trajectories in Fig. 6.7, the diffusion coefficient of the particle is be measured from the first 15 points of the MSD slope by fitting to a line as performed in optical experiments. This appears to be a lot of loss, however, the approximation being made, in the presence of no drift and perfect localization, is negligible [29]. The resulting diffusion coefficient is then compared to the baseline input parameter, plotted in Fig. 6.8.

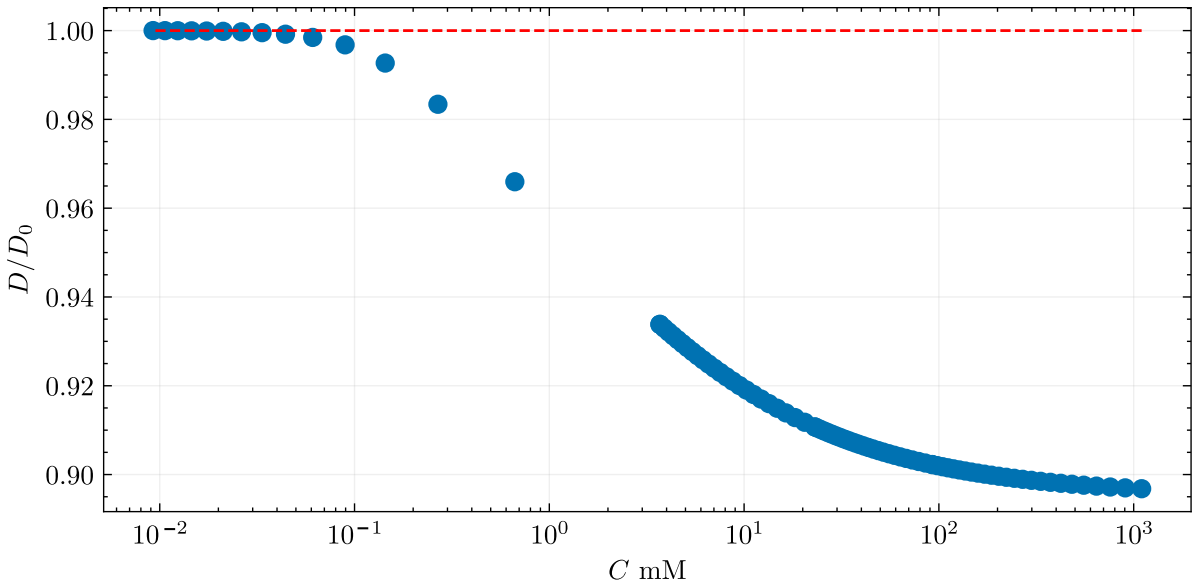

Figure 6.8: Effective diffusion D/D_0, or the measure diffusion coefficient compared to the input diffusion, plotted against the solution's Debye length expressed as concentration C of ions. The red dashed line represents the equality that the diffusion coefficient measured is the same as that input into the simulation. The blue dotted lines represent the ratio of diffusion coefficients measured using line-fitting of the first 3 points of the MSD_x. The particle parameters were chosen to be $a = 50\,\text{nm}, \zeta = -30\,\text{mV}$. The geometric conditions are: $\sigma_w = -2\,\text{mC/m}^2, L = 500\,\text{nm}, T = 300\,\text{K}, \Delta t_i = 0.2\,\mu\text{s}, \tau = 200\,\mu\text{s}, \eta = 1\,\text{mPa s}$.

The above shows that the presented simulations can be used to distinguish between classes of particles. The relationship between the particle's diffusion coefficient as a function of the particle charge and ion concentration is described by the curve in Fig. 6.8, allowing one to hypothetically distinguish particles based on where they lie. For instance, if a particle of the same charge, but different shape is measured, their diffusion coefficient and confinement length would theoretically differ from that same particle but at a different charge. This relationship offers future insights into how multi-labelled data can be used for particle identification. The scale of this difference notable at a 10% drop in the diffusion coefficient at high concentrations of ions. This is attributed to the particle being closer to the walls for a longer period of time due to the electrostatics having less effect. Intuitively, a larger diffusion volume would have larger contributions in frictional force due to the distance from the walls being smaller. This is contrast to highly electrostatically focused particles, which feel less drag as they are focused in the center of the double layer, maximizing their distance from the walls. By doing more simulations and testing for various particle types this relationship can be further investigated.

The effective confinement length compared to the zeta-potential of a particle at various Debye-lengths is simulated in the same geometric conditions as Fig. 6.7 and is plotted in Fig. 6.9. The figure presents the relationship between L, ζ, and the Debye-length, λ_D. Each curve shows the relationship between the values for a different ionic concentration. At high concentration, there is little to no effect on the confinement length, while at low concentration, there is a steep dependency near $\zeta \to 0$. As the particle becomes more negatively charged, the confinement length decreases towards the confinement length minima. When the particle becomes more neutral, the confinement length approaches the gap length between the two walls.

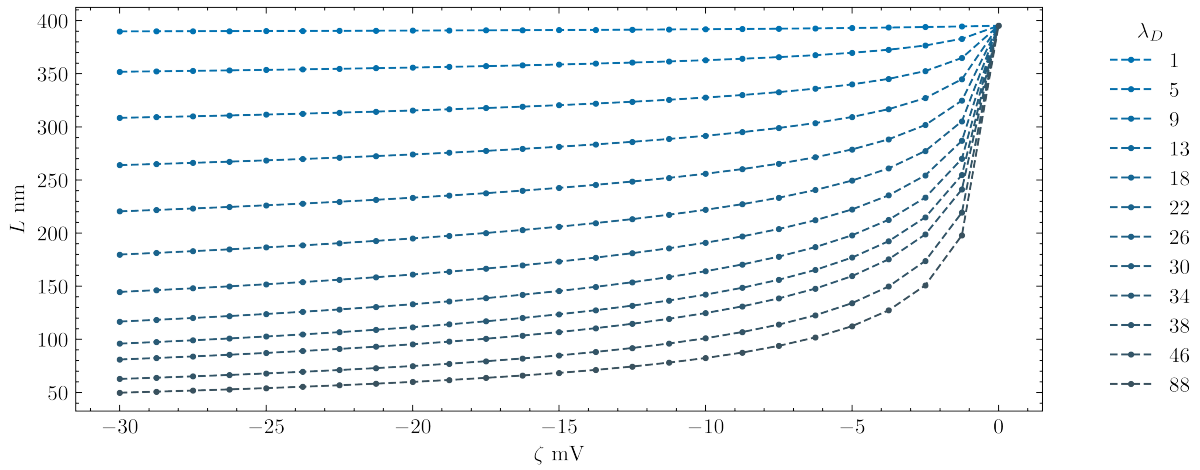

Figure 6.9: Effective confinement plotted against the particle's zeta-potential. Each curve represents one Debye-length labelled on the right-hand side. The zeta-potential of the particle is varied on the x-axis, while the recovered confinement length is plotted on the y-axis. Simulation parameters are identical to that of Fig. 6.7.

6.6 Surface Charge estimation

The surface charge of the confinement walls can be estimated based on the electrostatic potential. It can be first explicitly stated that by using the equivalence between a spatially varying diffusivity in the Langevin form and a drift in Itô form a drift amplitude can be statistically recovered using a maximum likelihood estimator function, or MLE of the Itô form.

Although the stochastic differential was defined in terms of a Langevin equation, the solution for the MLE was derived for the purpose of this thesis in the form and is presented in Eqn. 6.12. A detailed derivation is presented in Appendix. A. Recall Eqn. 6.8 as the form of the electrostatic potential. Defining the decoration $\hat{\cdot}$ as that of a statistical estimator, and $\phi(z)$ as the gradient of the electrostatic potential, the MLE can be defined as;

$$\hat{B} = \frac{\sum_n \phi(z)\left[\Delta z - D'(z)\Delta t\right]}{\sum_n D(z)\phi^2(z)\Delta t}, \quad \mathbb{V}[\hat{B}] = \frac{1}{2}\sum_{n=0}^{\infty} \beta^2 D(z)\phi^2(z), \tag{6.12}$$

where Δz is the random variable assigned to step displacement, $D(z)$ is the diffusion coefficient. Here, the gradient of the potential, $\phi(z)$ can be explicitly written out,

$$\phi(z) = e^{-\frac{H}{\lambda_D}} \sinh\left(\frac{z}{\lambda_D}\right). \tag{6.13}$$

where λ_D is the Debye-length and H is half of the width of the chamber. This estimator assumes that gravitational contributions are negligible. If the amplitude of the electrostatic interactions is known, one can invert the amplitude using relations in Eqn. 2.39 and Eqn. 2.40. This inversion requires at least the zeta-potential, ζ, of the particle, the radius, a, and Debye-length, λ_D.

The particle's exploration, demonstrated in Fig. 6.4 and Fig. 6.5, showed a clear dependence on a drift due to the electrostatics. This drift is a useful estimator of the particle-wall charge relationship. Using the aforementioned amplitude maximum-likelihood estimator function, the drift estimator can be extracted into meaningful, physical information.

To determine the effective accuracy of the MLE function, simulations are performed using the developed software. The Langevin simulation was ran for 20 ms with time-step of 0.2 µs. In this histogram there are 50 simulations with seeds set to $S_{\text{eed}} = i * 10 + 1$ for $i \in \{0, 49\}$ using the *numpy* seed parameter. $a = 50\,\text{nm}, L = 500\,\text{nm}, T = 300\,\text{K}, \Delta t_i = 0.2\,\mu\text{s}, \tau = 200\,\mu\text{s}, \eta = 1\,\text{mPa s}, \lambda_D = 10\,\text{nm}$ The surface charge is fixed at $\sigma_w = -2\,\text{mC/m}^2$. The distribution of the MLE is shown in Fig 6.10.

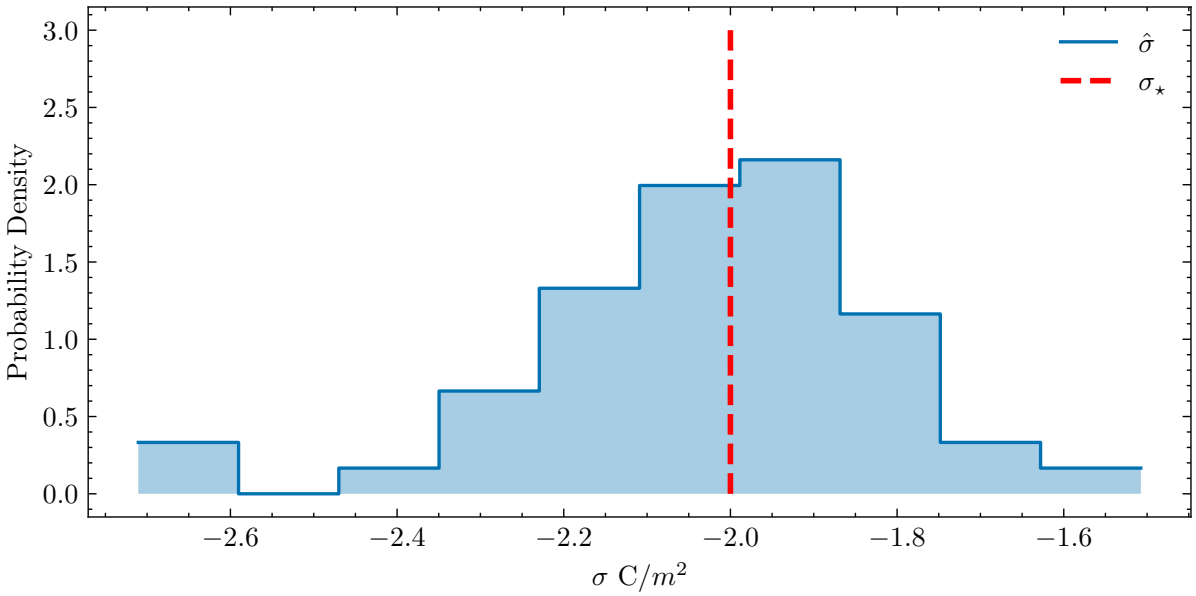

Figure 6.10: Distribution of $\hat{B}$ Re-parameterized for the surface charge. $\hat{\sigma} = 1/(A \cdot C)^2 \hat{B}$ where A, C are properly defined in Eqn. 2.39 and Eqn. 2.40 .

Chapter 7

iSCAT Axial Inference

This section presents a computational method for determining the axial (z-axis) position of a nanoparticle from iSCAT imaging data. The proposed approach is based on principles of light-scattering in the Rayleigh approximation, which govern the inference pattern as the nanoparticle moves along the optical axis.

In order to evaluate the accuracy and performance of the proposed method, Mie-theory simulations are used for for determining the an ideal nanoparticle's z-axis position. The precision and limitations are discussed in the context of recovering three-dimensional particle trajectories.

The raw images are acquired from a CMOS camera according the optical schematic illustrated in Figures 5.1 and 8.1. Each image consists of a two-dimensional interference recorded at a specific focal position z_{focus}, defined as the distance between the objective's focal distance and the objects in the field of view.

7.1 Method Overview

The method described in this section is based on computational image processing on images acquired using a CMOS camera. The camera has an array of 2048×2048 pixels. To acquire images at higher frame rates, only a portion of the CMOS array is used. This is set during the experimental details. Typical sizes used are 600×450 at 400 fps however, other fields of view can be used for specific experimental purposes. The highest frame-rate possible with this camera is 1000 fps at a FOV of 200×250.

The object sample is imaged through an oil-immersion objective lens. The optical system provides a magnification of $60\times$, with a tube lens of magnification $2\times$. The remaining is an digital zoom with a final resolution of 55 nm per pixel. The final magnification is then the ratio of the CMOS array length divided by the pixel size.

The image-processing workflow used to determine three-dimensional trajectory is shown in Fig. 7.1. The diagram summarized the sequence of image processing steps to extract each coordinate position.

1. **Acquisition** — Raw image captured on camera.
2. **Background subtraction** — Conversion of raw image to an iSCAT contrast image.
3. **Lateral tracking** — Transverse particle position, (x, y), located in each frame.
4. **Angular integration** — Extraction of the scattering data per frame.
5. **Nuisance-parameter estimation** — Fix iSCAT Amplitude, microscope decay parameter.
6. **Linking** — Axial position candidates, $z_{\min}$, linked by the Viterbi algorithm.
7. **Output** — Three-dimensional trajectory $\big[x(t), y(t), z(t)\big]$.

Figure 7.1: Outline of the 3D particle-tracking pipeline.

7.2 Raw image background subtraction

To obtain high-contrast interference signal from raw camera data, a post-processing background subtraction routine is introduced. This background subtraction measured a temporally averaged background using a window-filter and Fourier filtering. An example of this process is shown in Fig.7.2.

The full algorithm is presented in Alg. 1. The overview of the algorithm is to subtract the background noise in order to minimize Poisson, and electronic noise.

To reduce the contribution of the static background, the mean background is subtracted from each frame. Since the background illumination originates from the Gaussian illumination beam, which has a constant mean intensity over the acquisition period, this stationary background removes the background offset while preserving dynamic iSCAT signal. This just requires a good estimation of the background in order not to remove any iSCAT signal. The remaining shot noise, which follows Poisson statistics is reduced by temporal averaging during background estimation.

In order to suppress electronic noise from data-transfer, a row-wise averaging scheme is applied. Row-wise means are taken and normalized with respect to their row as the data-transfer from CMOS to computer is by horizontal lines. Thus, to ensure all lines are equally weighted, they are normalized with respect to their entire row. This is implemented

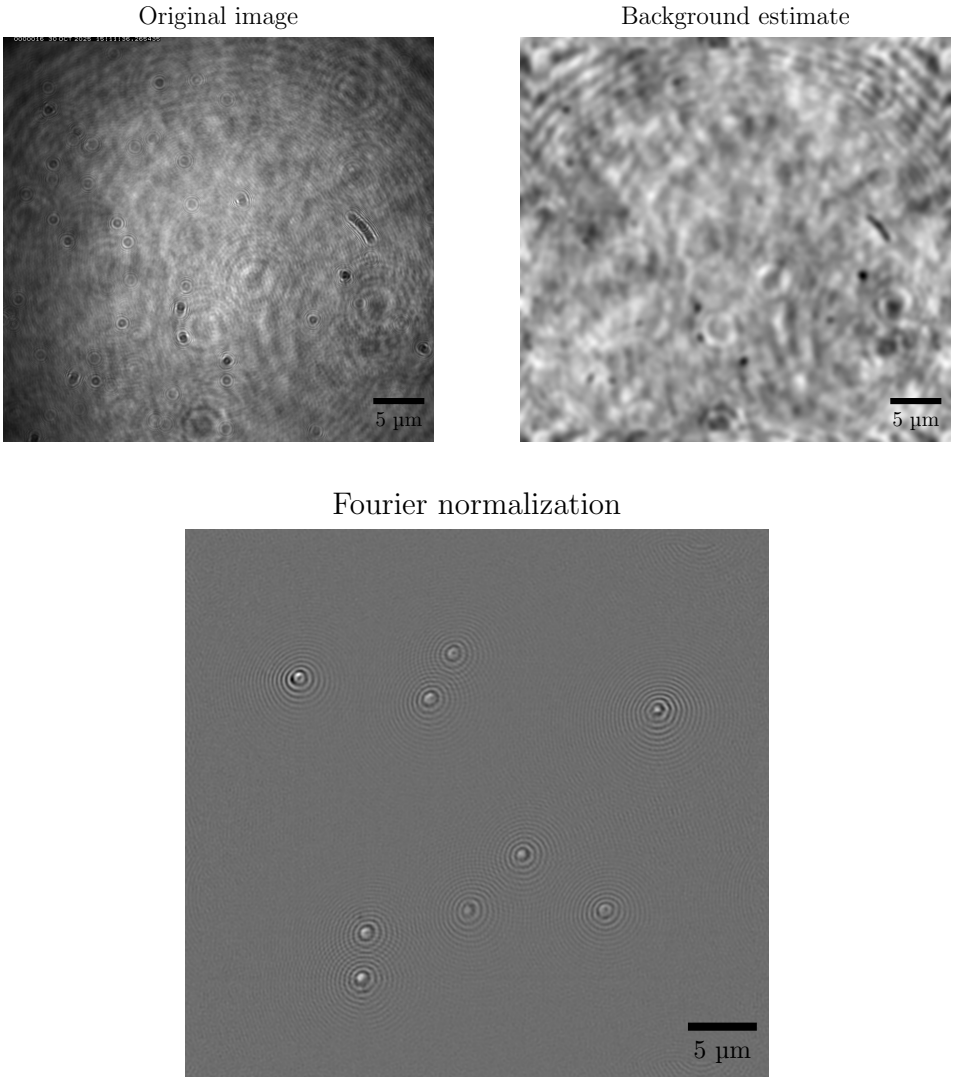

Figure 7.2: Raw image captured on camera to background subtracted iSCAT image. Single particles are tracked and extracted for further analysis.

as per prior iSCAT normalization methods by the Sandoghar laboratory [102].

The set of images, which consist of matrices of pixel-values of shape $F \times H \times W$, or F number of frames, W pixels wide, H pixels tall.

Before processing begins, the parameters of the Fourier-band-pass filter are determined from the image dimensions. The filter consists of an annular, donut mask of inner radius equal to 1% if the larger image dimension and outer radius equal to 25% of the larger image dimension, corresponding to $r_i = \max(H, W)/100, r_o = \max(H, W)/4$.

To minimize edge artifacts, a binary mask is applied to each frame. Pixels within 20 pixels of the boundary are assigned a value of 0 using a mask filter. On the interior pixels, the mask is set to 1, while on the pixels less than 20 pixels away from the edge are set to 0. This mask is used during normalization to prevent edge effects from biasing the inner

statistics.

For each frame in the stack, the image is first normalized using a row-wise normalization function. The function, titled AmpNorm in Alg. 1, first obtains the masked-image. The global normalized mean is stored. For each row, the mean is taken and the row of the global-mean is adjusted by the row-wise mean. The final image is the global mean divided by the row-mean for each row.

Once normalized, the background of the image is calculated by taking the forward and backward ΔL frames spaced by a spacer of Δf frames and taking the AmpNorm of these frames.

The iSCAT image then consists of a background subtraction between the normalized image and the background. Finally, once background subtracted, the image is Fourier-filtered using the pre-allocated band-pass filter. This process is demonstrated in Fig. 7.2

7.3 iSCAT (x, y) Tracking

To determine the transverse positions of any particles detected in the iSCAT method, the radial variance transform algorithm is used [103]. The original algorithm is briefly outlined below for completeness. The full code is written by the Sandoghdar group and can be found in Ref [103]. A demonstration of this method is presented in the Fig. 7.3.

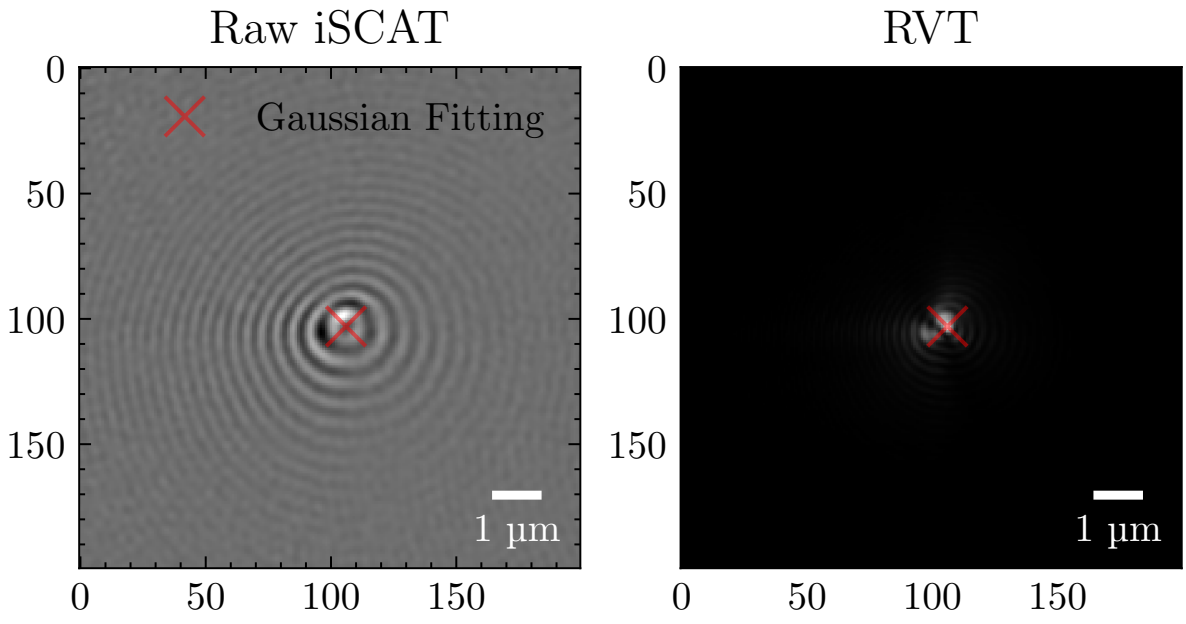

Figure 7.3: Raw image to RVT. Gaussian fitting of the RVT is shown on both plots.

The transverse localization algorithm, referenced as TransverseLocalization in 2 consists of transformations onto the iSCAT image from the previous section.

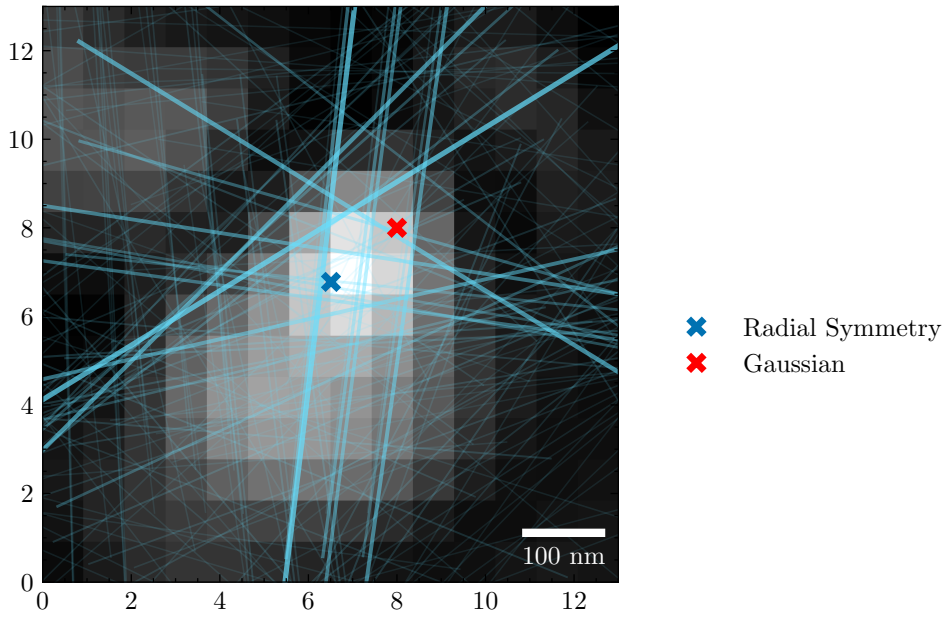

Figure 7.4: Demonstration of the Radial Symmetry method applied onto RVT of iSCAT data. Implemented from [104]. Figure adapted from [104].

The input image stack is first loaded and assumed to consist of radially symmetric interference fringes. The image is first transformed using the Radial Variance Transform [103]. This function converts a radial-fringe pattern into a point-like delta function when a radial symmetry is acted upon it.

Before proceeding, a frame-independent parameter for threshold SNR amplitude, number of pixels, and ROI cropping parameters are inputted. for a particle to be detected, the SNR of the RVT should be roughly 90% of the max iSCAT amplitude. An estimate of particle number is assumed, which acts as a prior for top n detections. Lastly, ROI for sub-pixel estimation is chosen.

For each frame, the iSCAT image is dilated. Particle positions that are considered maxima if the dilated image is equal to the RVT and the RVT is above the threshold for detection. Once candidates are chosen, a non-linear least squares Gaussian fitting to the RVT is performed to acquire higher-accuracy localization. To improve this to sub-pixel, a crop is done around each RVT Gaussian fitted point and a radial-symmetry algorithm is performed as per [104].

To ensure accurate linking, an implementation of the Munkres algorithm is used. This algorithm is chosen due to prior publication application, as well as a simple implementation within the author's lab codebase [90]. Briefly, a cost-matrix of distances is constructed and minimized in time to ensure that optimal displacements are chosen. This algorithm has limitations in cases where particles overlap. Ideally, experimental measure are taken to minimize this occurring. One such measure is to lower the concentration of particle in

the recorded images.

7.4 Angular Integral

Image data will have background misalignment due to iSCAT hardware setup. To optimize for fitting speed and misalignment property, radial scattering profile are assumed as in the Rayleigh regime model is symmetric. The radial integration method is described in Alg 3.

The discrete angular integral consists of summing angular terms on a radial grid. Mathematically, the output array obtained is described by the relation

$$g_t(\rho) \cong \frac{1}{\Delta\theta} \int_{\theta_0}^{\theta_0+\Delta\theta} I_t(\rho, \theta) d\theta, \tag{7.1}$$

where θ_0 is a constant, $\Delta\theta$ is the width of integration, $I_t(\rho, \theta)$ is the iSCAT image.

First, the cartesian image is Re-parameterized into polar coordinates using a change of basis transformation on the index grid of the image. Then, for each ring of indices, the image coordinates are summed up on this ring. The final array is an angular average that is indexed at radius r for some upper radial bound. This process is illustrated in Fig. 7.5.

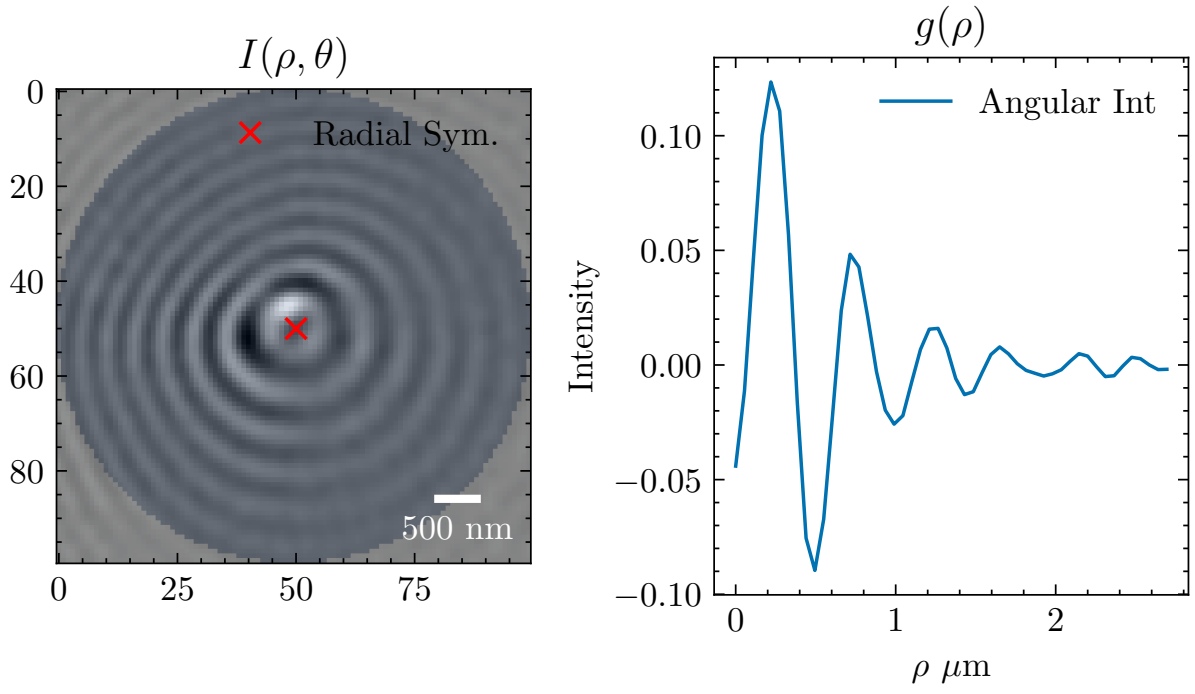

Figure 7.5: Angular Integral demonstration on experimental polystyrene image. The polar distance, ρ is plotted here where the red $\times$ symbol denotes the center of the image.

7.5 Nuisance parameter Estimation

In order to speed up computation and produce a more statistically relevant result, the method aims to fix the model parameters which should theoretically stay constant. These, for example, are the particle mass signal and optical aberration coefficients. These parameters, known as nuisance parameters, should be fixed before fitting to the more sensitive parameter of interest, the relative z position.

To obtain accurate estimation of the z position relative to nearby frames, nuisance estimation is done to focus on z fitting. To do this, L-BFGS-B Fitting is performed. alongside amplitude fitting. The theory of nuisance parameter estimation is described in [105].

The nuisance parameter method works on the premise of a model function. The iPSF model described in 4 is used as a weight function to compare to the data. First, bounds are placed for physical values of each parameter.

Here, it is important to note that the amplitude of the signal is proportional to the mass. The spatial decay of the aberration is modeled as a decaying exponential.

For each candidate decay parameter, the weighted correlation function is calculated. Let g be the radial profile, and G the theoretical iPSF model. The inner product of these two arrays, weighted in some integral of radiuses, is defined by $\langle G, g\rangle_w$. The correlation function is then written as

$$\rho(z,\gamma) = \frac{\langle G, g\rangle_w}{\sqrt{\langle G, G\rangle_w + \langle g, g\rangle_w + \varepsilon}}, \tag{7.2}$$

where ε is a small number to prevent division by 0.

The best-fit amplitude function here is analytical, given by the expression

$$\hat{E_0} = \frac{\langle G, g\rangle_w}{\langle G, G\rangle_w}. \tag{7.3}$$

The loss-function, $\mathcal{L}$, given the correlation, is given by the subtraction of ρ^2,

$$\mathcal{L} = 1 - \rho^2(z,\gamma). \tag{7.4}$$

Once a fitted $\gamma_0, \hat{E_0}$ are chosen, the optimal z for each frame is given by a non-linear bounded least squares which are acquired using the L-BFGS-B method with a Huber Loss ($\delta = 1/2$).

When computed for each frame in the stack, one has F optimally chosen γ_0, E_0. To obtain

the optimal parameter, a fit to distributions, inferred by [4], is used. These fits include a Gamma distribution for E_0 and a LogNorm distribution for γ. Together, these form nuisance estimation for E_0, γ_0.

7.6 2D Chain-link

Given an estimation of the nuisance parameters, a grid-search of the optimal z is simple. The physically best z position is a mean-squared error loss minima of the model subtracted with the data. However, due to the periodic nature of the data, there are up to 3 candidates for the optimal z position based on $\lambda/4$ having a high degree of similarity. Thus, to resolve this, a z-linking method was implemented to find the most likely trace based on diffusion probabilities.

To link consequent positions in space, a dynamic programming algorithm is used. The algorithm is known as the Viterbi method and is described in Alg. 5.

First, all minima for every frame must be collected. These are the candidate positions that trajectories are constructed from. The losses of these positions are also stored to ensure no spurious candidates are chosen.

For each position, the forward pass compares all chains moving in time. The loss is accumulated by a weighted Euclidean transition cost,

$$d_i(i,j) = \sqrt{w_x^2(x_{t-1}^i - x_t^j)^2 + w_y^2(y_{t-1}^i - y_t^j)^2}, \tag{7.5}$$

where x denote z motion, and y denotes the loss difference. This matrix compares all weighted transitions. The optimal transition $\pi_t(j)$ is then chosen to minimize this transition, and any prior transitions.

For each following forward pass, the cost is accumulated with a memory of 2 frames. The memory is chosen to ensure that an incorrect transition in a given frame is not penalized later on.

Thus linking method is thus a search algorithm for the shortest math based on the weighted Euclidean method. A short toy-model is shown in Fig. 7.6 to visualize the procedure.

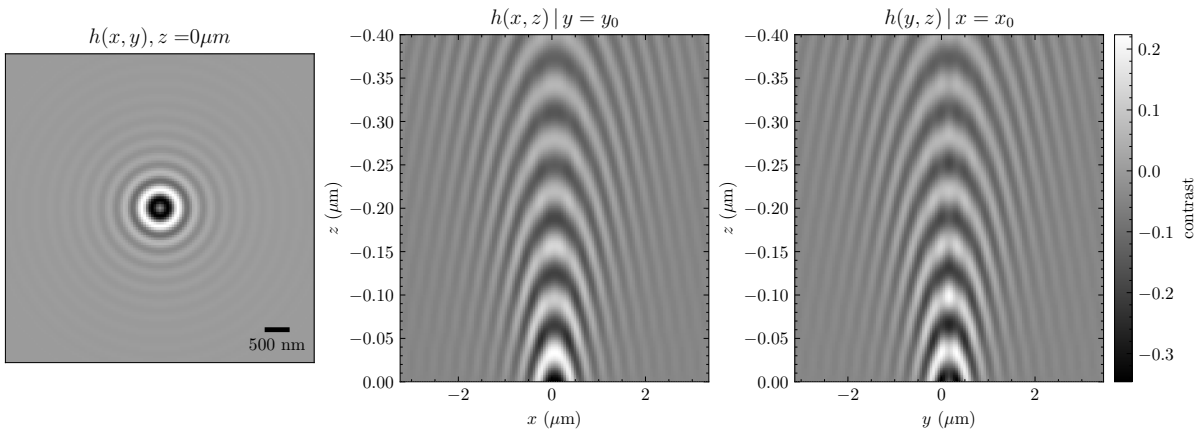

Figure 7.7: Simulated iPSF. Left. Projection of in-focus plane. Middle. x, z projection. Right. y, z projection. Focal scan is over 400 nm. pixel size is 55 nm/pixel. $\lambda = 405$ nm. Assumed particle radius $a = 50$ nm.

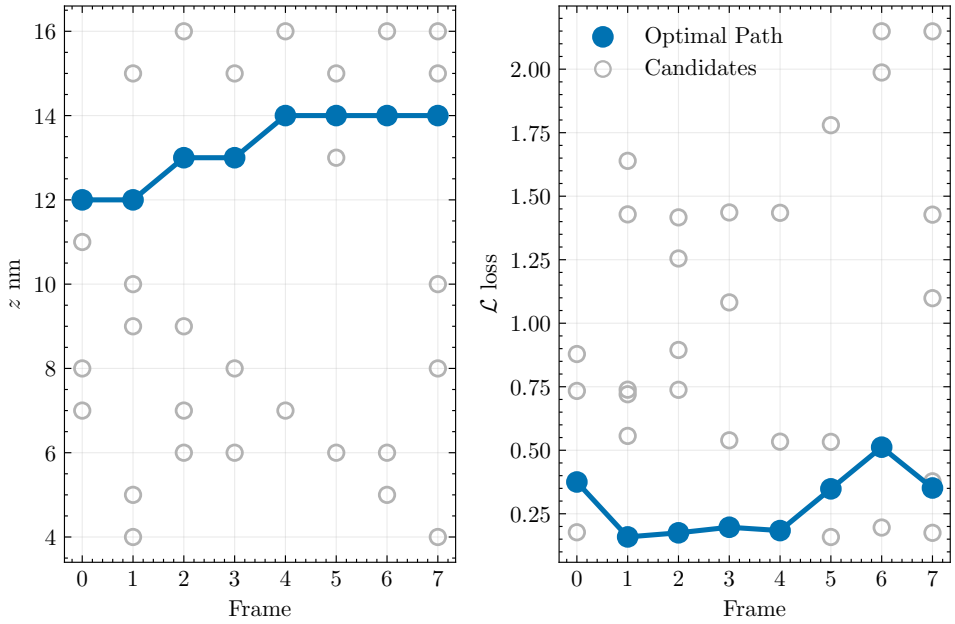

Figure 7.6: Viterbi Search algorithm toy example. Optimal path is chosen based on z similarity and lowest loss.

7.7 Simulated iPSF localization

To validate the proposed method, one performs simulated axial-estimation. Simulated iSCAT point spread functions were generated using the PySCAT library for $\lambda = 405$ nm. The iPSF was generated for the focal range $z_f \in [-1000, 0]$. The assumed particle is spherical with radius $a = 50$ nm. The iPSF is shown in Fig. 7.7

As mentioned, the localization pipeline utilizes a weighting scheme to account for objective aberration and misalignments to non-modeled regions of the data, such as the central

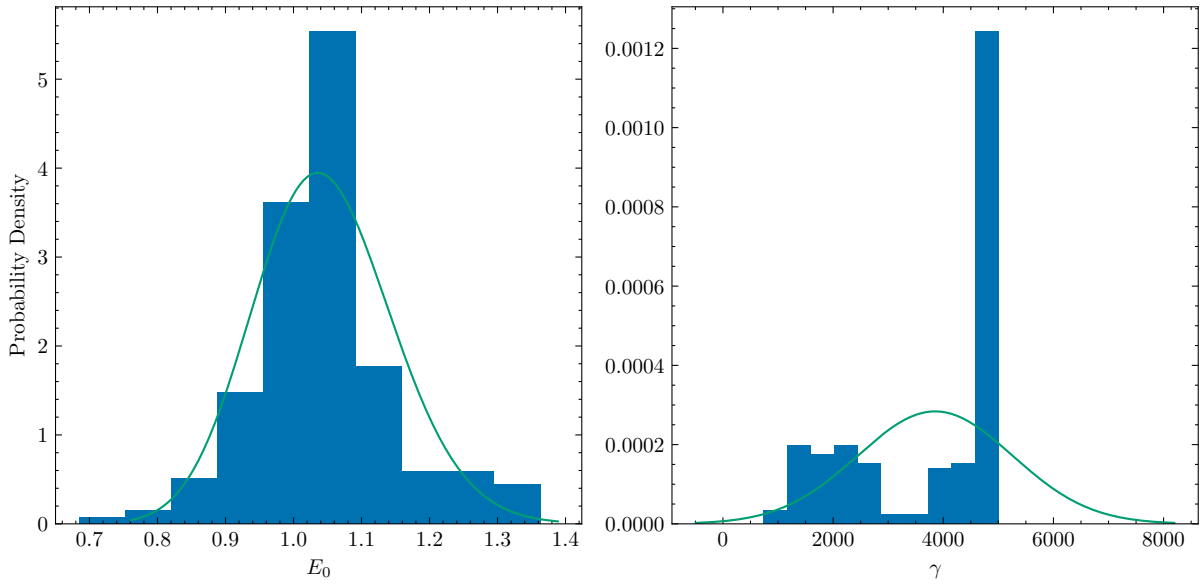

Figure 7.8: Nuisance Parameter estimation for iPSF fits. Left. Amplitude factor. right. exponential decay factor. The exponential decay is not of utmost importance as it often is very small. it is kept for stability purposes.

fringe. This is accounted for by using a binary weighting array, $\mathbf{w}$. In the case of these idealized simulations, the effect of changing the weighting scheme is demonstrated. This demonstrates a loss of absolute precision, however, maintains the relative precision, which is that of interest in measurements of axial trajectory.

7.8 3-parameter Estimation

As a reminder, the model for the iPSF is a multivariate function iPSF : $(z, E_0, \gamma) \rightarrow \mathbb{R}^2$. It has been shown that the statistical model of iPSF takes;

$$\begin{aligned} \hat{E_0} &\sim \text{Gamma}(\mu_{E_0}, \sigma_{E_0}) \\ \hat{\gamma} &\sim \text{logNorm}(\mu_\gamma, \sigma_\gamma) \end{aligned} \tag{7.6}$$

The so-called 3-parameter estimation is a sampling of nuisance parameters in the model. The nuisance parameters here are $\hat{E_0}, \hat{\gamma}$, where $\hat{E_0}$ is an amplitude factor, and γ is an exponential decay factor used to account for spatial decay in signal. The amplitude factor can be analytically solved for using a grid-search sweep along z. The γ factor is fitted to using the L-BFGS-B minimization algorithm. This nuisance parameter fit is shown in the histograms of these two parameters shown in Fig. 7.8.

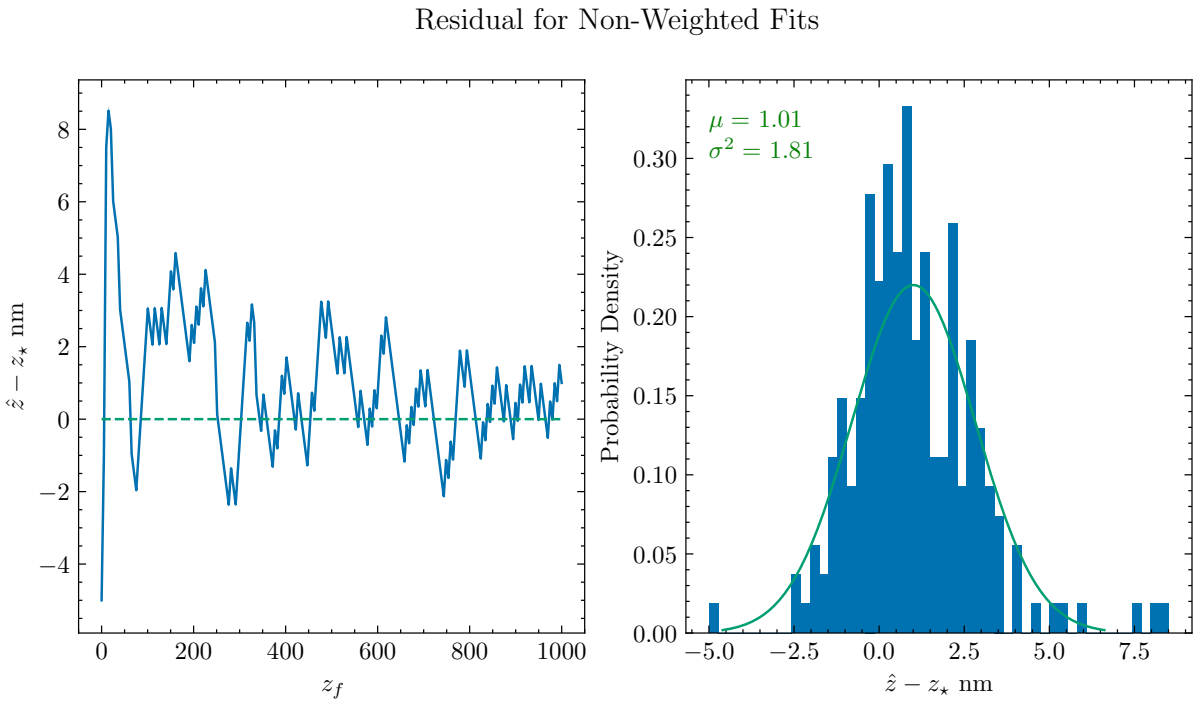

Figure 7.9: Fit residuals using no weighting scheme. reporting an absolute bias of 1 nm with variance of 2 nanometers.

7.8.1 Grid-Search Estimation

Once the nuisance parameters are chosen, a grid-search with a dynamic linking is performed. Each frame is assumed to have a maximal diffusion time. This is in place to optimize for noise and have coherent, physical traces without outliers. The grid search takes the difference between the data and a scan over the model with the nuisance parameters in place. The best loss is chosen. In the case of linking, the minimal loss over the path is chosen. The full algorithm is discussed elsewhere. the residuals of this using no fitting are shown in Fig. 7.9.

7.9 Conclusions

In conclusion, the pseudocode for the method was outlined explicitly. The methodology was tested on simulation data with a high degree of accuracy. The method shows an accuracy of ± 1 nm with a forward bias of 1 nm.

Chapter 8

Experimental Results

To validate the iSCAT imaging technique and computational methods described in previous sections, three major groups of experiments were conducted. One group focused on performing validation of confined diffusion of electrostatic motion of polystyrene beads. The second was investigating the transient mean-squared displacement behavior. The third was done using lipid-nanoparticles in a PEG-passivated environment to observe diffusion in biologically-relevant conditions.

The effects of electrostatic interaction of nanoparticle diffusion were investigated in various ionic concentrations. The salt concentration regulates the particle's effective confinement length*. The change in confinement length directly affects how size-inferencing is performed due to the change in particle mobility near confinement boundaries [27].

The transient behavior and confined mean-squared displacement relations were performed on polystyrene bead diffusion by adding a glycerol solution to slow down the step-length of the particle's motion. Glycerol was chosen as the viscous reagent as the water-glycerol relation is well defined [106] and has a lipid-membrane permeability [107, 108].

To apply the theoretical models and statistical tools to a current, applicable biological system, the diffusion of lipid-nanoparticles was investigated within glass flow-cells. The effects of flow-cell poly-ethylene glycol passivation were measured. LNP size measurement techniques were compared and contrasted. LNP transient behavior was measured using glycerol. The complete list of samples used, as well as number of video and trajectories obtained from the data, is listed in Tab. 8.1.

*As shown in Chap. 6.

Sample	Buffer	Gap Length	N_{vid}	N_{traj}
Polystyrene Beads	PBS 0.1%	225 nm	10	8
Polystyrene Beads	45% Glycerol, PBS 0.1%	225 nm	3	7
Polystyrene Beads	PBS 0.01%	225 nm	10	53
Polystyrene Beads	PBS 0.001%	225 nm	6	13
Polystyrene Beads	PBS 0.00001%	225 nm	10	62
LNP - Liposomal	1× PBS	225 nm	10	low-signal
LNP - Liposomal	75% Glycerol, 1× PBS	225 nm	10	low-signal
LNP - Onpattro - Standard - 25 mM NaOAc	1× PBS	225 nm	10	14
LNP - Onpattro - Standard - 25 mM NaOAc	75% Glycerol, 1× PBS	225 nm	10	3
LNP - Onpattro - Bleb-like - 300 mM NaOAc	1× PBS	225 nm	10	7
LNP - Onpattro - Bleb-like - 300 mM NaOAc	75% Glycerol, 1× PBS	225 nm	10	2

Table 8.1: List of samples used in experiments for this thesis.

8.1 Experimental Setup

The experiments described in this chapter were conducted using the wide-field iSCAT optical system as described in Chap. 5. The samples were placed within two planar glass coverslips to create a flow-cell. The confinement system, convex-lens induced confinement, was used to control the height between the two coverslips. The experimental schematics and apparatuses are described below.

8.1.1 Optical System Schematic

The schematic of the optical system used in the laboratory experimentation is shown in Fig. 8.1. To illuminate the samples, an illumination laser with wavelength of 405 nm was used. This wavelength laser was selected to have the lowest cost and provide illumination within the visible light spectrum. The optical power of the laser was 240 mW. The optical filter drops after the spatial filter to 120 mW which is ∼50% of the initial laser output.

The microscope objective was oil based with a 100× optical magnification[†]. The images were captured by a sCMOS camera[‡]. The system was controlled by a custom computer software written in the LabView programming language. The complete parts list of the optical system was presented in Tab. 5.1.

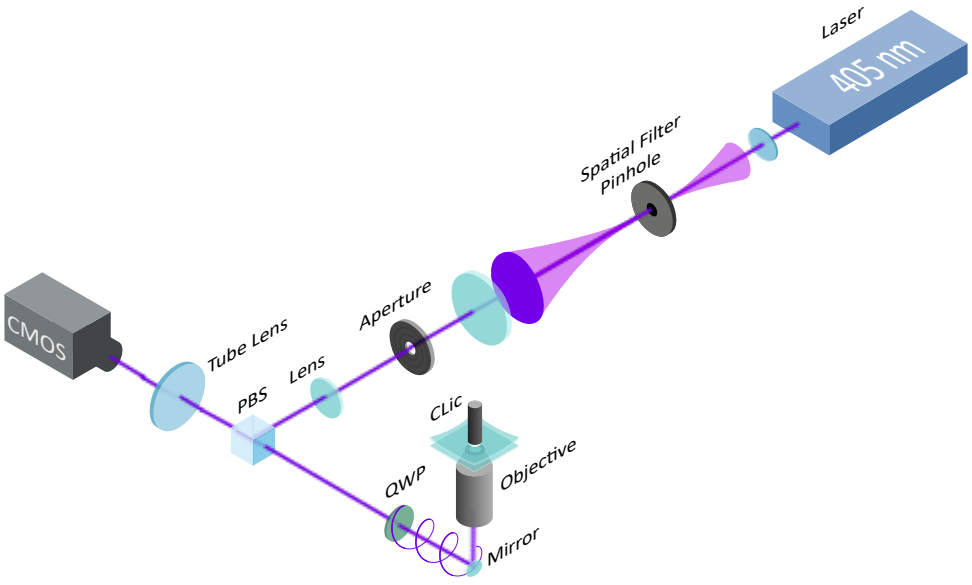

Figure 8.1: Schematic of iSCAT system from Chap. 5. See 5.1 for details.

[†](Nikon, Apochromat TIRF 100× oil immersion objective N.A. 1.49, W.D. 0.12 mm, F.O.V 22 mm)
[‡](PCO.EDGE 4.2)

8.1.2 Glass Flow-cell Preparation

The sample flow-cell device consisted of two glass borosilicate substrates [§], separated by a 30 µm thick, double-sided adhesive spacer [¶] as shown in Fig. 8.2. The bottom coverslip has no etched features, while the top coverslip contains two 1 mm holes at opposite corners for fluidic injection through the chip.

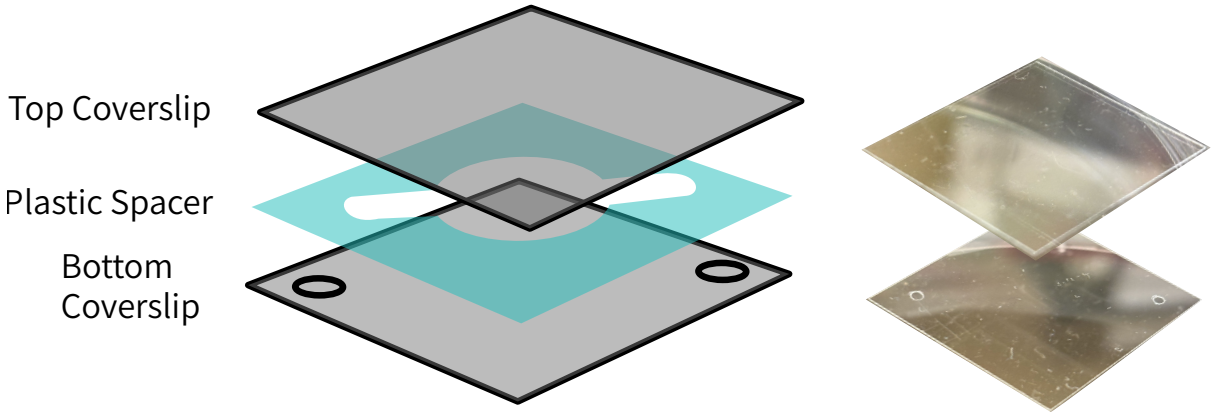

Figure 8.2: Flow cell schematic. Left, diagram of flow-cell with sticker. Right, top and bottom coverslips of the flow cell separated.

The flow-cell is held in place by a metal plate holder and an acrylic chuck. The chuck had a printed shape which contained two fluidic inlets, screw holes that mounted overtop the metal chuck, and a central hole for the viewing. The metal chuck has a groove for the coverslip, as well as tapped holes for the mounted plastic chuck. The flow-cell is assembled using two M5 screws. The assembled system is shown in Fig. 8.3.

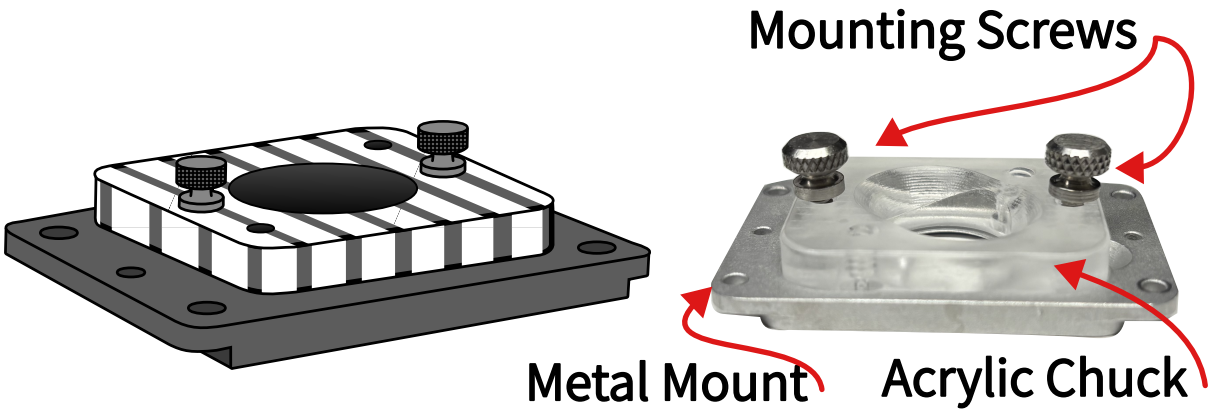

Figure 8.3: flow-cell schematic and assembled flow-cell photograph.

To prepare the glass coverslips, the cover-glass was first cleaned of organic matter with piranha solution. The solution consisted of a 7 : 3 sulfuric acid [‖], hydrogen peroxide [**]

[§](Ted-Pella, product no.260372, 145 ± 15 µm thick, 25 × 25 mm cover glass)
[¶](Nitto-Denko, product no. 5603)
[‖](H2SO4, CAS: 7664-93-9, Thermo Fisher Scientific)
[**](H2O2, CAS: 7722-84-1, Thermo Fisher Scientific)

mixture. Prior to piranha solution exposure, the cover-glass was exposed to toluene to rinse off excess organic compounds. An isopropyl alcohol soak was used to remove excess toluene film. Alternatively, an acetone soak followed by a Hellmanex detergent could be used in the place of toluene. Before exposure to piranha, the glass was washed with de-ionized water three consecutive times, or until the solution was without bubbles from the detergent. The cover-glass was soaked in the piranha for 1 hour. For experiments where PEG passivation was not performed, the cover-glass was assembled within 12 hours of piranha cleaning and using in imaging.

The passivation technique used for lipid-nanoparticle measurements was the 5 kDa PEG passivation as per [33]. After the piranha cleaning, the cover-glass was etched in 1M KOH for 30 min at room temperature. The cover-glass was placed into a desiccator together with an APTES[††] solution. The vacuum desiccator was set to 1.5 Torr for 30 min at room temperature. The cover-glass was removed and placed on a hotplate set to 120 °C. The cover-glass was then exposed to succinimidyl-valerate (SVA) functionalized methoxy-PEG-valeric acid [‡‡] in a 0.9 M sodium sulfate [§§] solution for 30 min. Before imaging, the coverslips were thoroughly rinsed with deionized (DI) water and dried with high-purity nitrogen gas. Pegylated cover-glass was assembled and stored at −80 °C for up to 2 months.

8.1.3 Convex-lens Induced Confinement

To confine the nanoparticles within the field of view of the objective, a convex-lens induced confinement, referred to as CLiC, was used. This method was developed by the Leslie Lab, described in detail in the seminal publication referenced as per [66].

A convex pusher lens is used together with a piezo linear actuator to displace the top coverslip of the flow-cell in a parabolic displacement. The piezo actuator is manually controlled during imaging and set to a specific, controlled height. The height is determined using the inference fringes of Newton's rings. The full method outline is visualized in Fig. 8.4.

8.1.4 Sample Preparation Steps

To obtain experimental results, samples must be prepared for imaging. In the electrostatic and transient diffusion experiments, carboxylate modified polystyrene beads were used. They were obtained from Thermofisher company[¶¶]. The polystyrene beads had a bulk-

[††]((3-aminopropyl)-triethoxysilane, CAS: 919-30-2, ThermoFisher Scientific)
[‡‡](5K mPEGSVA, Laysan Bio)
[§§](Na2SO4, CAS: 7757-82-6, Thermo Fisher Scientific)
[¶¶](0.1 µm FluoSpheres™carboxylate-modified microspheres, Catalog No. F8799)

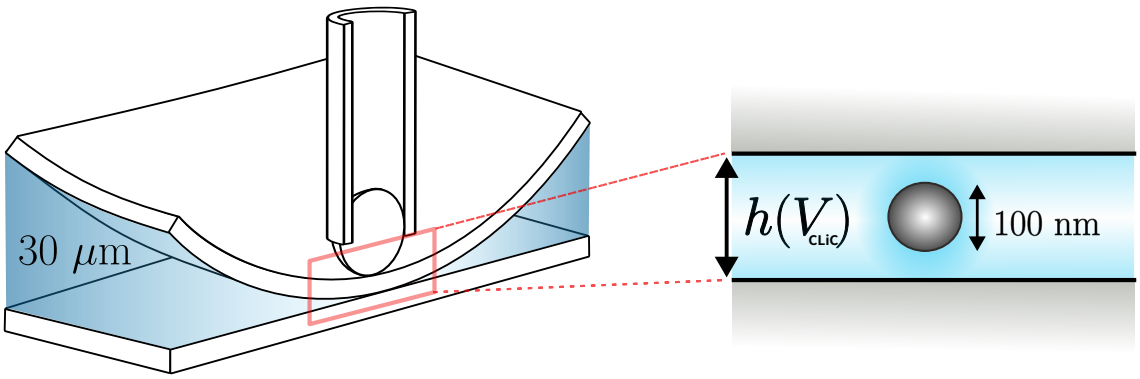

Figure 8.4: Convex-lens induced confinement schematic. Two glass coverslips, separated by a 30 µm spacer are confined with a convex-lens pusher. The height between the coverslips is voltage-regulated to nanometer precision. Nanoparticle of radius 50 nm pictured for reference.

average size estimate of 0.1 µm. The polystyrene beads were fluorescently labelled of 500 µm. The beads were measured in de-ionized water*** as well as a concentration series of phosphate-buffered saline. Fluorescent labelled was not used during the experiments.

The polystyrene bead bulk-size was experimentally verified using dynamic light scattering measurements using the Malvern DLS†††. The bulk size and zeta potential were characterized, as shown in Fig. 8.5 and Tab. 8.2.

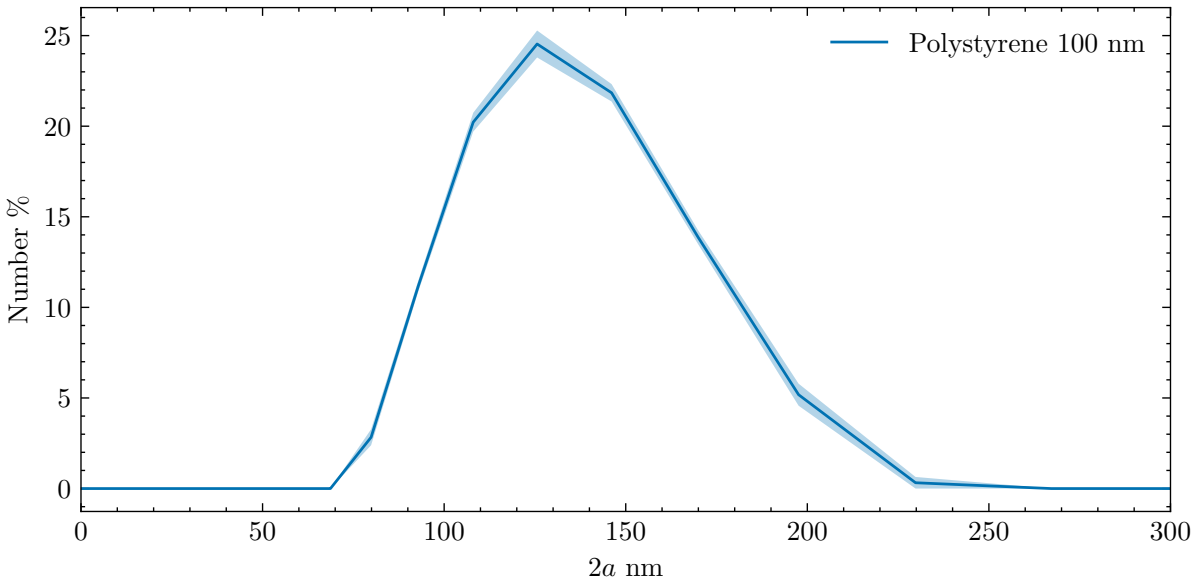

Figure 8.5: DLS results for Polystyrene beads in MilliQ water.

The figure shows the bulk-average size to be around $2a = 120\,\text{nm}$ with a log-normal distribution.

***(MilliQ)

†††(Malvern Panalytical Ltd., Malvern, UK; $\lambda = 633\,\text{nm}$, $\theta = 173°$ backscatter)

Sample	$\langle\zeta\rangle$ (mV)	σ_ζ (mV)	RSD (%)	ζ_{min} (mV)	ζ_{max} (mV)
DLS Standard -40	-42.695	0.000	0.00	-42.695	-42.695
Polystyrene Beads	-36.863	2.065	5.602	-39.246	-35.612

Table 8.2: Electrophoretic zeta-potential statistics from three consecutive measurements per sample. The average zeta-potential $\langle\zeta\rangle$, standard deviation σ_ζ, relative standard deviation RSD, minimum ζ_{min}, and maximum values ζ_{max} are displayed for each sample.

For the lipid nanoparticle motion experiments, the lipid-nanoparticles were formulated by the Cullis lab as referenced in [51, 109]. The lipid nanoparticle formulations used were of the liposomal type, Onpattro oil-core type, and the Onpattro bleb-like type. The liposomal formulation used contained 20% norMC3 lipids. The Onpattro formulation consisted of 50% norMC3 lipids. The standard oil-core was dialyzed with 25 mM NaOAc. The bleb-like was dialyzed with 300 mM NaOAc.

8.2 Electrostatics Effects on Polystyrene Bead Diffusion in PBS

The effect of electrostatic interaction on particle diffusion was investigated. The experiments performed and presented in this section are shown in Tab. 8.3.

Sample	Buffer	Gap Length	N_{vid}	N_{traj}
Polystyrene Beads $a = 50\,\text{nm}$	PBS 0.1%	225 nm	10	8
Polystyrene Beads $a = 50\,\text{nm}$	PBS 0.01%	225 nm	10	53
Polystyrene Beads $a = 50\,\text{nm}$	PBS 0.001%	225 nm	6	13
Polystyrene Beads $a = 50\,\text{nm}$	PBS 0.00001%	225 nm	10	62

Table 8.3: List of experiments.

For each experiment, a video was obtained and analyzed to get the three-dimensional trajectories using Python software, described in 7. The videos are stored in binary format and had around $\sim 30\,\text{GB}$ each. The output from each video was a collection of trajectories of particle diffusion. From the data, the three-dimensional trajectories were reconstructed, as shown in the density plot in Fig. 8.6.

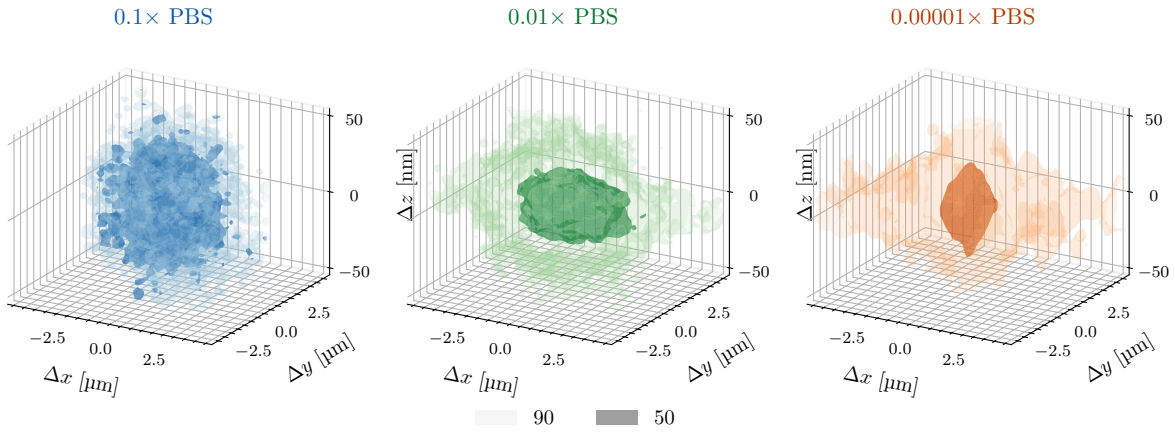

Figure 8.6: Spatial density of the average trajectory of particles.

The figure shows that for high concentration of PBS, the trajectory is contained in the largest space, which indicates large transverse movement. As the concentration decreases, the trajectory is confined into a smaller space.

These effects were captured by the spatial probability density for the three different concentrations of PBS. The positional probability density of particle motion is shown in Fig. 8.7.

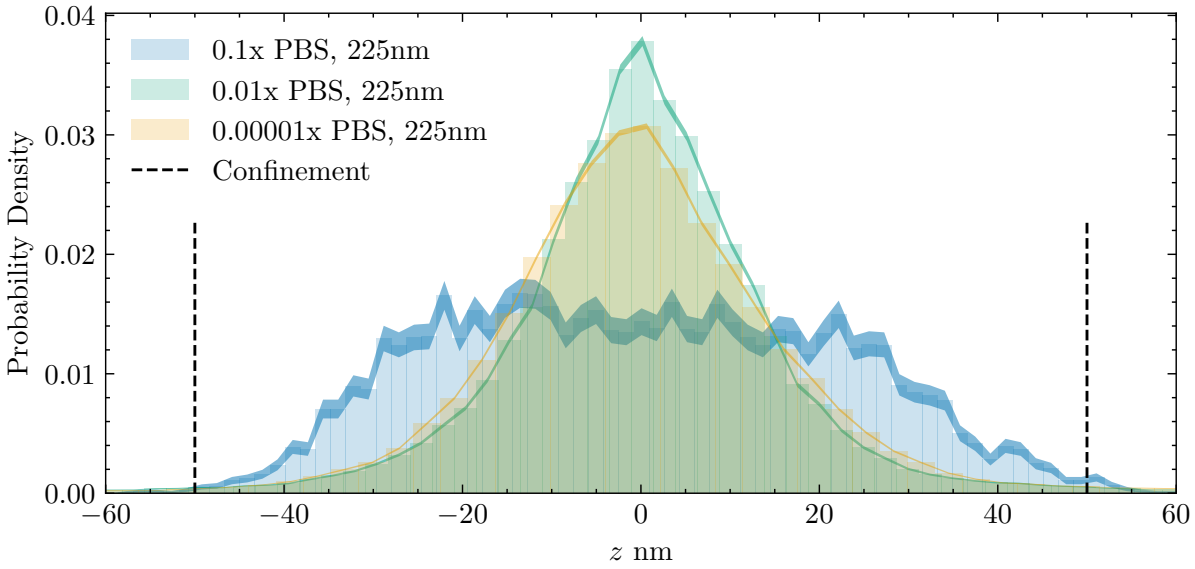

Figure 8.7: Probability density for three different concentrations of PBS. The blue line has a concentration of 0.1% PBS, green has a concentration of 0.01% PBS, and orange has a concentration of 0.00001% PBS. The horizontal axis is the z-axis position of the particle. The vertical axis is the frequency of position measurement.

The probability density of 0.01× and 0.00001× appear to follow same distribution. The difference between peaks is a normalization artifact. The probability density of 0.1× does not follow the same distribution, but rather has a flat-top profile. The electrostatic effects in 0.01× and 0.00001× are highly present, where the particle does not explore the full volume of the well. This is in contrast to 0.1× where it appears the particle explores the full volume. This experiment qualitatively agrees with the simulation results performed previously presented in Fig. **??**.

The individual FWHM of the particle movement probability density is compared to the transverse size measurements as shown in Fig. 8.8.

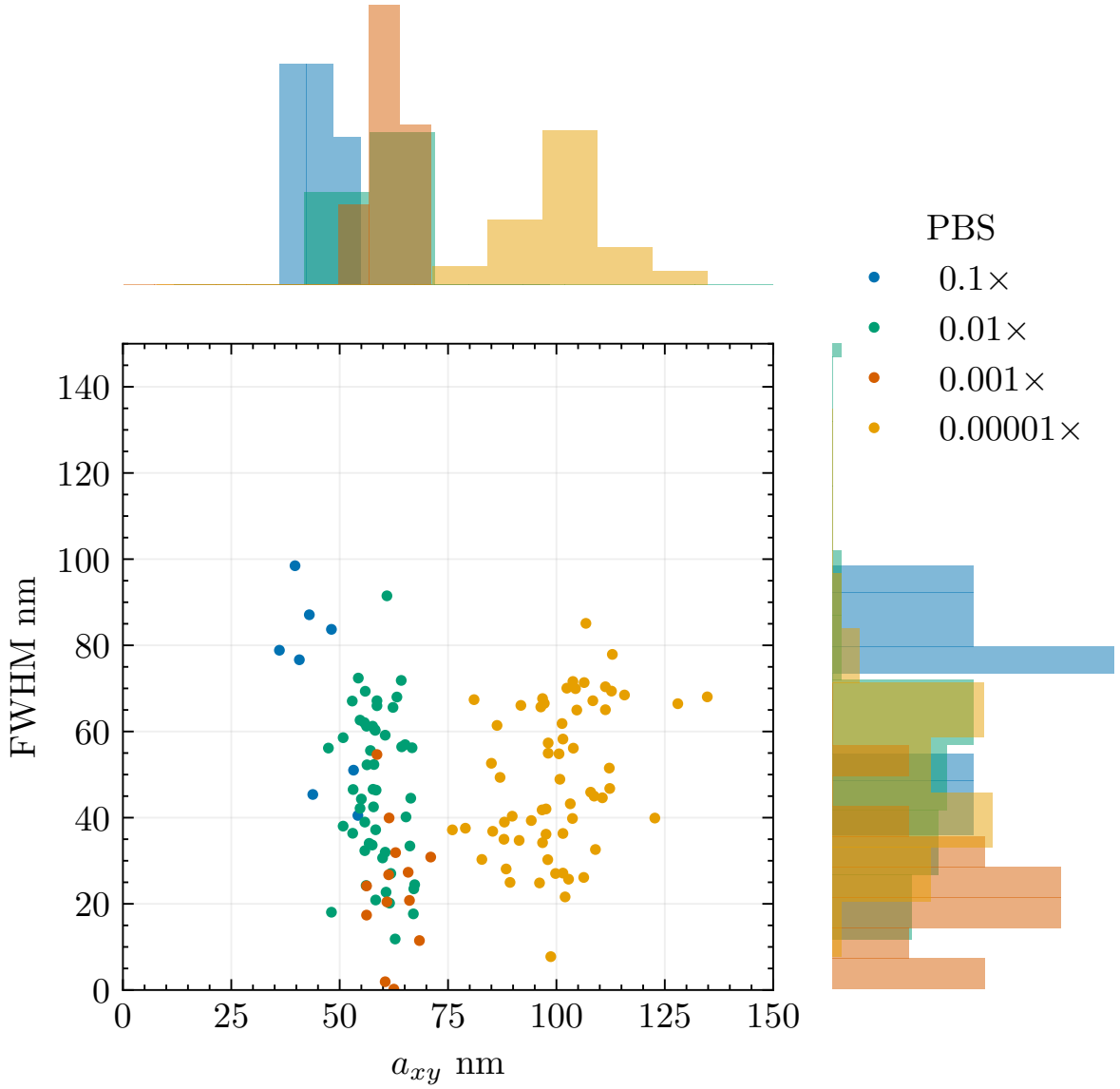

Figure 8.8: Relationship between the FWHM and a_{xy}, where FWHM is the full-width half maximum of the positional histogram, and a_{xy} is the particle radius.

The figure shows the relationship between the width of the probability distribution, quantized by the full-width half maximum, and the radius measured by the transverse trajectory for four different PBS concentrations. It can be seen that for concentrations larger than 0.00001× the particle radius distribution appears to be centered just over 50 nm. For the particles in the 0.00001× concentration, the size estimate appears to be double the expected size. The FWHM appears to be larger in the 0.1× case than compared to the lower concentrations.

The transverse diffusion measurements were constructed using the MSD of the first 15 points. The confinement coefficient was determined using the self-consistency relation.

Since the particle is a sphere and is contained between a known height, the radius of the

particle can be extracted from the axial MSD plateau of the trajectory. This size estimate, labelled as a_z was compared to the size estimate from transverse diffusion estimation. This relationship is shown in Fig. 8.9.

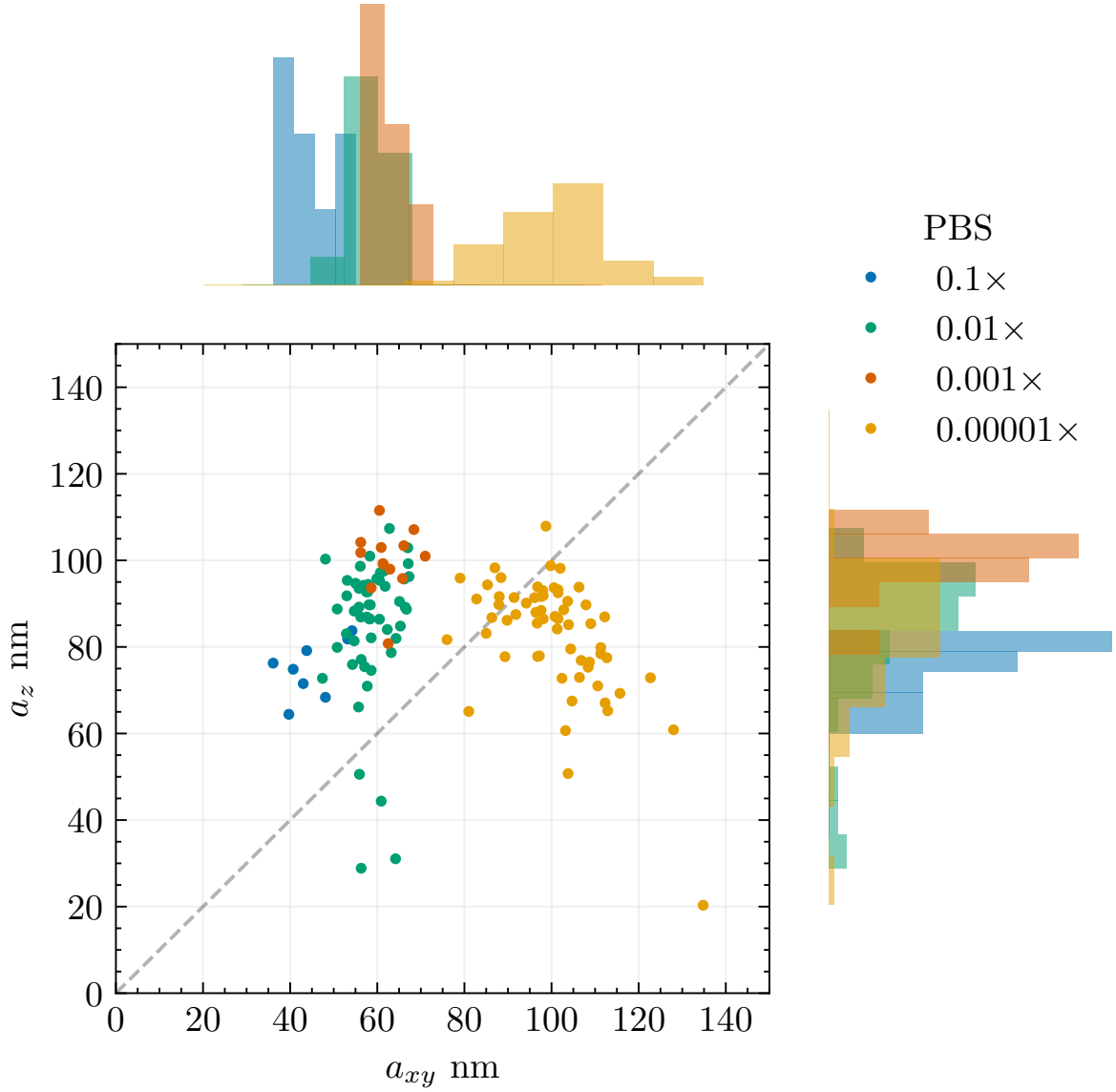

Figure 8.9: Plotted is the radius estimated by axial exploration compared to inferred transverse size of particles for various ionic concentrations.

The dotted line shows the $a_z = a_{xy}$ plane, where both methods obtain measure the same particle radius. For all concentrations, the axial method measures a consistent size estimate. As shown in the previous figure, the transverse estimation overestimates the size of the 0.00001× concentration.

For the case of 0.00001× PBS, the surface charge of the glass was measured using the relation derived in 2. The zeta-potential used in surface charge inference was measured by the zeta-sizer shown in Tab. 8.2. The surface charge of the walls is estimated by Eqn. 6.12 and the distribution for 62 particles traces is shown in Fig. 8.10.

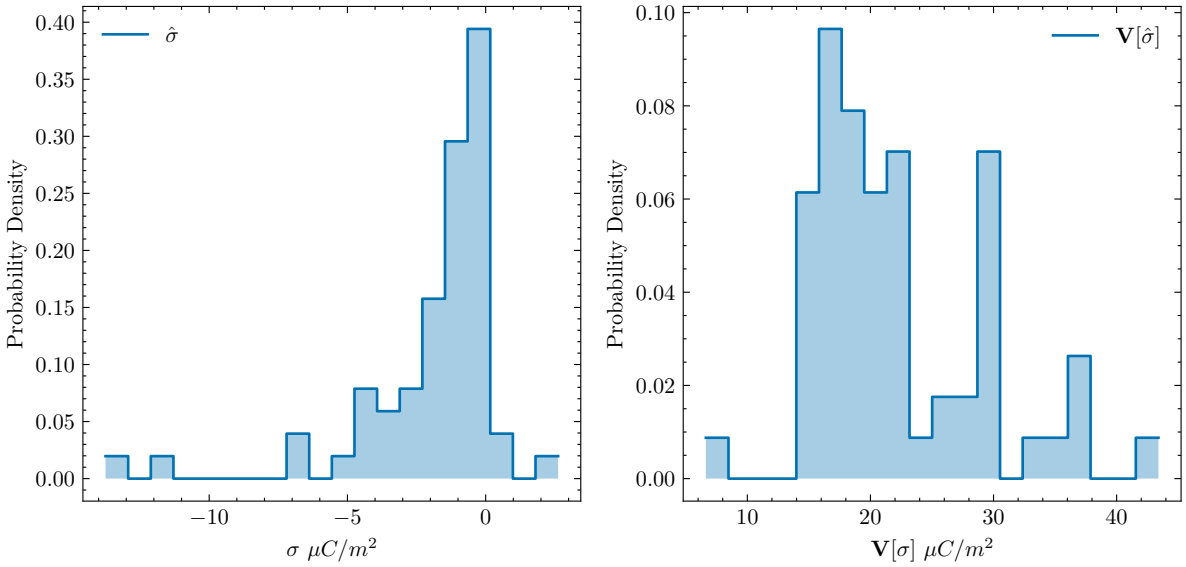

Figure 8.10: Surface charge estimation via MLE of 100 nm polystyrene beads in 0.00001% PBS solution. Imaging conducted at 400 fps.

The left plot shows the distribution of charge densities obtained from the MLE. The right plot shows the distribution of the variances as calculated by the MLE. The surface charge appears to be greatly underestimated with a mean around $-2\,\mu C/m^2$. The literature presents the surface charge for piranha-cleaned glass at around $-2\,C/m^2$ [110].

8.3 Polystyrene Transient Diffusion

To measure the transient diffusion in laboratory experiments, the experimental results for the samples listed in the table below were performed.

Sample	Buffer	Gap Length	N_{vid}	N_{traj}
Polystyrene Beads	45% Glycerol, PBS 1×	225 nm	3	7

Table 8.4: List of samples.

To obtain transient diffusion trajectories, the beads were suspended in 45% glycerol and 1× phosphate-buffered saline solutions. Three videos, of length 10000 frames each, were obtained according to the method described in 8.1.

The glycerol solution was adjusted to ensure that the diffusion coefficient was around 10× smaller than the previous experiment. The mean step length was $\sim 10\,\text{nm}$. the concentration of the solution was obtained by the relationship presented in Fig. 8.11.

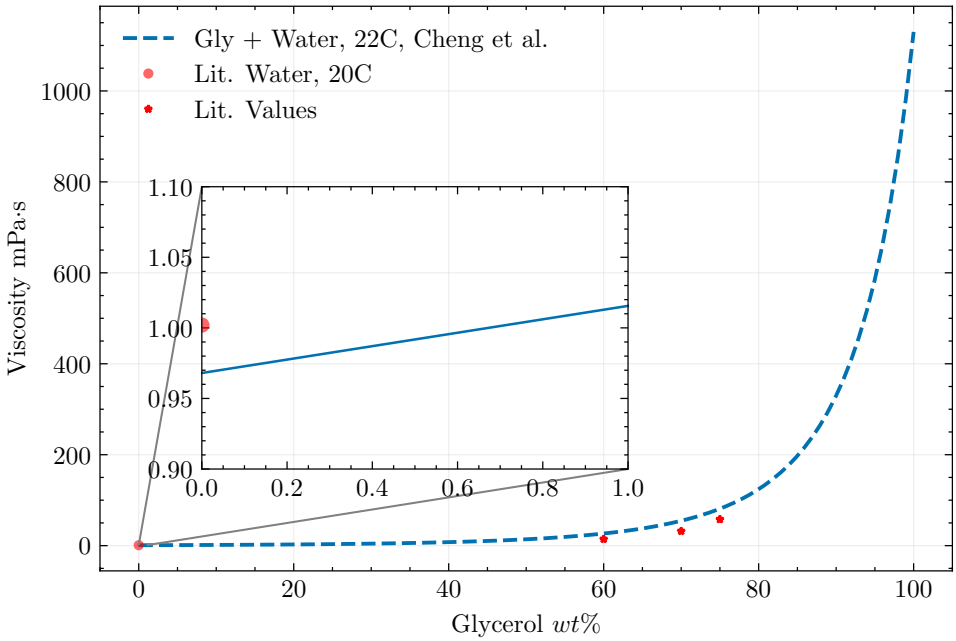

Figure 8.11: Glycerol-water curve. The theoretical line plotted is used from Cheng et al. [106].

The dotted blue line presents the empirically determined equation for the viscosity of a glycerol-water solution obtained from Ref. [106]. The red dots are the literature values for the viscosities. The inset frame shows the magnified region near 0, where the approximation is very close.

The experiment was performed with the parameters listed in Tab. 8.4 and the video was captured. From the video, a long-range trajectory was obtained. This trajectory is shown in Fig. 8.12

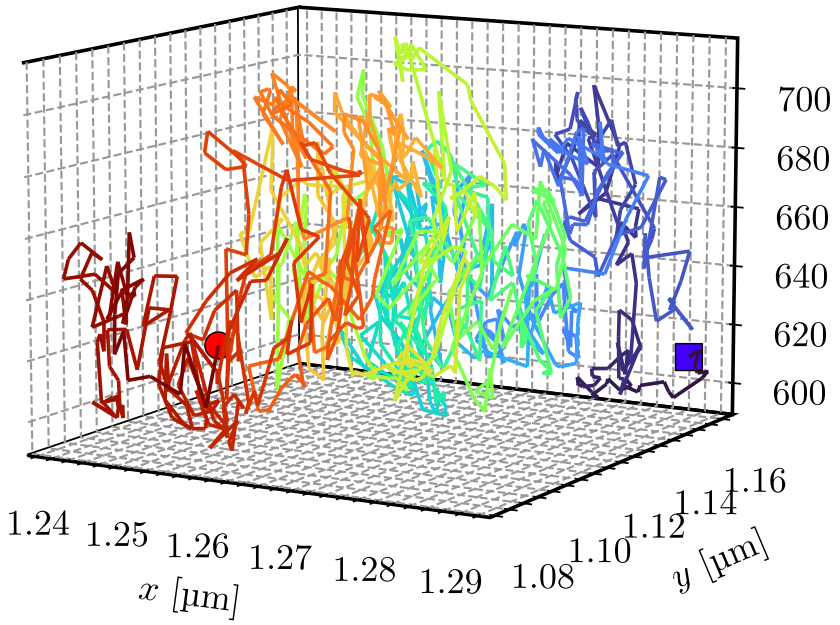

Figure 8.12: Trajectory of a polystrene bead in glycerol. The red dot and blue square show the initial and final position respectfully.

From this trajectory, the mean-squared displacement of the transverse and axial motion is plotted in Fig. 8.13.

The figure on the left shows the transverse mean squared displacement as a function of the lag-time between frames. The dotted line is the ensemble mean MSD between the 7 trajectories, while the filled area is the standard deviation. The figure on the right shows the axial mean squared displacement as a function of the lag-time between frames. The dotted green line shows the ensemble mean, while the filled green shows the standard deviation. The horizontal red lines show the theoretical plateau values for confined diffusion of various radii, a, particles between a 220 nm slit. The MSD of both trajectories appear within a narrow range of each other.

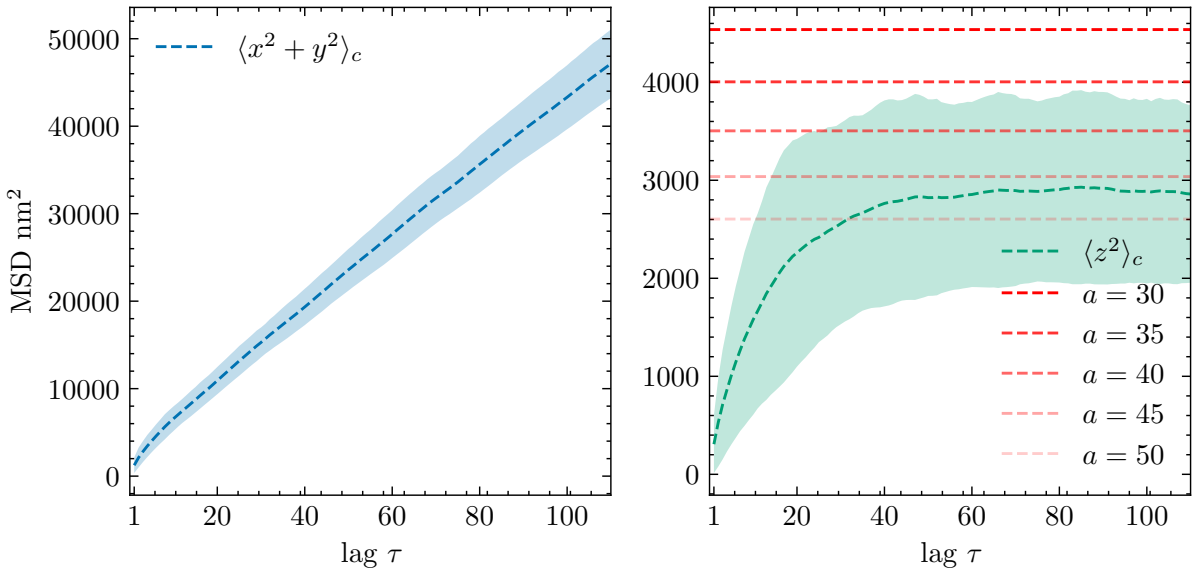

Figure 8.13: Mean-squared displacement of glycerol-slowed polystyrene beads. Axial MSD is clearly convergent to the confinement length. The confinement length is fixed at 225 nm and the particle's estimated size is $a = 50\,\text{nm}$. The lag time, τ is plotted in frame indices. Each unit of τ represents one frame at 580 fps.

In addition to these curves, the diffusivity across the confinement channel can be recovered by partitioning the trajectory into axially separated slices. The trajectory was converted into diffusivity values utilizing the first two points of the MSD and was plotted as a function of the axial position, as shown in Fig. 8.14.

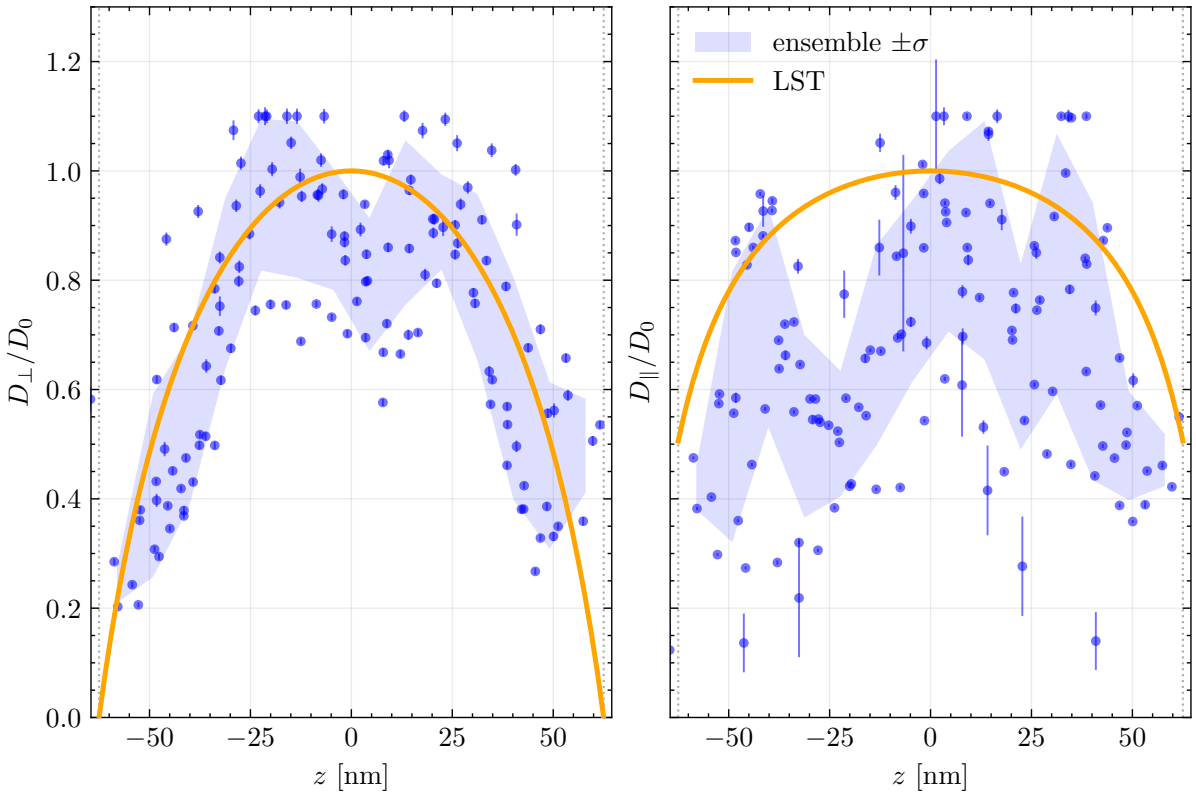

Figure 8.14: Theoretical diffusivity shown in orange using the linear-superposition theory (LST). Measured diffusivity of 7 particle traces are plotted in blue. Each point is a portion of a single trajectory. The ensemble distribution is plotted in light blue.

The left plot shows the theoretical diffusivity curve for axial movement in a 220 nm slit plotted in solid orange. The measured axial diffusivity values are shown with error bars in solid blue. The ensemble standard deviation is shown in light blue. The right plot presents the theoretical diffusivity curve for transverse movement in a 220 nm slit plotted in solid orange. The measured axial diffusivity values are shown with error bars in solid blue. The ensemble standard deviation is shown in light blue.

The left hand side plot follows the theoretical expected value, while the data on the right hand side is not. This is because of an increased uncertainty obtained from the video trajectories.

8.4 LNP Axial Diffusion

The lab experiments to characterize the diffusion of lipid nanoparticles were conducted for three LNP species to evaluate the electrostatic and transient motion. The electrostatic motion was obtained using a setup as described previously with the exception of utilizing a PEG-passivation. The test conditions are listed in the Tab. 8.5.

Sample	Buffer	Gap Length	N_{vid}	N_{traj}
LNP - Liposomal	1× PBS	225 nm	10	low-signal
LNP - Liposomal	75% Glycerol, 1× PBS	225 nm	10	low-signal
LNP - Onpattro - Standard - 25 mM NaOAc	1× PBS	225 nm	10	14
LNP - Onpattro - Standard - 25 mM NaOAc	75% Glycerol, 1× PBS	225 nm	10	3
LNP - Onpattro - Bleb-like - 300 mM NaOAc	1× PBS	225 nm	10	7
LNP - Onpattro - Bleb-like - 300 mM NaOAc	75% Glycerol, 1× PBS	225 nm	10	2

Table 8.5: List of LNP species

For each LNP species, the bulk-average size and zeta-potential were characterized using the Malvern DLS and zeta-sizer. The experimental size distribution is plotted in Fig. 8.15. The experimental zeta-potential measurements are summarized in Tab. 8.6.

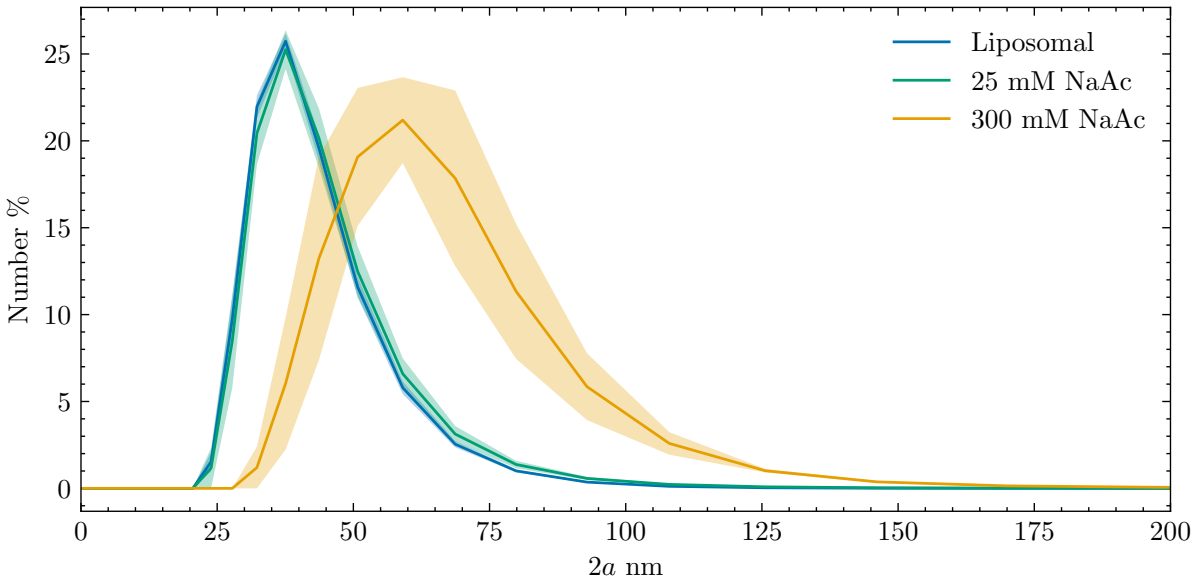

Figure 8.15: DLS measurements of LNP samples. The blue, green and orange plot denote the liposomal, 25mM NaAC, and 300mM NaAC LNP formulations respectfully.

The liposomal and oil-core LNPs, shown in the blue and green solid lines, have similar

bulk-average size distributions centered around 40 nm with a log-normal distribution. The bleb-like LNP, shown in the orange solid line, is centered around 60 nm with a log-normal distribution. All three are below the size of 125 nm.

Sample	$\langle\zeta\rangle$ (mV)	σ_ζ (mV)	RSD (%)	$\zeta_{\min}$ (mV)	$\zeta_{\max}$ (mV)
DLS Standard -40	−42.695	0.000	0.00	−42.695	−42.695
Liposomal LNP	−7.739	0.089	1.153	−7.816	−7.642
25 mM NaAc Onpattro	−6.425	0.466	7.25	−6.906	−5.976
300 mM NaAc Onpattro	−6.569	1.414	21.52	−7.423	−4.938

Table 8.6: Electrophoretic zeta-potential statistics from three consecutive measurements per sample. The average zeta-potential $\langle\zeta\rangle$, standard deviation σ_ζ, relative standard deviation RSD, minimum $\zeta_{\min}$, and maximum values $\zeta_{\max}$ are displayed for each sample.

The zeta-potentials of all three LNP species have a similar mean zeta-potential around −7 mV. The size and the negative charge produce results that are consistent with movement within the full confinement.

The experimental data consisted of videos of LNPs diffusing in a 220 nm slit. The particle positions were measured using the iSCAT optical system and the trajectories were calculated from the video data. The spatial probability density histogram is shown in Fig. 8.16.

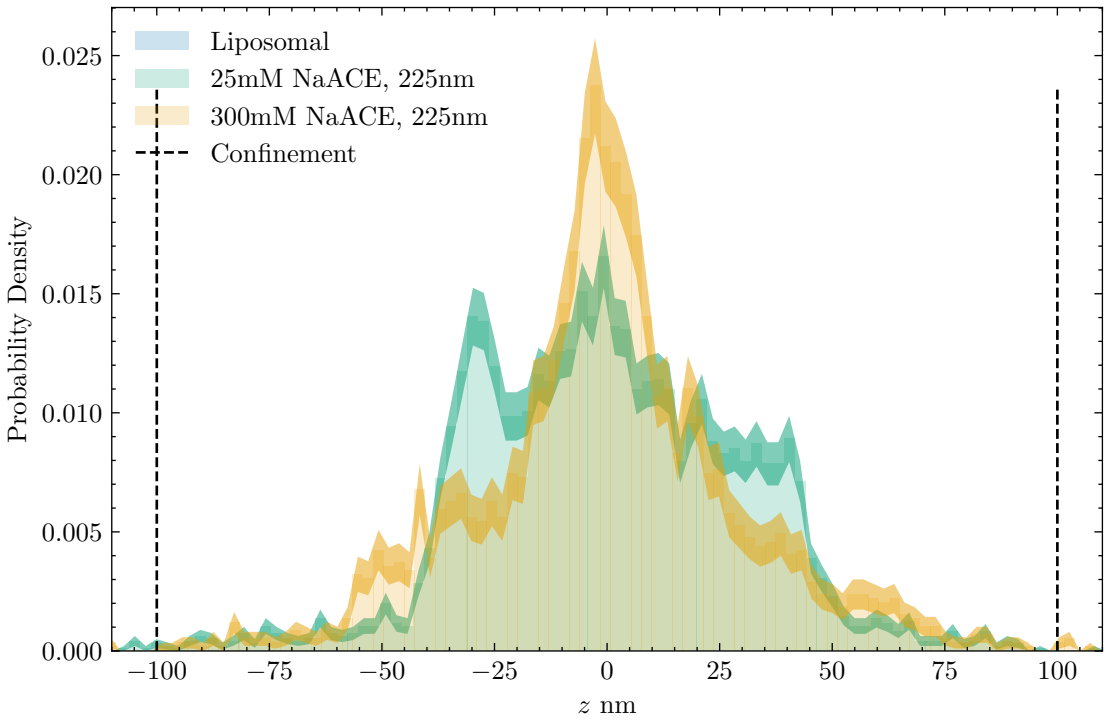

Figure 8.16: Probability histogram of LNP axial position. Plotted in green and orange are the 25mM NaAC, and 300mM NaAC LNP formulations respectfully. The liposomal, LNPs were too small to be tracked over a long-range and are hence omitted.

The liposomal LNPs were not consistently detectable to allow for 2D tracking. The low detectability of the data may be related to the smaller size and index of refraction being closely related to that of the solution. The oil-core and bleb-like LNPs had enough detectability to calculate trajectory information. The probability density of the oil-core resembles a flat-top distribution indicating motion across the whole channel. The bleb-like lNPs appear to have a flat-top with a secondary central peak in the distribution indicating motion across the channel with a potential secondary interaction.

To characterize the transient behavior, the LNPs were suspended in a 75% glycerol solution and the video from the optical system was obtained. The three samples were imaged and their ensemble MSD was measured. This is shown in Fig. 8.17.

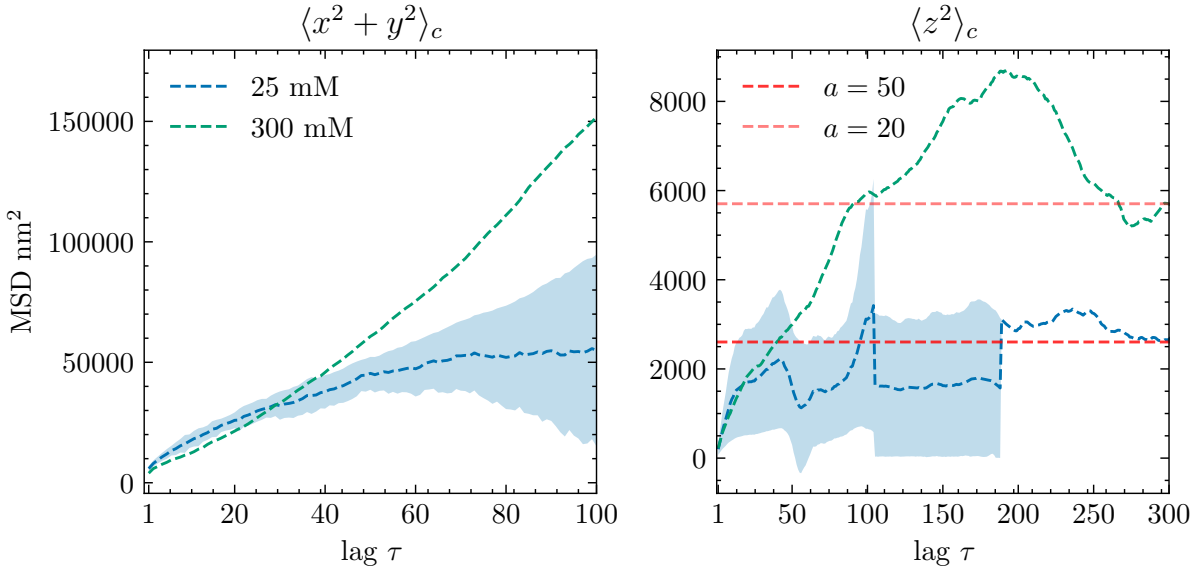

Figure 8.17: Ensemble MSD of LNP formalations. Plotted in blue and green are the 25mM NaAC, and 300mM NaAC LNP formulations respectfully. The lag time, τ is plotted in frame indices. Each unit of τ represents one frame at 400 fps. The red line in the right-plot display the expected MSD plateaus for particles of radius $20, 50$nm in order of increasing boldness. The 25mM NaAC traces contain ensemble information of 3 traces. The 300mM NaAC traces contain ensemble information of 2 traces.

The left plot depicts the transverse MSD of the oil-core and bleb-like LNPs. The dotted blue line shows the oil-core LNP transverse diffusion. The solid light blue indicates the standard deviation. The dotted green line shows the bleb-like LNP transverse diffusion. The solid light green indicates the standard deviation. The right plot depicts the axial MSD of the oil-core and bleb-like LNPs. The dotted blue line shows the oil-core LNP axial diffusion. The solid light blue indicates the standard deviation. The dotted green line shows the bleb-like LNP axial diffusion. The solid light green indicates the standard deviation. The dotted red lines show the theoretical plateau values for particles of radius, a, in a 220 nm slit. The transverse MSD is acceptable as it has a linear form near the first

20 lag-times. The axial MSD, however, contains non-linear behavior and sporadic jumps. This noise indicates non-accurate axial tracking due to poor image data.

Chapter 9

Conclusion

9.1 Conclusion

This thesis presents the current state of research in the fields of nanoparticle diffusion and single-particle tracking.

The theory of thermal motion was reviewed based on existing research publications, which provides the theoretical foundation for the topics discussed in this thesis. Particular emphasis is placed on Brownian motion, which describes the random motion of larger particles suspended in a fluid as a result of collisions with thermally moving molecules. Brownian motion forms the basis for the analysis of nanoparticle diffusion, as shown in Chapter 2.

As nanoparticles are present in nature almost everywhere, it is of interest to study their behavior. Special interest in nanoparticles existing in biological forms was formed in recent years leading to development of vaccines. Using Brownian motion, properties of biological nanoparticles can be determined as described in Chapter 3.

Interferometric scattering principles for scattering microscopy were presented in Chapter 4 It was demonstrated that scattering microscopy can be used to reconstruct three-dimensional trajectories of nanoparticles from sequences of images. The trajectories can be used to calculate various properties of nanoparticles.

An interferometric scattering (iSCAT) microscope optical system was constructed to observe diffusion of nanoparticles in real time, as described in Chapter 5 The system was used to perform nanoparticle tracking under laboratory conditions and to obtain experimental data. Three-dimensional particle trajectories were reconstructed from the iSCAT images for nanoparticles of various sizes and fluid concentrations. The statistical parameters obtained from the collected data confirmed that the system was able to achieve

the desired results and can be fundamental in further study in this field.

Mathematical description of Brownian motion was used to develop diffusion simulation of nanoparticles in confined spaces. The simulation model is a valuable tool for predicting nanoparticle behavior without the need for conducting lengthy and expensive experiments. The simulation software and results are presented in Chapter 6 The simulations can be conducted on various confinements with inclusion of external influences like electric field.

Analysis of the reconstructed trajectories in three-dimensions allows extraction of important nanoparticle physical parameters, including the mean squared displacement (MSD), diffusion coefficient, drift velocity, interaction forces, particle size, and confinement effects, as described in Chapter 7. The software developed for this purpose was able to analyze large number of videos autonomically and present results in required form. The processing time was in order of hours, which is the area of improvement. Using C++ code instead of Python should be investigated.

Chapter 8 presents the experimental results using the iSCAT imaging technique for specific nanoparticles of interests. Polystyrene beads of 100nm size were used for the characterization of the system. The data shows that the developed technique provides confirmation of the hardware and software correct operation and validate the mathematical framework implemented in the image analysis software. The experimental results demonstrated the accuracy and reliability of the proposed methodology for nanoparticle size estimation. As a final step, lipid nanoparticles were subjected to the experimental characterization. The test results showed that the technique was able to provide information about the electrostatic motion effects and transient motion of the particles. Some limitations were noticed for nanoparticle size of 40 nm as they presented a weak image signal. To improve this, the optical system should be further optimized and more data collected.

Overall, the presented results constitute good foundational method for tracking polystyrene beads and biological nanoparticles of sizes above 60 nm. The passivation methods and chemical solutions used also were validated as they provided good samples for the tests. The system is relatively inexpensive and can be replicated with relative ease.

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Appendix A

Surface Charge Estimator Derivation

This is a section for the derivation of the surface charge estimator used in the axial diffusion estimation.

A.1 $\hat{B}$ Maximum Likelihood Estimator

The Force used in the modeling is given by;

$$V'(z_n) = B\phi(z_n) = B\frac{k_b T}{\lambda_D} \cdot \exp(-H/\lambda_D) \cdot (\exp(-z_n/\lambda_D) - \exp(z_n/\lambda_D)) \tag{A.1}$$

The dynamics are governed by the equation, written in the Itô convention [111],

$$\Delta z = D'(z)\Delta t - \beta D(z)V'(z)\Delta t + \sqrt{2D(z)}W_t. \tag{A.2}$$

Re-writing in terms of drift and diffusion coefficients;

$$\Delta z = \left[D'(z) - \beta B D(z)\phi(z)\right]\Delta t + \sqrt{2D(z)}W_t. \tag{A.3}$$

This is a Markov-process that has a propagator probability from the Onsager-Machlup path integral representation [112, 113],

$$P(z_{n+1}|z) = \frac{1}{4D(z)\Delta t}\exp\left[-\frac{\left(\Delta z_n - D'(z)\Delta t + \beta B D(z)\phi(z)\Delta t\right)^2}{4D(z)\Delta t}\right]. \tag{A.4}$$

To obtain an estimator for B, take the log-likelihood and differentiate, maximizing it.

$$
\begin{aligned}
l(B) &= \sum_{n=0}^{\infty} \left[-\frac{\left(\Delta z_n - D'(z)\Delta t + \beta B D(z)\phi(z)\Delta t\right)^2}{4D(z)\Delta t} \right] + \text{const} \\
&= -\frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{2D(z)\Delta t} \left(\Delta z_n - D'(z)\Delta t + \beta B D(z)\phi(z)\Delta t\right)^2 + \text{const.}
\end{aligned}
$$

Define $r_n = \Delta z_n - D'(z)\Delta t$ and $w_n = \frac{1}{2D(z)}$. Find the maximum of the log-likelihood by taking the derivative of l with respect to B.

$$
l'(B) = -\sum_{n=0}^{\infty} \frac{w_n}{\Delta t} \left(r_n + \beta B D(z)\phi(z)\Delta t\right) \beta D(z)\phi(z)
$$

Setting this to 0 and solving for $\hat{B}$,

$$
\begin{aligned}
0 &= -\sum_{n=0}^{\infty} \frac{w_n}{\Delta t} \left(r_n + \beta B D(z)\phi(z)\Delta t\right) \beta D(z)\phi(z) \\
0 &= \sum_{n=0}^{\infty} \frac{w_n}{\Delta t} r_n D(z)\phi(z) + \sum_{n=0}^{\infty} \frac{w_n}{\Delta t} \beta B D^2(z)\phi^2(z)\Delta t \\
\frac{1}{\Delta t} \sum_{n=0}^{\infty} w_n r_n D(z)\phi(z) &= -\beta B \sum_{n=0}^{\infty} w_n D^2(z)\phi^2(z) \\
B &= \frac{\sum_{n=0}^{\infty} w_n r_n D(z)\phi(z)}{\sum_{n=0}^{\infty} w_n D^2(z)\phi^2(z)\Delta t}.
\end{aligned}
$$

Simplifying the terms, the estimator is written in the form over a sum of steps z;

$$
\hat{B} = \frac{\sum_n \phi(z) \left[\Delta z - D'(z)\Delta t\right]}{\sum_n D(z)\phi^2(z)\Delta t} \tag{A.5}
$$

A.2 $\hat{B}$ Variance

The variance of this can be found using the fisher information matrix;

$$\begin{aligned}\mathcal{I}(B) &= -l''(B) \\ &= \sum_{n=0}^{\infty} \frac{w_n}{\Delta t} \Delta t \beta^2 D^2(z) \phi^2(z) \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \beta^2 D(z) \phi^2(z)\end{aligned}$$

In the univariate case, the variance is proportional to the inverse of the fisher information, which is also proportional to the denominator of the MLE.

$$\mathbb{V}[\hat{B}] = \frac{1}{2} \sum_{n=0}^{\infty} \beta^2 D(z) \phi^2(z) \tag{A.6}$$

Appendix B

Self-Consistent Diffusion Recovery

The self-consistency relation for confined diffusion assumes that the measured quantity is the confined diffusion in a known height. The measured confined diffusion is self-consistent if the following condition is true,

$$\hat{D}_{\parallel}(\hat{a}) = \mu_{\parallel}(a^*, h) D_0^*(a^*). \tag{B.1}$$

where μ is the Faxén mobility coefficient. Rewriting this equation in terms of the radius:

$$\begin{aligned}
\hat{D}_{\parallel}(\hat{a}) &= \mu_{\parallel}(a^*, h) D_0^*(a^*), \\
\frac{k_b T}{6\pi\eta\hat{a}} &= \mu_{\parallel}(a^*, h) \frac{k_b T}{6\pi\eta a^*}, \\
\frac{1}{\hat{a}} &= \mu_{\parallel}(a^*, h) \frac{1}{a^*}.
\end{aligned}$$

This final relation must be true if $\hat{a} = a^*$. It should be noted here that μ is a monotonic function. This is shown in Fig. 2.1. The monotonicity of this allows to ensure that a minima exists under mean-squared error. However, since μ is an infinite series, it is not-analytical. Thus, one can resort to a truncation of terms and a solution via a means of numerical techniques. An example of this is shown in Fig. 2.2. This thesis utilizes the Newton-Rhapson method due to its computational availability;

$$a^* = \arg\min_{a^*} \left| \hat{a}\mu_{\parallel} - a^* \right|. \tag{B.2}$$

Appendix C

Axial Estimation Algorithm Pseudocode

This section of the appendix lists all detailed algorithms and computational methods presented in the axial-estimation section of this project.

The following algorithms are presented:

Function	Desc.
Raw2iScat:	Camera image to iSCAT contrast stack.
TransverseLocalization:	(x, y) localization function with sub-pixel resolution.
RadialIntegral:	Radial integral function.
Nuisance-parameter:	Search algorithm with a grid-scan and raw-python implemented least-squares for Model.
Axial Chain-Link:	Linking algorithm for axial trajectories. Analogous to Munkre's algorithm in 2D tracking.

Table C.1: List of functions presented in appendix. Function names and descriptions are listed.

Each algorithm is presented in language-general mathematical pseudocode. Verbal explanations are described in chapter 7. The appendix serves as a more-detailed explanation so that the interested reader may implement the software themselves and have exact computational details.

A Python implementation, written by the author and open-source contributions, can be produced upon request.

Algorithm 1 Raw image stack → iSCAT contrast stack

Require: $I_{\text{raw}} \in \mathbb{R}^{F\times H\times W}$ ▷ F frames of $H \times W$ pixels
Ensure: $I_{\text{iSCAT}} \in \mathbb{R}^{F\times H\times W}$

Parameters

$R_1 \leftarrow \max(H,W)/100, \quad R_2 \leftarrow \max(H,W)/4$ ▷ band-pass radii in k-space [px]
$\Delta f \leftarrow 25, \quad \Delta L \leftarrow 15$ ▷ temporal gap; background half-window, empirical for data conditions.
$N \leftarrow$ border width ▷ pixels discarded at the edges

Pre-computation (frame independent)

$(k_x, k_y) \leftarrow \textsc{Meshgrid}\big(\big[-\frac{H}{2}, \frac{H}{2}\big), \big[-\frac{W}{2}, \frac{W}{2}\big)\big)$ ▷ centred spatial frequencies
$h \leftarrow \mathbf{1}\big[\, R_1^2 < k_x^2 + k_y^2 < R_2^2 \,\big]$ ▷ annular band-pass filter
$M_{pq} \leftarrow \mathbf{1}[\, N \le p < H - N \ \wedge \ N \le q < W - N \,]$ ▷ edge mask

function $\textsc{AmpNorm}(I, M)$ ▷ row-wise amplitude normalisation of one frame. Row-wise means to avoid readout noise from the camera. Mean over rows to remove line artifacts. (From Vahid's Method)
    $I^{(1)} \leftarrow I \odot M$ ▷ $\odot$: element-wise (Hadamard) product
    $I^{(2)} \leftarrow I^{(1)} / \langle I^{(1)} \rangle$ ▷ $\langle\cdot\rangle$: mean over all pixels
    for each row $p \in \{0, \dots, H-1\}$ **do**
        $a_p \leftarrow \langle M_{p,:} \rangle / \langle I^{(2)}_{p,:} \rangle$ ▷ row means
        $I^{(3)}_{p,:} \leftarrow I^{(2)}_{p,:} / a_p$
    end for
    return $I^{(3)}$
end function

Main loop

for $i = 0$ **to** $F - 1$ **do**
    $I_0 \leftarrow \textsc{AmpNorm}(I_{\text{raw}}[i],\, M)$ ▷ signal frame

    $\mathcal{B}_i \leftarrow \big[\, i - \Delta f - \Delta L,\ i - \Delta f \,\big] \ \cup \ \big[\, i + \Delta f,\ i + \Delta f + \Delta L \,\big]$ ▷ background frame indices
    $\mathcal{B}_i \leftarrow \mathcal{B}_i \cap [\, 0, F \,)$ ▷ clip at the ends of the stack
    $\bar{B} \leftarrow \dfrac{1}{|\mathcal{B}_i|} \displaystyle\sum_{j \in \mathcal{B}_i} \textsc{AmpNorm}(I_{\text{raw}}[j],\, M)$ ▷ mean background
    assert $\bar{B}$ contains no NaN

    $I_m \leftarrow \big(I_0 \oslash \bar{B}\big) \odot M$ ▷ $\oslash$: element-wise division
    $I_{\text{iSCAT}}[i] \leftarrow \mathcal{F}^{-1}\{h \odot \mathcal{F}\{I_m\}\}$ ▷ band-pass filtering
end for

return I_{iSCAT}

Algorithm 2 Transverse Localization algorithm

Inputs
I_{iSCAT} ▷ from Alg. 1
$I_{\text{RVT}} \leftarrow \text{RVT}[I_{\text{iSCAT}}]$

Pre-computation (frame independent)
$t \leftarrow 0.9 \cdot \max I_{\text{iSCAT}}$ ▷ threshold
$n \leftarrow 30$ ▷ number of particles per frame to assume, empirical
$x_{\text{ROI}}, y_{\text{ROI}} \leftarrow 5$ ▷ ROI cropping for radial symmetry algorithm
Fitting Routine
for $i = 0$ **to** $F - 1$ **do**
    $I_{\text{dil}} \leftarrow \textsc{ImDialate}(I_{\text{RVT}}[i])$
    $\{x, y\} \leftarrow \textsc{Where}\,[I_{\text{dil}} == I_{\text{RVT}} \wedge I_{\text{RVT}} > t]$ ▷ Candidate Positions
    $\{x_1, y_1\}|_n = \textsc{GaussianFit}(x, y)$ ▷ Non-linear least squares fitting, take best n fits
    $\{x_\star, y_\star\} = \textsc{RadialSymmetry}(x_1, y_1, x_{\text{ROI}}, y_{\text{ROI}})$ ▷ from [104]
end for
2D Trace Linking
Traces $\leftarrow \textsc{Munkres}[x_\star, y_\star]$ ▷ Minimization of distance cost-matrix
ROI $= I_{\text{iSCAT}}[Traces]$ ▷ Cropping of image around every frame for each trace.

Algorithm 3 Discrete radial profile (angular average over a wedge)

Require: $I \in \mathbb{R}^{F \times H \times W}$ ▷ stack of F image ROIs
Require: $\mathbf{c} = (c_p, c_q)$ ▷ ROI centre (may be sub-pixel)
Require: θ_0, $\Delta\theta$ ▷ start angle and angular width of the wedge
Require: R ▷ outer radial bound, in pixels
Ensure: $g \in \mathbb{R}^{F \times R}$, where

$$g_t(\rho) \;\simeq\; \frac{1}{\Delta\theta} \int_{\theta_0}^{\theta_0 + \Delta\theta} I_t(\rho, \theta)\, \mathrm{d}\theta$$

is the angular mean of frame t over the wedge, as a function of radius.

Pre-computation (frame independent)

$\rho_{pq} \leftarrow \big\|(p,q) - \mathbf{c}\big\|_2, \quad \theta_{pq} \leftarrow \operatorname{arctan2}\big(p - c_p,\; q - c_q\big) \bmod 2\pi$ ▷ polar coordinates of each pixel

for $r = 0$ **to** $R - 1$ **do** ▷ annular-wedge bins
  $\mathcal{A}_r \leftarrow \big\{\, (p,q) \;:\; r \le \rho_{pq} < r + 1 \;\wedge\; \big(\theta_{pq} - \theta_0\big) \bmod 2\pi \;<\; \Delta\theta \,\big\}$
  $N_r \leftarrow |\mathcal{A}_r|$ ▷ $N_r \approx (r + \frac{1}{2})\,\Delta\theta$
end for

Main loop

for $t = 0$ **to** $F - 1$ **do**
  for $r = 0$ **to** $R - 1$ **do**
    if $N_r = 0$ **then**
      $g_t(r) \leftarrow \textsc{NaN}$ ▷ empty bin: undefined, not zero
    else
      $g_t(r) \leftarrow \dfrac{1}{N_r} \displaystyle\sum_{(p,q) \in \mathcal{A}_r} I_t[p,q]$ ▷ angular average at radius r
    end if
  end for
end for

return g

Algorithm 4 Nuisance-parameter estimation by profiled grid search + local refinement

Require: $g \in \mathbb{R}^M$ ▷ radial profile, sampled at radii r_i, $i = 1, \dots, M$
Require: $r \in \mathbb{R}^M$, $w \in \mathbb{R}^M_{\geq 0}$ ▷ radii [nm] and fit weights
Require: $k = 2\pi n_0/\lambda$ ▷ optical wave-vector in the medium
Require: $[z_-, z_+]$, $[E_-, E_+]$, $[\gamma_-, \gamma_+]$ ▷ parameter bounds
Require: $\Delta_\lambda = 32$

iPSF Model

$G_i(z,\gamma) \;=\; \frac{q(r_i,z)}{\varrho(r_i,z)^2} \cdot \cos\big[\, k\,(z + \varrho(r_i,z))\,\big]\, e^{-r_i/\gamma}, \quad \varrho(r,z) = \sqrt{r^2+z^2}$

$\chi^2(E_0, z, \gamma) \;=\; \big\langle\, g - E_0 G,\; g - E_0 G \,\big\rangle_w, \quad \langle a, b\rangle_w \;\equiv\; \sum_{i=1}^M w_i\, a_i\, b_i$

Pre-computation

$\delta_z \leftarrow 2\pi/(k\,\Delta_\lambda)$ ▷ $= (\lambda/n_0)/\Delta_\lambda$: fringe period / Δ_λ
$\mathcal{Z} \leftarrow \{\, z_- + m\,\delta_z \;:\; m = 0, \dots, N_z - 1 \,\}, \quad N_z = \lceil (z_+ - z_-)/\delta_z \rceil$
$\Gamma \leftarrow \{\, \gamma_1, \dots, \gamma_{N_\gamma} \,\} \subset [\gamma_-, \gamma_+]$ ▷ decay-length grid
$\langle g, g\rangle_w \leftarrow \sum_i w_i\, g_i^2$

Profiled grid search

for $\gamma \in \Gamma$ **do**
  for $z \in \mathcal{Z}$ **do**
    $\varrho_i \leftarrow \sqrt{r_i^2 + z^2} \quad \forall\, i$
    $G_i \leftarrow \dfrac{q(r_i, z)}{\varrho_i^2}\, \cos\big[k(z + \varrho_i)\big]\, e^{-r_i/\gamma} \quad \forall\, i$
    $\langle G, G\rangle_w \leftarrow \sum_i w_i\, G_i^2, \quad \langle G, g\rangle_w \leftarrow \sum_i w_i\, G_i\, g_i$
    $\rho(z,\gamma) \leftarrow \dfrac{\langle G, g\rangle_w}{\sqrt{\langle G, G\rangle_w\, \langle g, g\rangle_w \;+\; \varepsilon}}$ ▷ weighted correlation coefficient
    $\hat{E}_0(z,\gamma) \leftarrow \textsc{Clip}\left[\, \dfrac{\langle G, g\rangle_w}{\langle G, G\rangle_w},\; E_-,\; E_+ \right]$ ▷ profiled amplitude
    $\mathcal{L}(z,\gamma) \leftarrow 1 - \rho^2(z,\gamma)$ ▷ $\chi^2_{\min}/\langle g, g\rangle_w$
  end for
end for

Refinement

$(z^\star, \gamma^\star) \leftarrow \underset{(z,\gamma)\,\in\,\mathcal{Z}\times\Gamma}{\arg\min}\; \mathcal{L}(z,\gamma)$ ▷ coarse global minimum
$E_0^\star \leftarrow \hat{E}_0(z^\star, \gamma^\star)$
$(\hat{E}_0, \hat{z}, \hat{\gamma}) \leftarrow \text{L-BFGS-B}\big[\, \sum_i \mathcal{H}_\delta\big(g_i - E_0 G_i(z,\gamma)\big);\; \text{init } (E_0^\star, z^\star, \gamma^\star);\; \text{bounds } [E_\pm], [z_\pm], [\gamma_\pm] \,\big]$
▷ Huber loss $\mathcal{H}_\delta$

return $(\hat{E}_0,\, \hat{z},\, \hat{\gamma})$

Algorithm 5 Dynamic program for the optimal 2D chain of candidate points, Viterbi Algorithm

Require: Candidate sets $\mathcal{C}_t = \left\{ \mathbf{p}_t^{(i)} = \left(x_t^{(i)},\, y_t^{(i)}\right) \right\}_{i=1}^{n_t},\ t = 0, \ldots, F-1$
where $x_t^{(i)} \equiv \texttt{min_idx}[t][i]$ (position) and $y_t^{(i)} \equiv \texttt{min_vals}[t][i]$ (value)
Require: Weights $\mathbf{W} = \mathrm{diag}(w_x, w_y)$, default $\mathbf{W} = \mathbf{I}$
Require: Maximum step $s_{\max} \in (0, \infty]$ ▷ $s_{\max} = \infty \Rightarrow$ unconstrained
Ensure: Chain $\mathbf{k} = (k_0, \ldots, k_{F-1})$, $k_t \in \{1, \ldots, n_t\}$, minimizing

$$J(\mathbf{k}) \;=\; \sum_{t=1}^{F-1} \left\| \mathbf{W}\left(\mathbf{p}_{t-1}^{(k_{t-1})} - \mathbf{p}_t^{(k_t)}\right) \right\|_2 \qquad \text{s.t.} \quad \left\| \mathbf{W}\left(\mathbf{p}_{t-1}^{(k_{t-1})} - \mathbf{p}_t^{(k_t)}\right) \right\|_2 \le s_{\max}$$

together with the corresponding x- and y-chains.

Forward pass ▷ $C_t(j)$: cost of the best chain ending at $\mathbf{p}_t^{(j)}$

$C_0(i) \leftarrow 0 \quad \forall\, i \in \{1, \ldots, n_0\}$
for $t = 1$ **to** $F-1$ **do**
▷ Weighted Euclidean transition cost, $n_{t-1} \times n_t$
$d_t(i,j) \leftarrow \sqrt{w_x^2\left(x_{t-1}^{(i)} - x_t^{(j)}\right)^2 + w_y^2\left(y_{t-1}^{(i)} - y_t^{(j)}\right)^2} \quad \forall\,(i,j)$
$d_t(i,j) \leftarrow \infty$ **if** $d_t(i,j) > s_{\max}$ ▷ forbid over-long steps

▷ Bellman recursion
$\pi_t(j) \leftarrow \arg\min_i \left[\, C_{t-1}(i) + d_t(i,j) \,\right] \quad \forall\, j \in \{1, \ldots, n_t\}$
$C_t(j) \leftarrow C_{t-1}\left(\pi_t(j)\right) + d_t\left(\pi_t(j),\, j\right)$

if $C_t(j) = \infty \ \forall\, j$ **then**
return $\varnothing$ ▷ no admissible chain survives frame t
end if
end for

Backtracking
$k_{F-1} \leftarrow \arg\min_j C_{F-1}(j)$ ▷ cheapest terminal node
for $t = F-1$ **down to** 1 **do**
$k_{t-1} \leftarrow \pi_t(k_t)$
end for

return $\left\{k_t\right\}, \quad \left\{x_t^{(k_t)}\right\}, \quad \left\{y_t^{(k_t)}\right\}$ for $t = 0, \ldots, F-1$